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Brunner et al.

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(54) **RNA COMPOSITIONS AND METHODS FOR INHIBITING ANGPTL8**

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C12N 15/113 (2010.01)

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CPC **C12N 15/113** (2013.01); **C12N 2310/14** (2013.01); **C12N 2310/321** (2013.01); **C12N 2310/322** (2013.01); **C12N 2310/351** (2013.01); **C12N 2310/3515** (2013.01); **C12N 2320/30** (2013.01)

(58) **Field of Classification Search**

CPC **C12N 15/113**; **C12N 2310/14**; **C12N 2310/321**; **C12N 2310/322**; **C12N 2310/351**; **C12N 2310/3515**; **C12N 2320/30**

See application file for complete search history.

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(57) **ABSTRACT**

The present disclosure relates to dsRNAs targeting ANGPTL8, methods of inhibiting ANGPTL8 gene expression, and methods of treating one or more conditions associated with ANGPTL8 gene expression.

16 Claims, 17 Drawing Sheets

Specification includes a Sequence Listing.

FIG. 1A

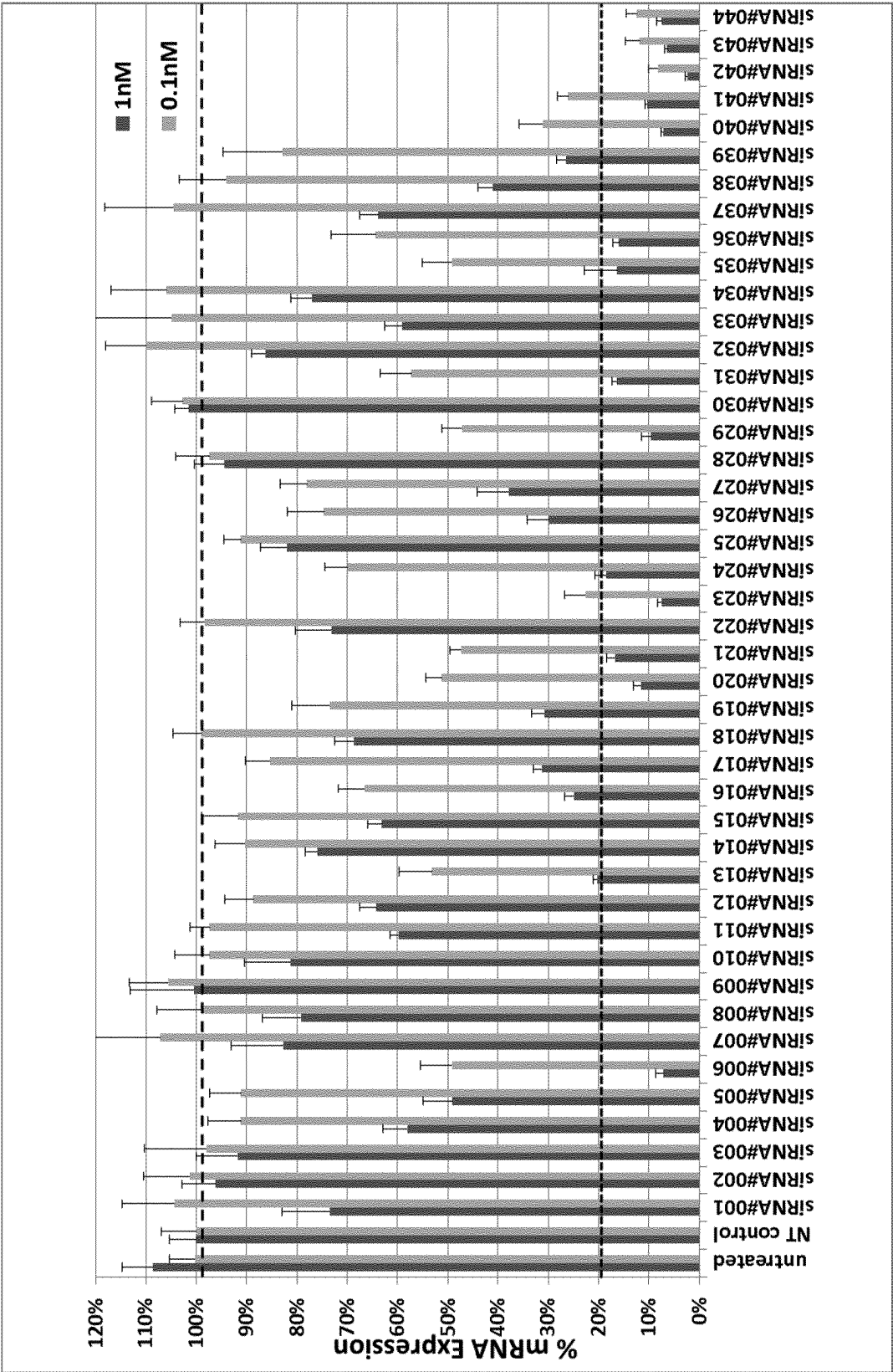


FIG. 1B

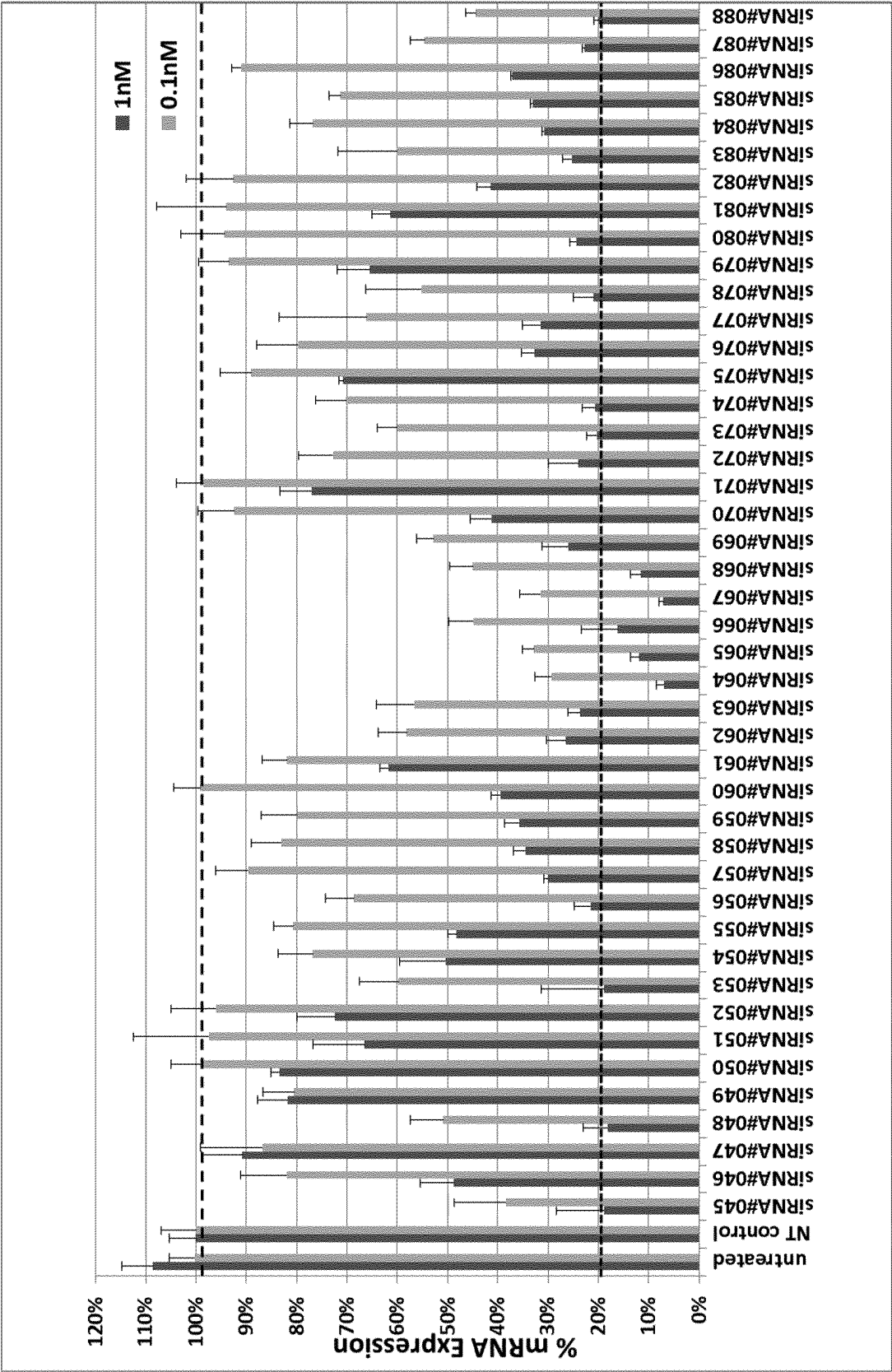


FIG. 1C

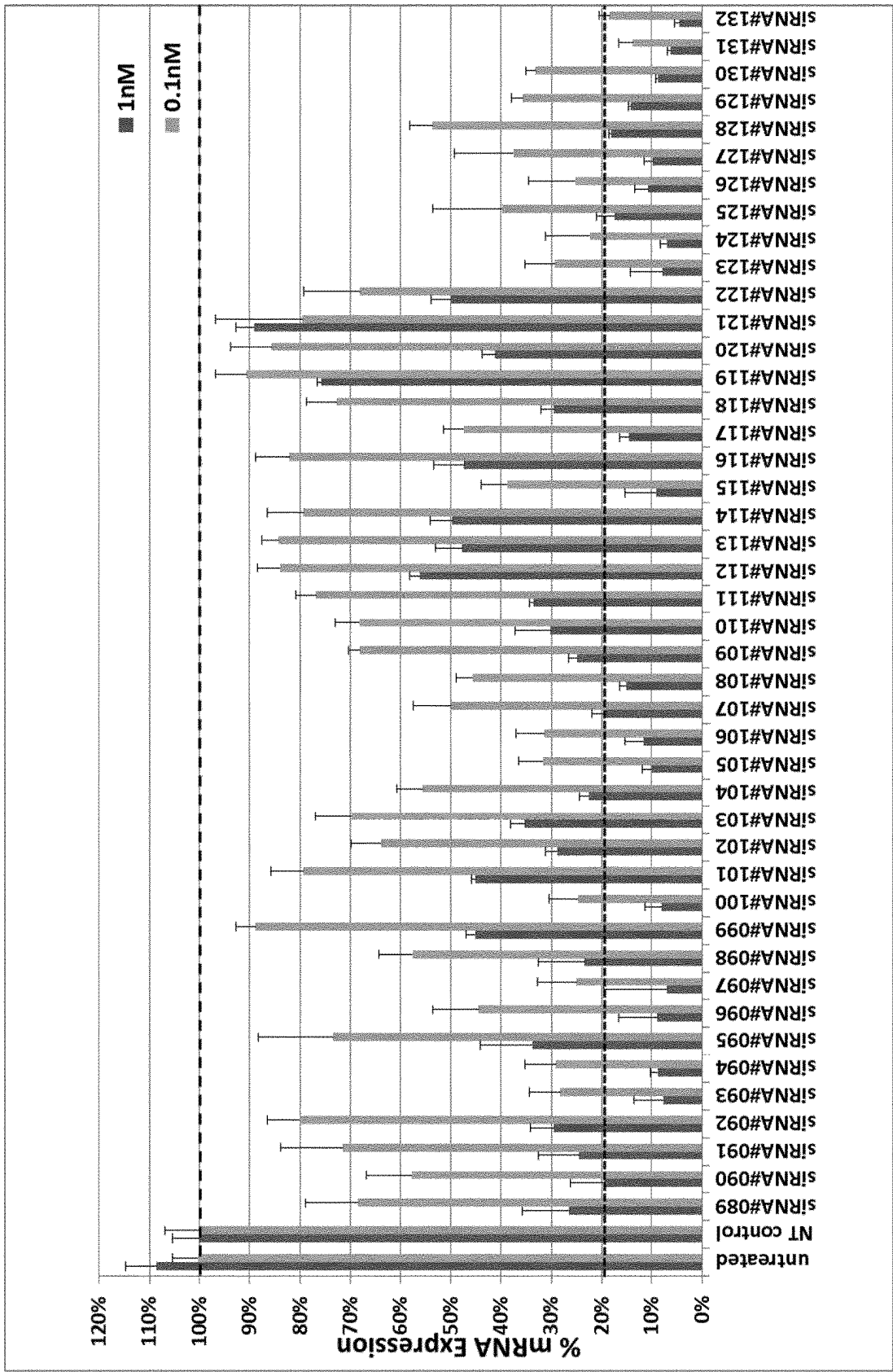


FIG. 2

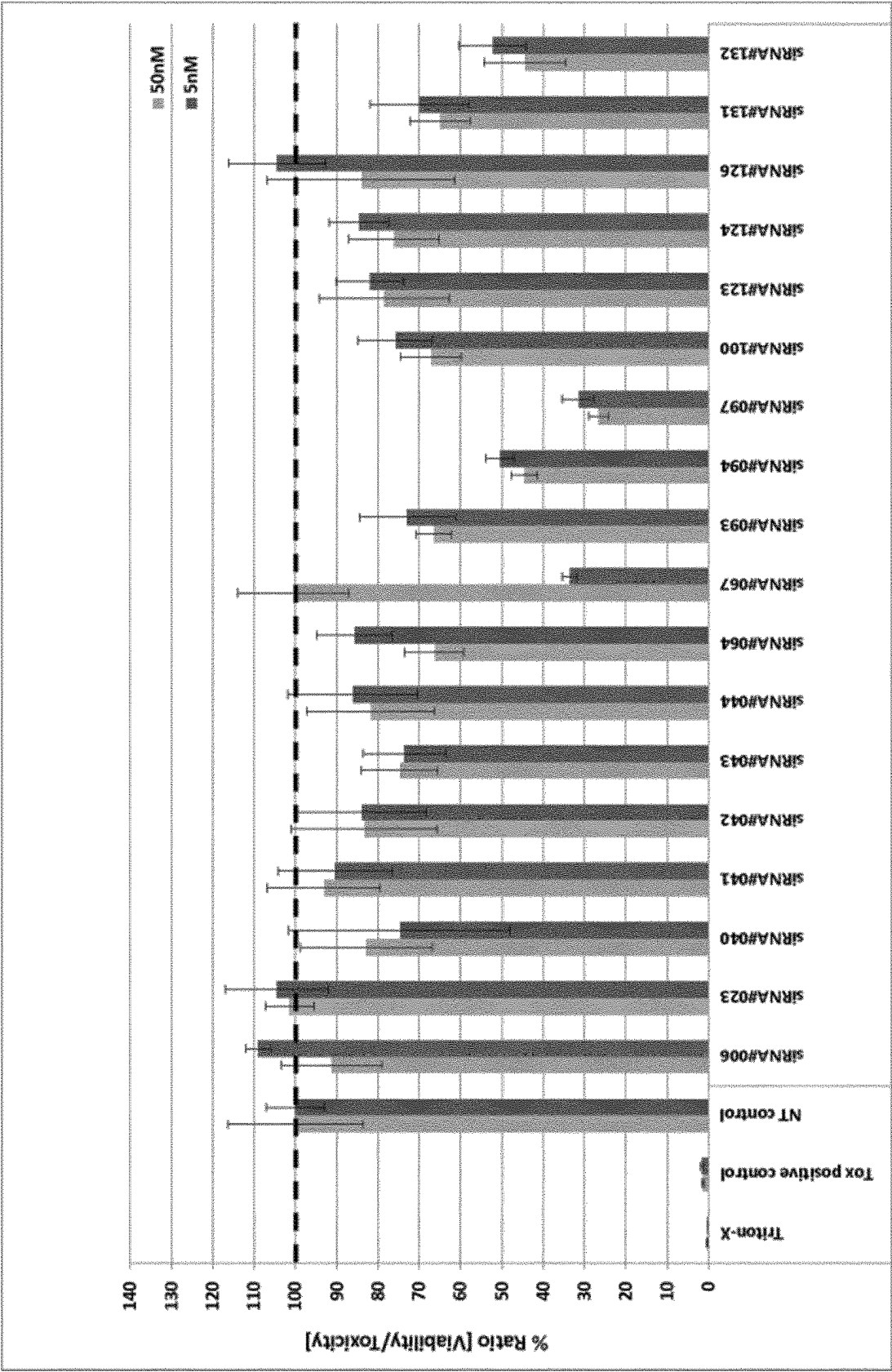


FIG. 3

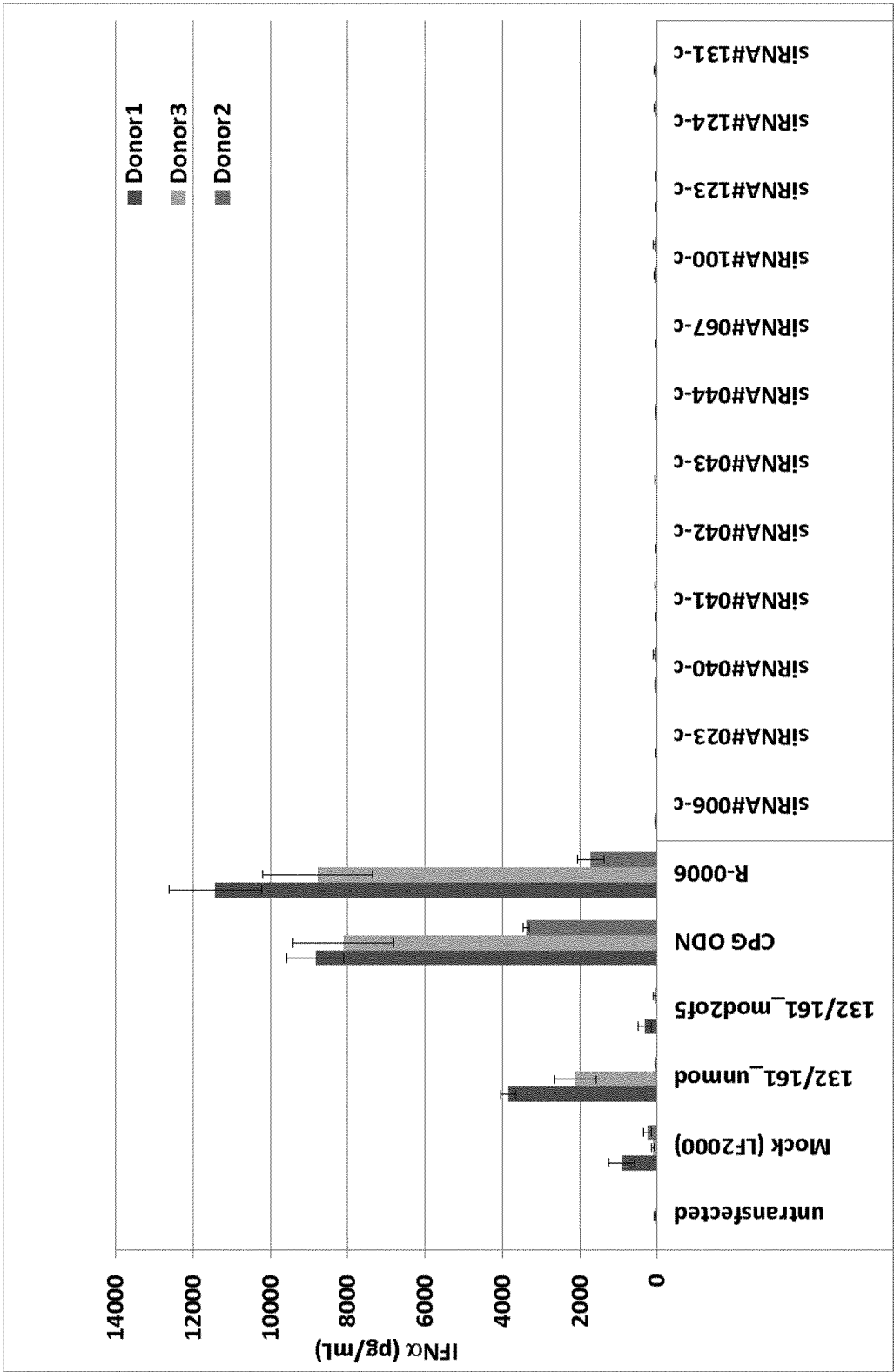


FIG. 4

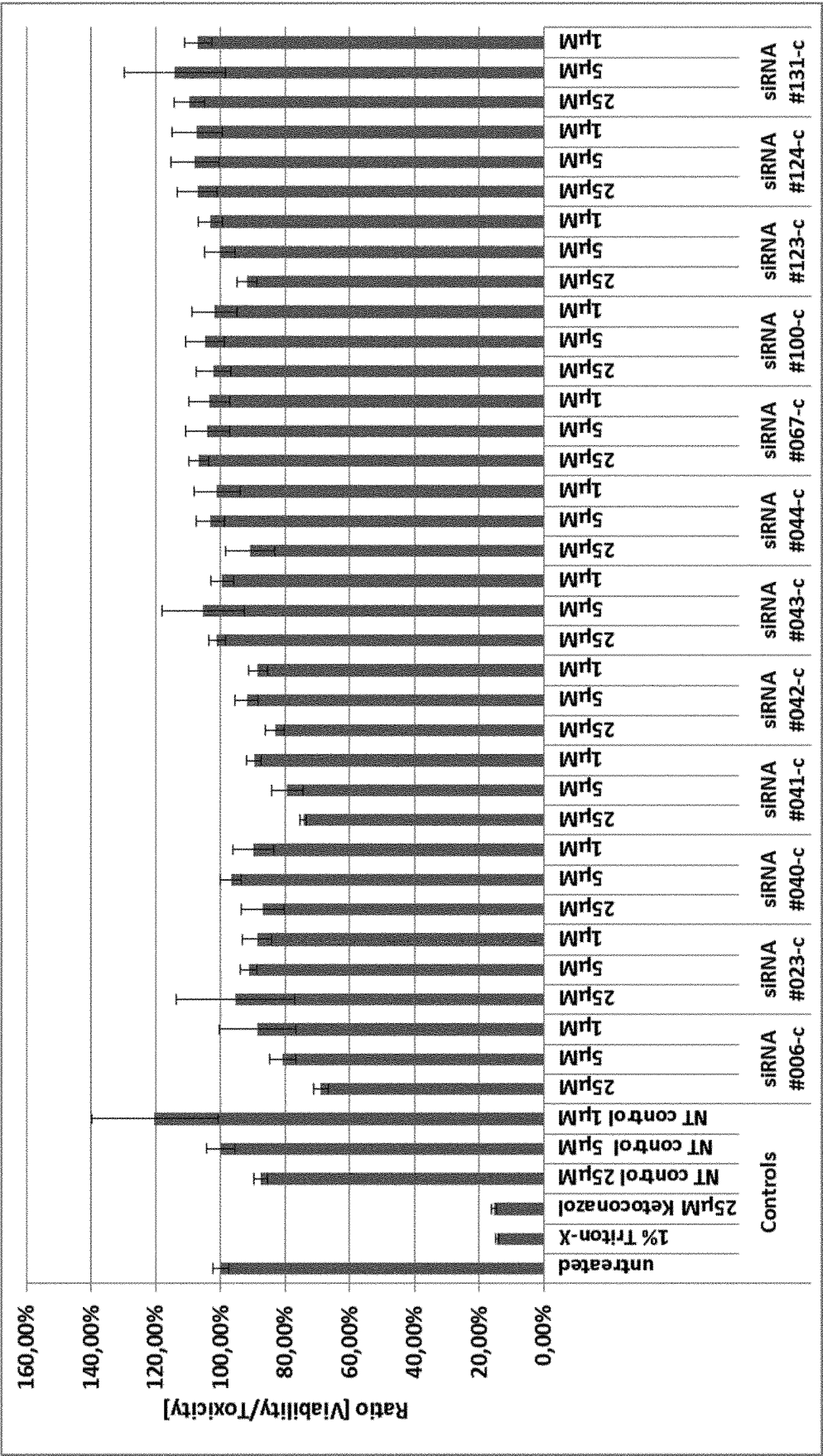


FIG. 5A

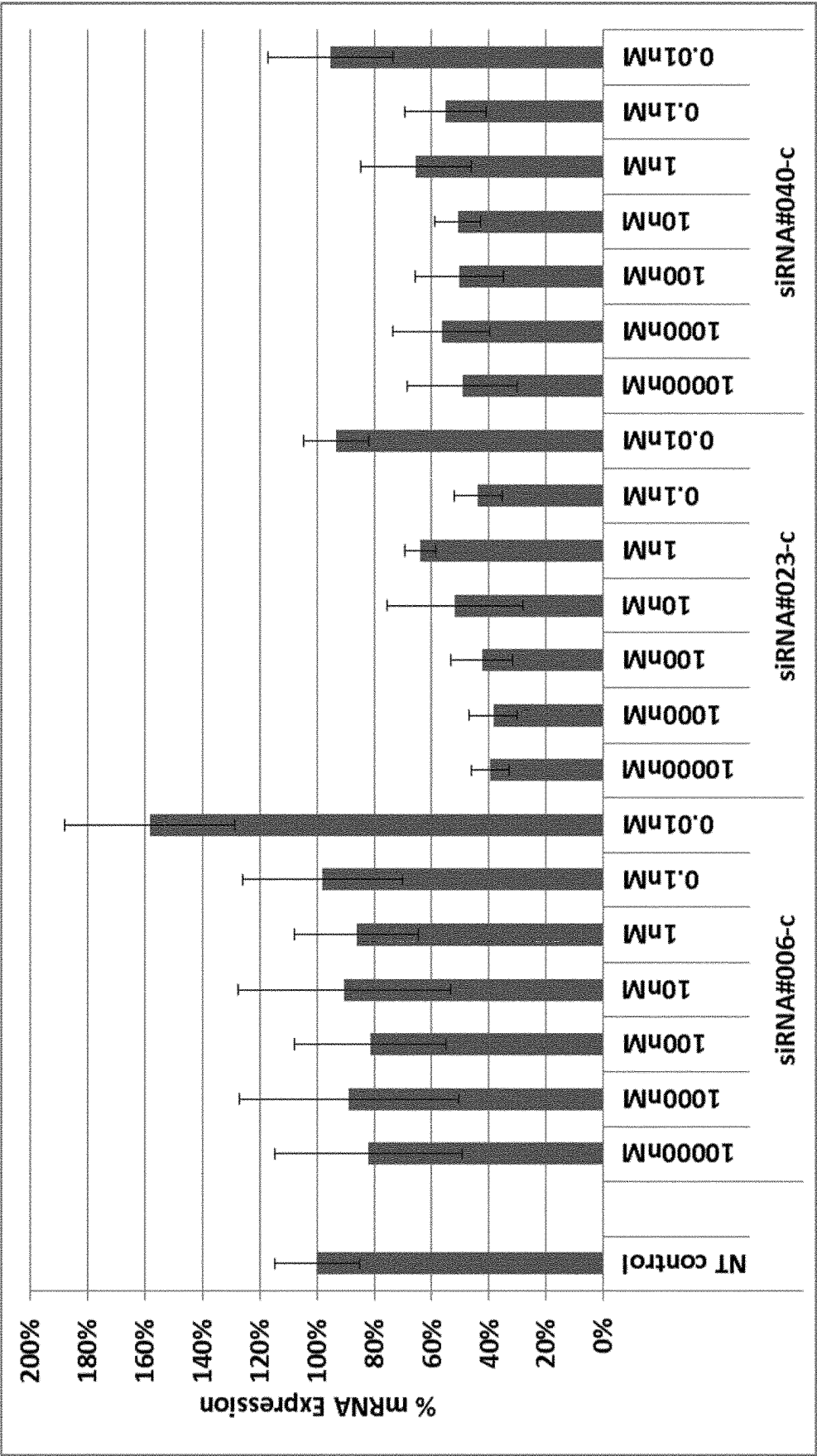


FIG. 5B

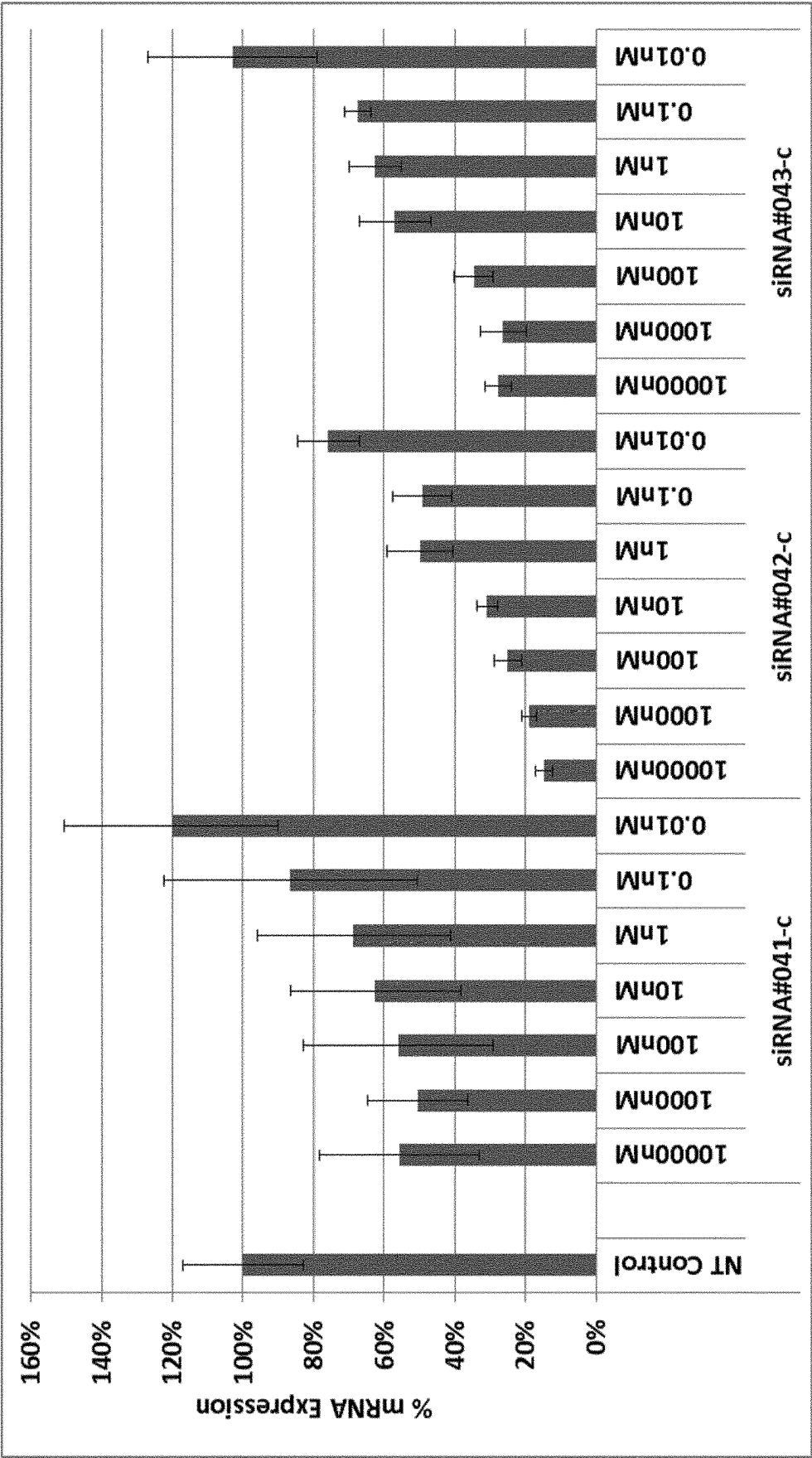


FIG. 5C

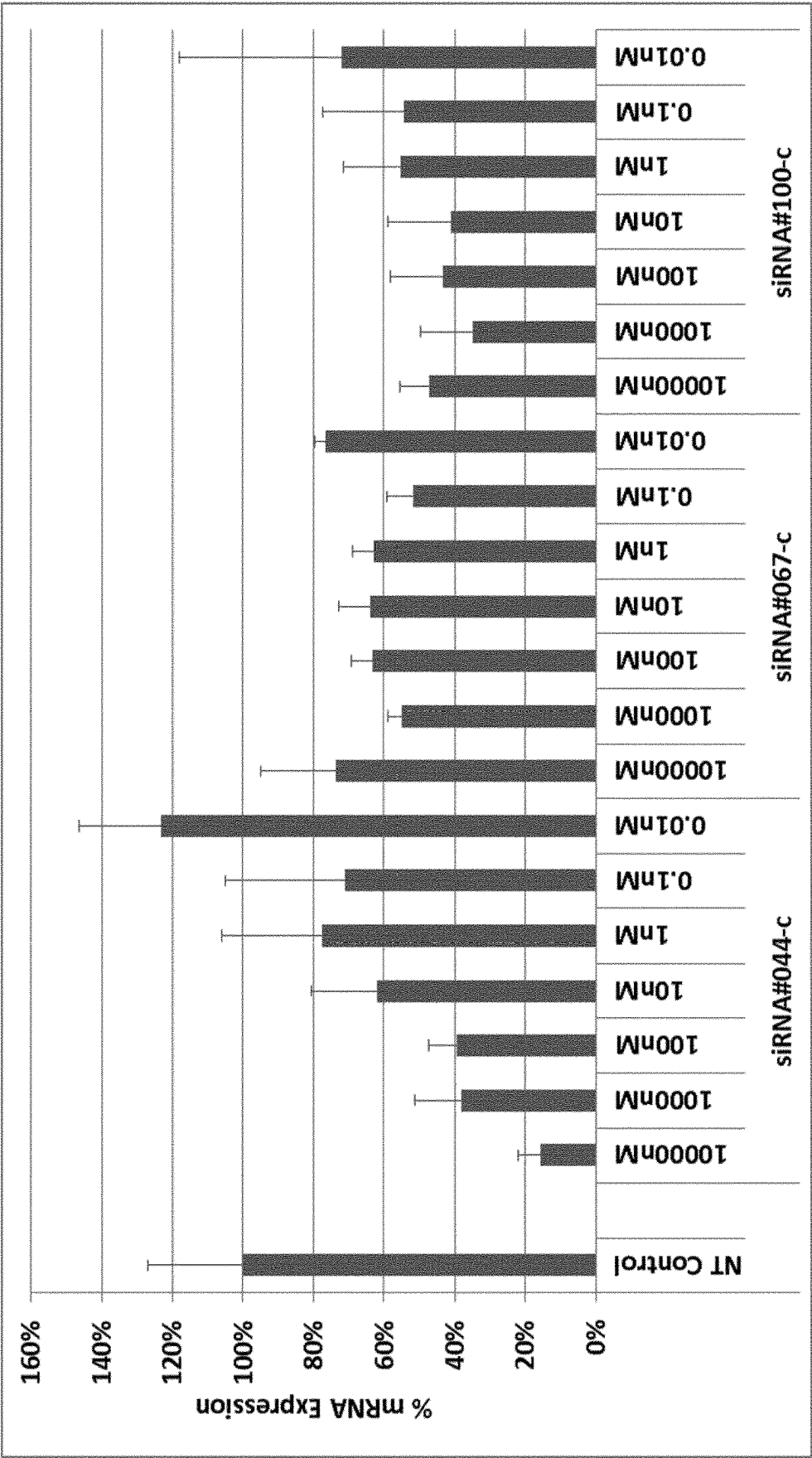


FIG. 5D

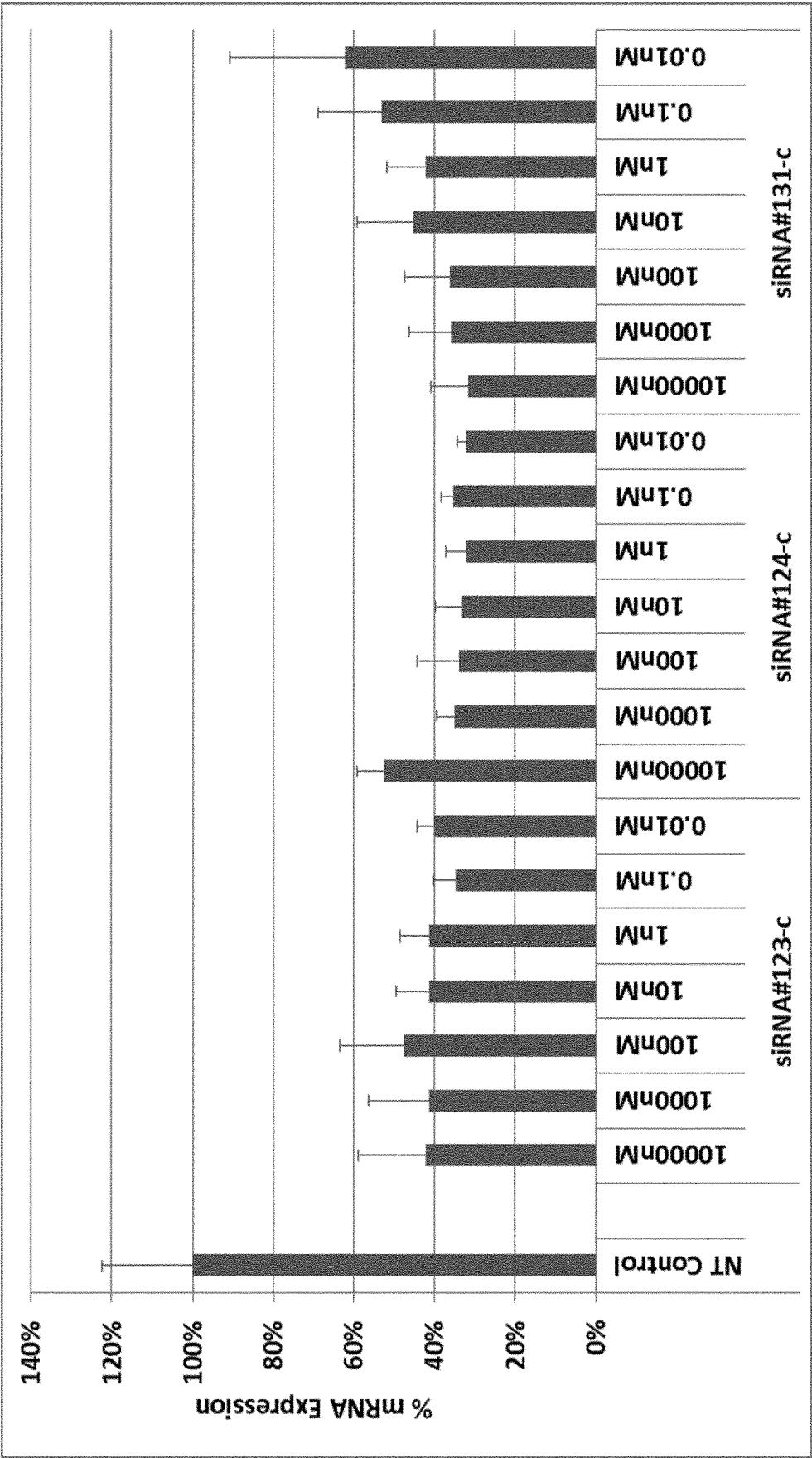


FIG. 6

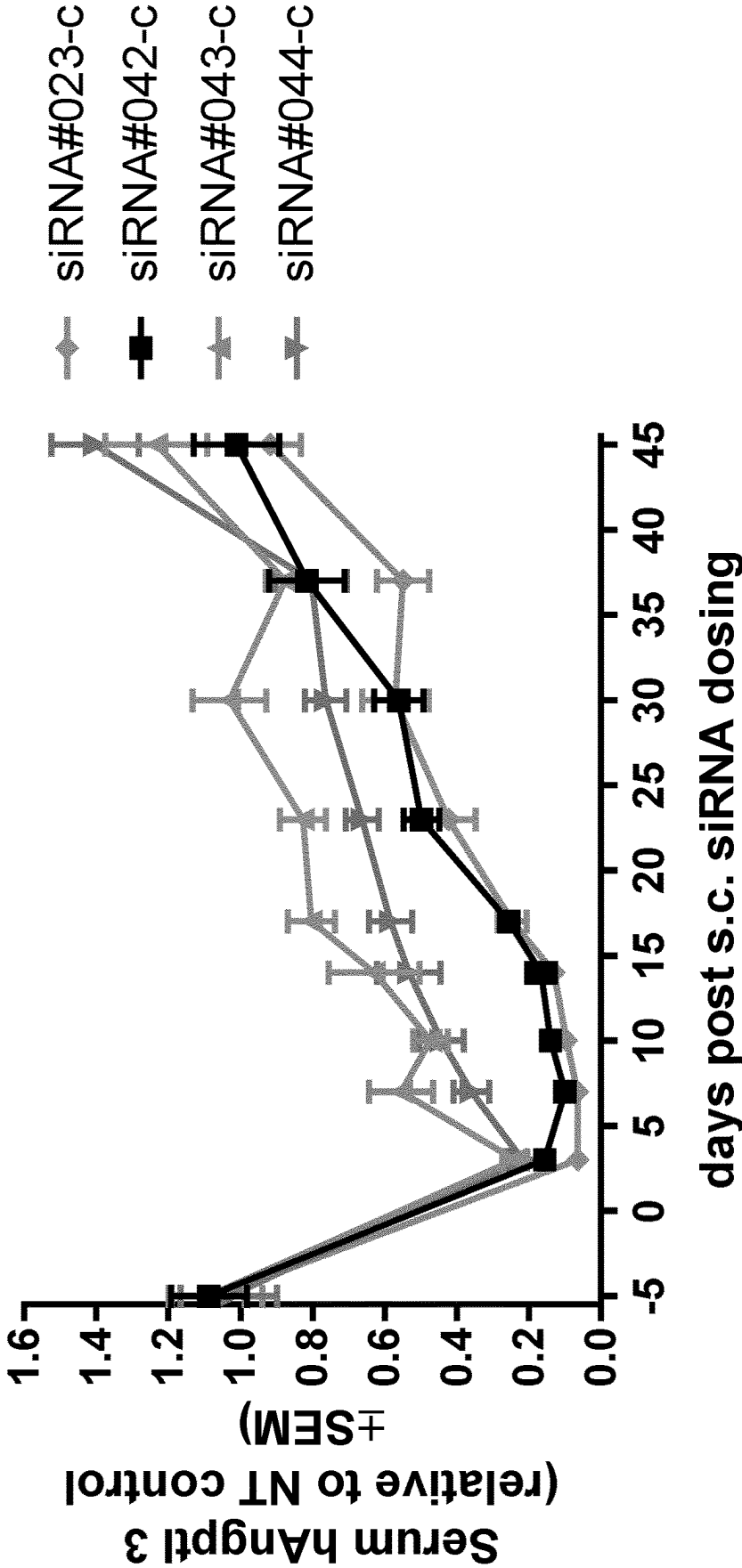


FIG. 7A

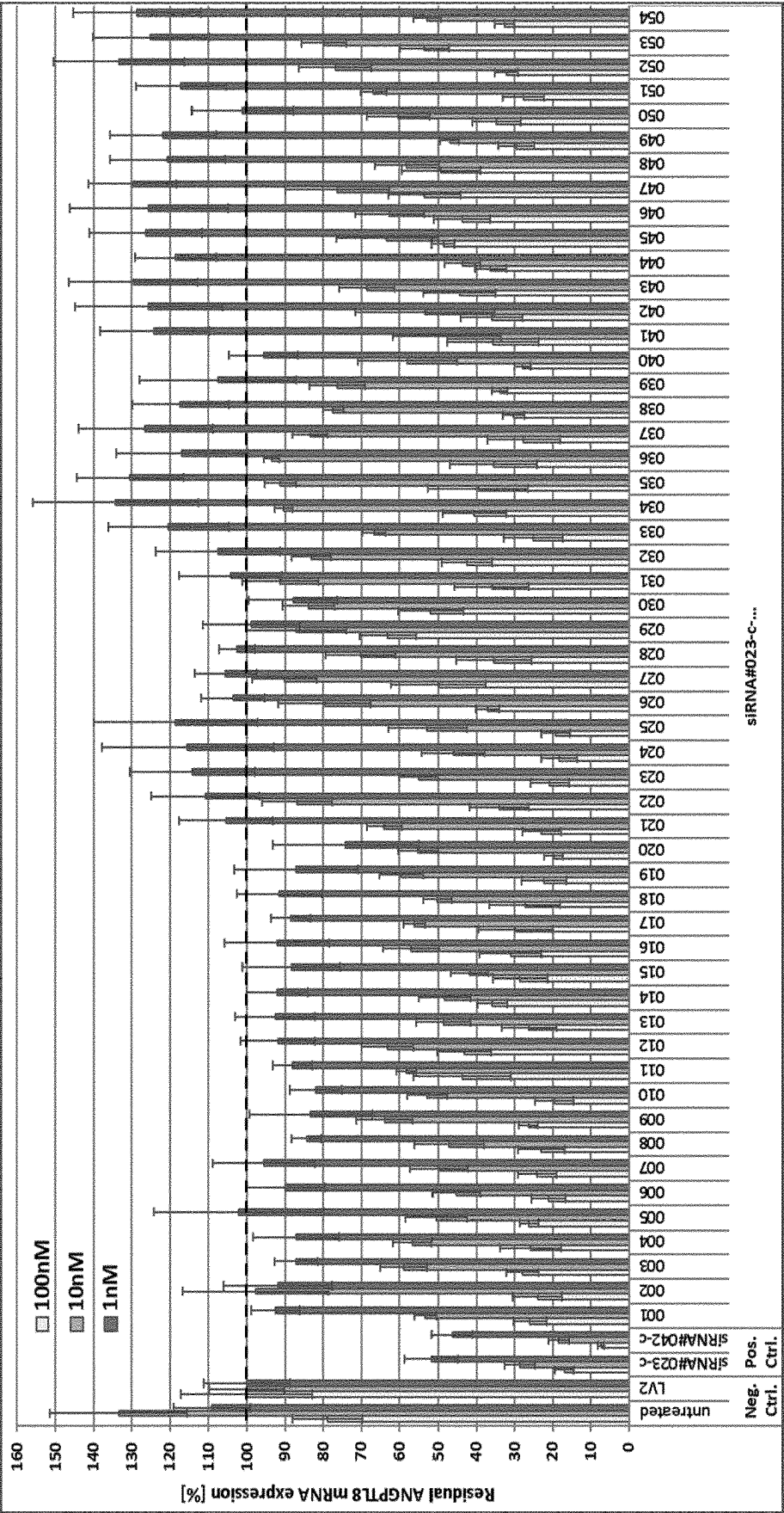


FIG. 7B

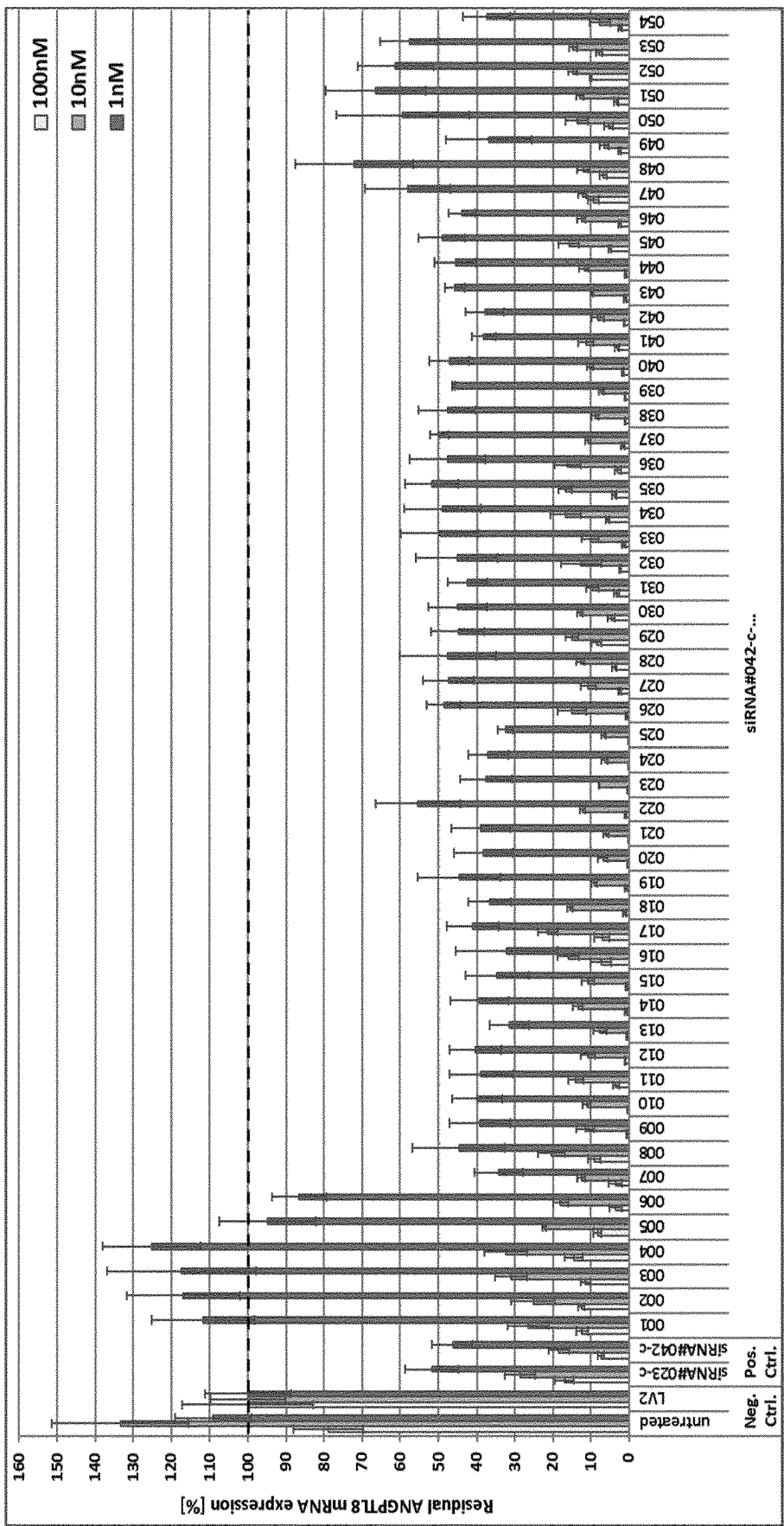


FIG. 8

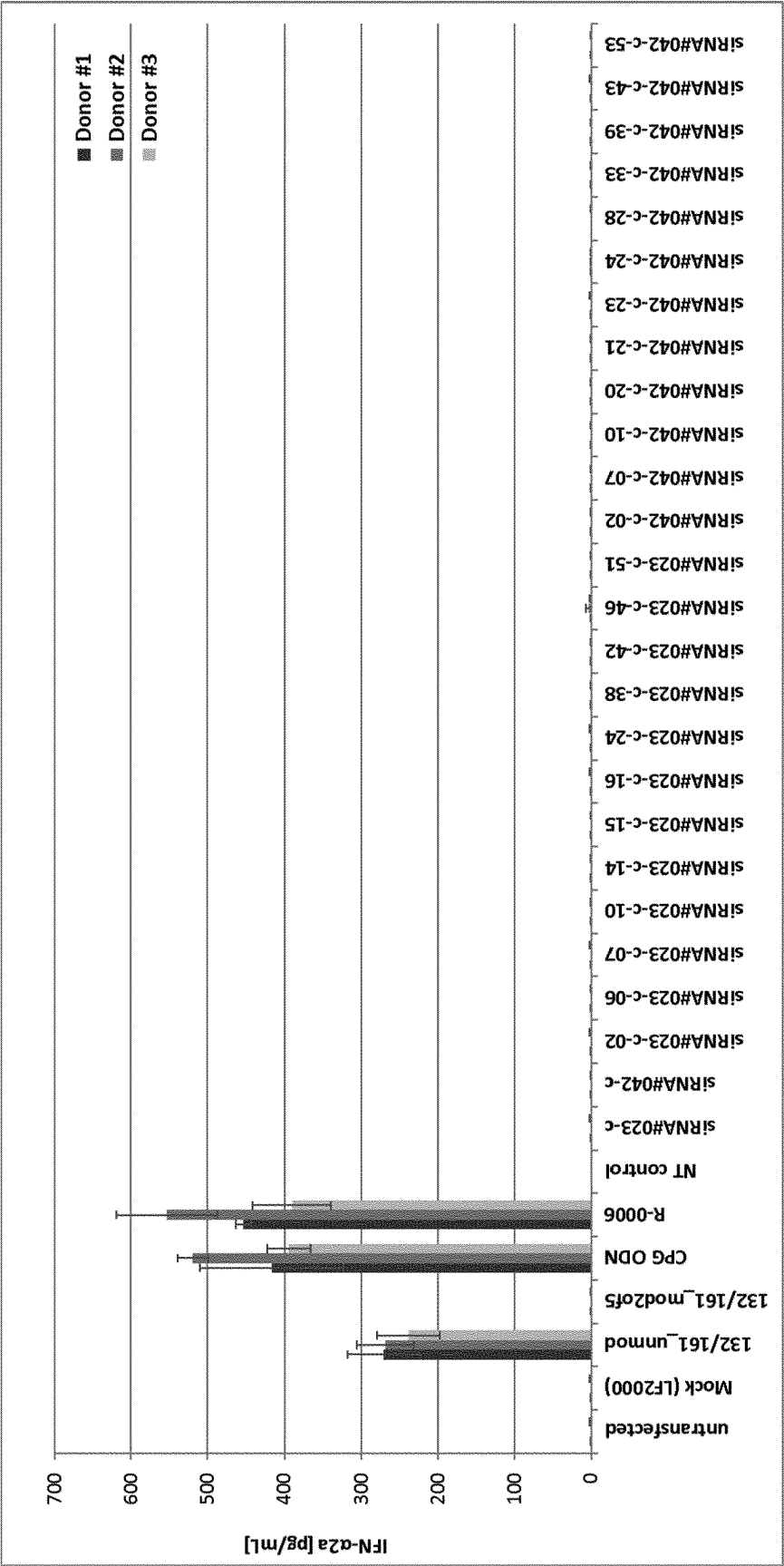


FIG. 9

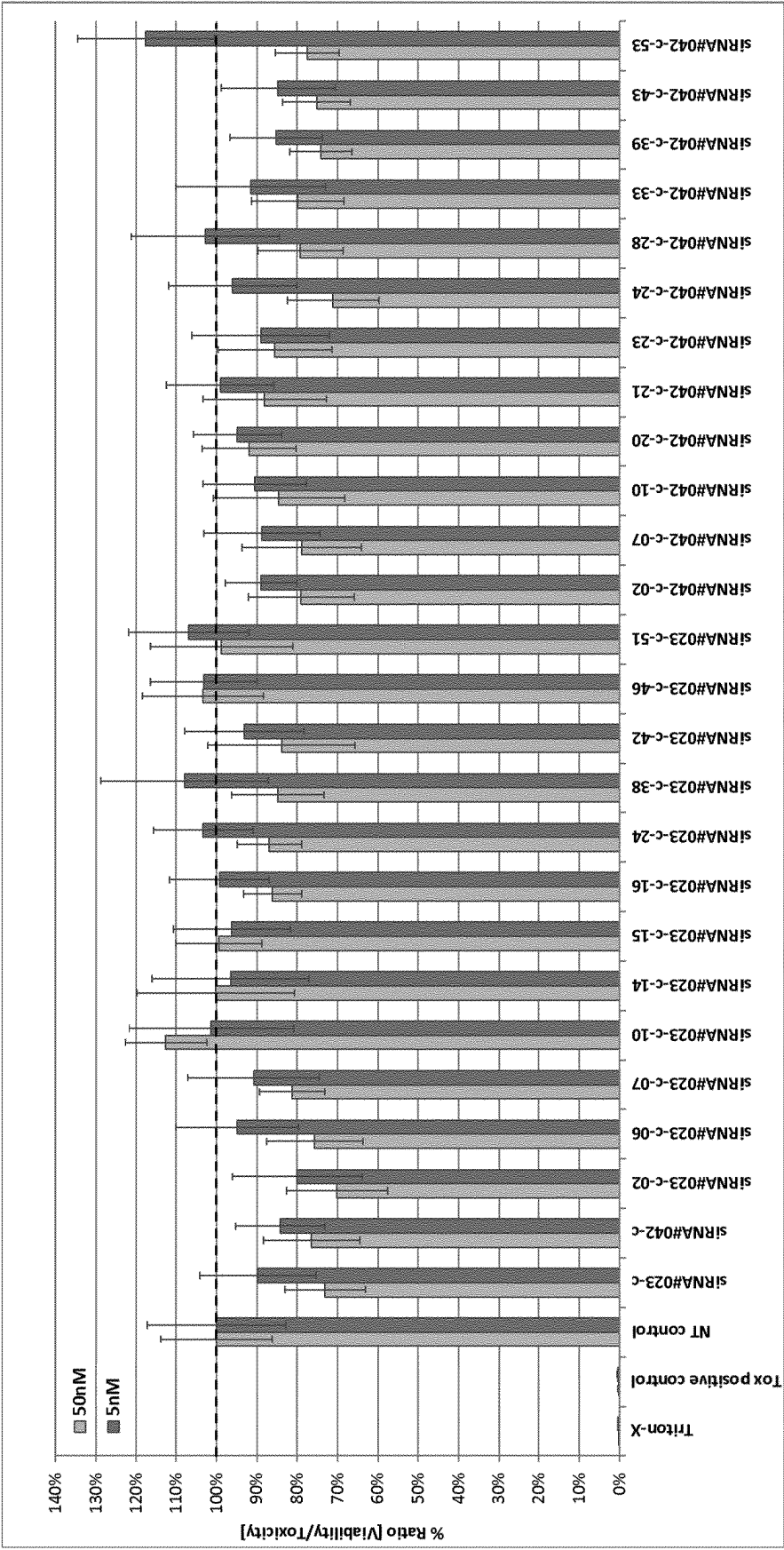


FIG. 10A

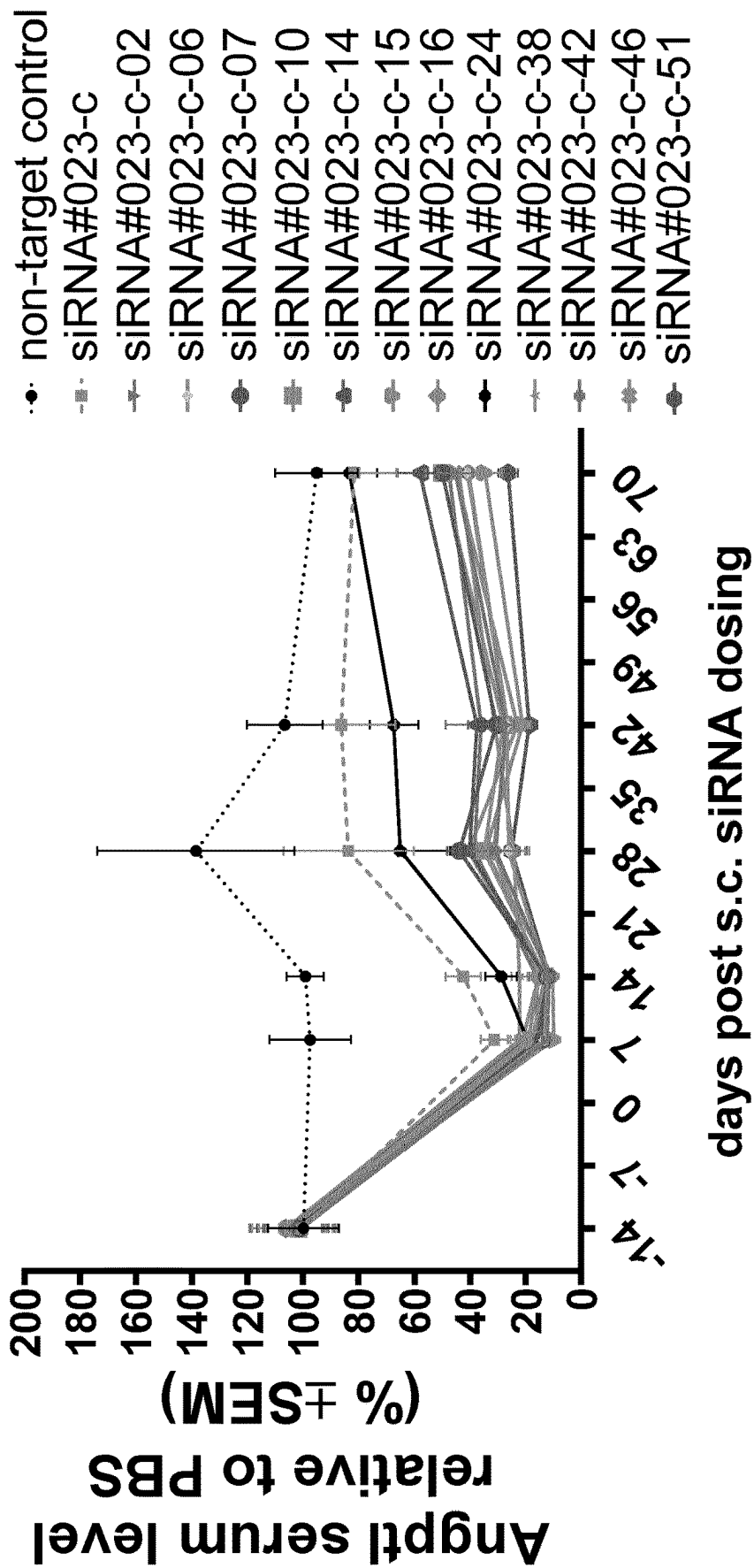
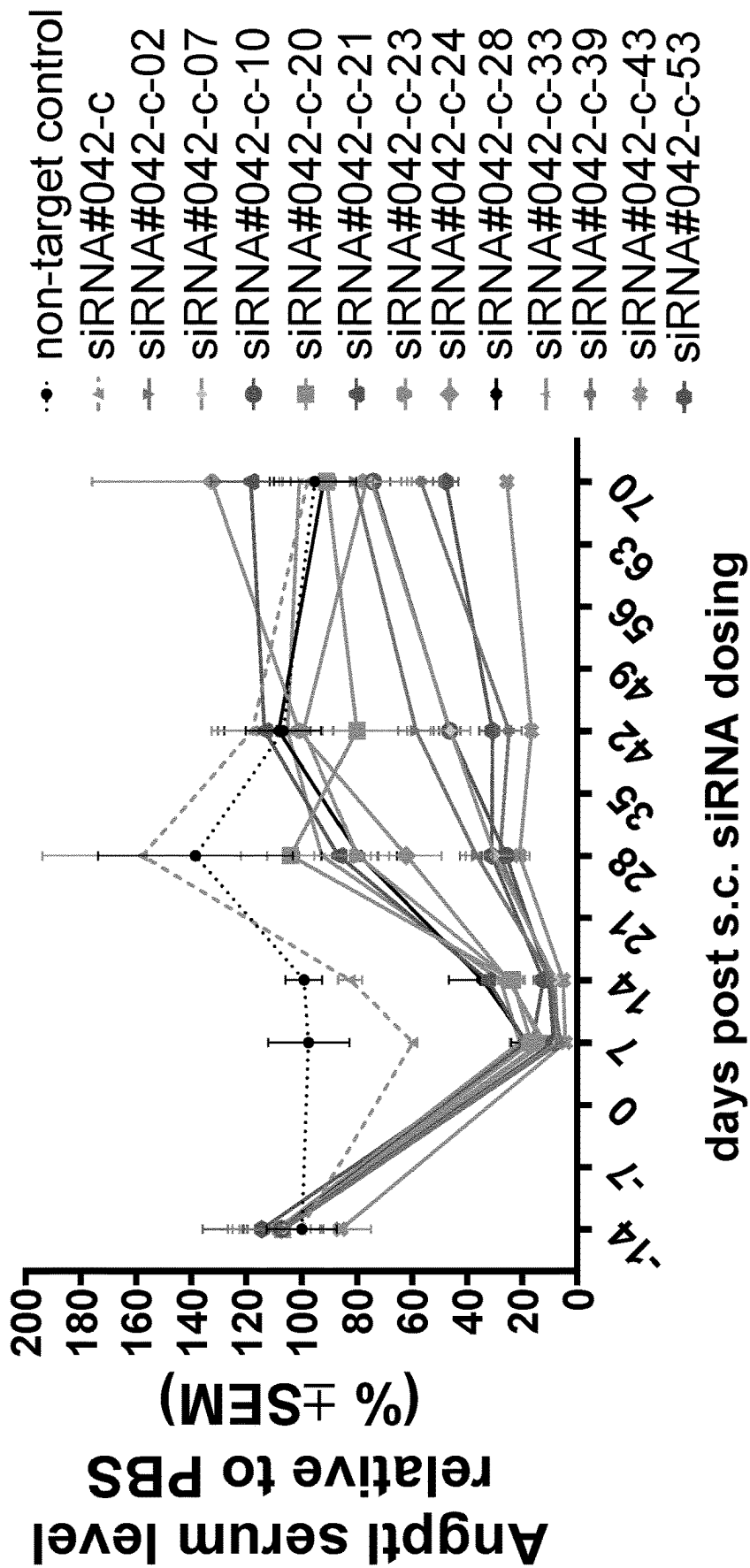


FIG. 10B



RNA COMPOSITIONS AND METHODS FOR INHIBITING ANGPTL8

CROSS-REFERENCE TO RELATED APPLICATIONS

The present application is a national stage application under 35 U.S.C. § 371 of International Patent Application No. PCT/EP2019/082216, filed Nov. 22, 2019, which claims priority to EP patent application Ser. No. 18/306,562.2, filed Nov. 23, 2018. The contents of the aforementioned priority applications are incorporated herein by reference in their entirety.

SEQUENCE LISTING

Nucleic acid sequences are disclosed in the present specification that serve as references. The same sequences are also presented in a sequence listing formatted according to standard requirements for the purpose of patent matters. In case of any sequence discrepancy with the standard sequence listing, the sequences described in the present specification shall be the reference. The instant application contains a Sequence Listing that has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on 15 Nov. 2019, is named 022548_P1058_SL.txt and is 244,191 bytes in size.

BACKGROUND OF THE INVENTION

Angiopietin-like protein 8 (ANGPTL8) is an ANGPTL family member believed to be involved in lipid and potentially glucose metabolism. ANGPTL8, a 22 kDa protein expressed primarily in liver tissue in humans, is also known as betatrophin, lipasin, refeeding-induced fat and liver protein, hepatocellular carcinoma-associated protein TD26, LOC55908RIFL, and C19ORF80. Levels of circulating ANGPTL8 are positively correlated with triglyceride levels. Overexpression of ANGPTL8 in the liver is associated with hypertriglyceridemia, whereas its inactivation reduces plasma triglyceride levels (Quagliarini et. al., *Proc. Natl. Acad. Sci.* (2012) 109(48):19751-6; Wang et. al., *Proc. Natl. Acad. Sci.* (2013) 110:16109-14). Elevated levels of blood triglycerides (e.g., 150 mg/dL or higher) significantly increase the risk of cardiovascular conditions such as heart disease, heart attack, stroke, and atherosclerosis, e.g., by contributing to risk factors such as obesity, hypertension, high cholesterol levels, high blood sugar, and metabolic syndrome. Very high levels of blood triglycerides (e.g., 500 mg/dL or higher) significantly increase the risk of pancreatitis.

Double-stranded RNA molecules (dsRNAs) have been shown to block gene expression in a highly conserved regulatory mechanism known as RNA interference (RNAi). This appears to be a different mechanism of action from that of single-stranded oligonucleotides such as antisense oligonucleotides, antimiRs, and antagomiRs. In RNA interference technology, double-stranded RNAs, such as small interfering RNAs (siRNAs), bind to the RNA-induced silencing complex ("RISC"), where one strand (the "passenger strand" or "sense strand") is displaced and the remaining strand (the "guide strand" or "antisense strand") cooperates with RISC to bind a complementary RNA (the target RNA). Once bound, the target RNA is cleaved by RNA endonuclease Argonaute (AGO) in RISC and then further degraded by RNA exonucleases. RNAi has now been used to develop

a new class of therapeutic agents for treating disorders caused by the aberrant or unwanted expression of a gene.

Due to the importance of ANGPTL8 in regulating triglyceride and lipid metabolism, and the prevalence of diseases associated with elevated triglyceride levels, there is a continuing need to identify inhibitors of ANGPTL8 expression and to test such inhibitors for efficacy and unwanted side effects such as cytotoxicity.

SUMMARY OF THE INVENTION

Provided herein are dsRNAs useful for inhibiting expression of an ANGPTL8 gene. Compared to currently available treatments for conditions associated with elevated levels of triglycerides, the dsRNAs provided herein may provide a superior clinical efficacy. The RNA agents of the present disclosure may reduce elevated triglyceride levels into normal ranges, or maintain normal triglyceride levels, resulting in overall improved health. The RNA agents of the present disclosure may be used to treat conditions such as lipid metabolism disorders characterized in whole or in part by elevated triglyceride levels (e.g., hypertriglyceridemia and associated diseases such as pancreatitis).

Accordingly, provided herein is a double-stranded ribonucleic acid (dsRNA) that inhibits expression of the human angiopoietin-like protein 8 (ANGPTL8) gene by binding to a target sequence on an RNA transcript of the ANGPTL8 gene. The dsRNA contains a sense strand comprising a sense sequence, and an antisense strand comprising an antisense sequence, wherein the sense sequence is at least 90% identical to the target sequence. In some embodiments, the sense strand and the antisense strand of the present dsRNA are complementary to each other over a region of 15-25 contiguous nucleotides. In some embodiments, the sense strand and the antisense strand are no more than 30 nucleotides in length.

In some embodiments, the target sequence of the present dsRNA is nucleotides 211-229, 315-333, 455-473, 456-474, 457-475, 458-476, 459-477, 573-591, 648-666, 843-861, 844-862, 851-869, 455-474, 455-475, 455-476, 455-477, 456-475, 456-476, 456-477, 457-476, or 457-477 of SEQ ID NO: 529. In further embodiments, the target sequence is nucleotides 315-333, 457-475, 458-476, or 459-477 of SEQ ID NO: 529. In some embodiments, the target sequence of the present dsRNA is nucleotides 315-333, 457-475, 458-476, or 459-477 of SEQ ID NO: 529. As used herein, a target sequence defined as the range "x-y" of SEQ ID NO: Z consists of the target sequence beginning at the nucleotide in position x and ending at the nucleotide in position y of the nucleic acid sequence of SEQ ID NO: Z. Illustratively, for the sake of clarity, the target sequence defined as the range "211-229" consists of the target sequence beginning at the nucleotide in position 211 and ending at the nucleotide in position 229 of the nucleic acid sequence of SEQ ID NO: 529.

In some embodiments, the dsRNA comprises an antisense sequence that is at least 90% identical to a nucleotide sequence selected from the group consisting of SEQ ID NOs: 138, 155, 172-176, 199, 232, 255, 256, and 263.

In some embodiments, the sense sequence and the antisense sequence of the present dsRNA are complementary, wherein a) the sense sequence comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 6, 23, 40-44, 67, 100, 123, 124, and 131; or b) the antisense sequence comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 138, 155, 172-176, 199, 232, 255, 256, and 263.

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In some embodiments, the sense strand and antisense strand of the dsRNA respectively comprise the nucleotide sequences of: a) SEQ ID NOs: 6 (sense strand) and 138 (antisense strand); b) SEQ ID NOs: 23 and 155; c) SEQ ID NOs: 40 and 172; d) SEQ ID NOs: 41 and 173; e) SEQ ID NOs: 42 and 174; f) SEQ ID NOs: 43 and 175; g) SEQ ID NOs: 44 and 176; h) SEQ ID NOs: 67 and 199; i) SEQ ID NOs: 100 and 232; j) SEQ ID NOs: 123 and 255; k) SEQ ID NOs: 124 and 256; or l) SEQ ID NOs: 131 and 263. In some embodiments, the sense strand and antisense strand of the dsRNA respectively comprise the nucleotide sequences of: a) SEQ ID NOs: 23 and 155; b) SEQ ID NOs: 42 and 174; c) SEQ ID NOs: 43 and 175; or d) SEQ ID NOs: 44 and 176.

In some embodiments, the dsRNA comprises one or more modified nucleotides, wherein at least one of the one or more modified nucleotides is 2'-deoxy-2'-fluoro-ribonucleotide, 2'-deoxyribonucleotide, or 2'-O-methyl-ribonucleotide. In further embodiments, the dsRNA comprises two or more 2'-O-methyl-ribonucleotides and two or more 2'-deoxy-2'-fluoro-ribonucleotides (e.g., in an alternating pattern). In some embodiments, the sense sequence and the antisense sequence comprise alternating 2'-O-methyl ribonucleotides and 2'-deoxy-2'-fluoro ribonucleotides.

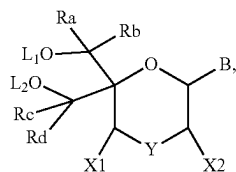
In some embodiments, the dsRNA comprises an inverted 2'-deoxyribonucleotide at the 3'-end of its sense or antisense strand.

In some embodiments, one or both of the sense strand and antisense strand of the present dsRNA comprise a) a 5' overhang comprising one or more nucleotides; and/or b) a 3' overhang comprising one or more nucleotides. In further embodiments, an overhang in the dsRNA comprises two or three nucleotides. In certain embodiments, an overhang in the dsRNA comprises one or more thymines.

In some embodiments, in the present dsRNA, the sense strand comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 270, 287, 304-308, 331, 364, 387, 388, and 395; and/or the antisense strand comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 402, 419, 436-440, 463, 496, 519, 520, and 527. In further embodiments, the sense strand and antisense strand of the dsRNA respectively comprise the nucleotide sequences of: a) SEQ ID NOs: 270 and 402; b) SEQ ID NOs: 287 and 419; c) SEQ ID NOs: 304 and 436; d) SEQ ID NOs: 305 and 437; e) SEQ ID NOs: 306 and 438; f) SEQ ID NOs: 307 and 439; g) SEQ ID NOs: 308 and 440; h) SEQ ID NOs: 331 and 463; i) SEQ ID NOs: 364 and 496; j) SEQ ID NOs: 387 and 519; k) SEQ ID NOs: 388 and 520; or l) SEQ ID NOs: 395 and 527.

In some embodiments, the present dsRNA is conjugated to one or more ligands with or without a linker. The ligand may, for example, be a cholesterol derivative or a lipophilic moiety. The present dsRNA may also be conjugated to one or more N-acetylgalactosamine (GalNAc) groups.

In some embodiments, one or both strands of the present dsRNA comprise one or more compounds having the structure of



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wherein:

B is a heterocyclic nucleobase,

one of L1 and L2 is an internucleoside linking group linking the compound of formula (I) to said strand(s) and the other of L1 and L2 is H, a protecting group, a phosphorus moiety or an internucleoside linking group linking the compound of formula (I) to said strand(s),

Y is O, NH, NR1 or N—C(=O)—R1, wherein R1 is:

a (C1-C20) alkyl group, optionally substituted by one or more groups selected from an halogen atom, a (C1-C6) alkyl group, a (C3-C8) cycloalkyl group, a (C3-C14) heterocycle, a (C6-C14) aryl group, a (C5-C14) heteroaryl group, —O—Z1, —N(Z1)(Z2), —S—Z1, —CN, —C(=J)—O—Z1, —O—C(=J)—Z1, —C(=J)—N(Z1)(Z2), and —N(Z1)—C(=J)—Z2, wherein

J is O or S,

each of Z1 and Z2 is, independently, H, a (C1-C6) alkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group,

a (C3-C8) cycloalkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group,

a group [C(=O)]_m-R2-(O—CH₂—CH₂)_p-R3, wherein

m is an integer meaning 0 or 1,

p is an integer ranging from 0 to 10,

R2 is a (C1-C20) alkylene group optionally substituted by a (C1-C6) alkyl group, —O—Z3, —N(Z3)(Z4), —S—Z3, —CN, —C(=K)—O—Z3, —O—C(=K)—Z3, —C(=K)—N(Z3)(Z4), or —N(Z3)—C(=K)—Z4, wherein

K is O or S,

each of Z3 and Z4 is, independently, H, a (C1-C6) alkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group, and

R3 is selected from the group consisting of a hydrogen atom, a (C1-C6) alkyl group, a (C1-C6) alkoxy group, a (C3-C8) cycloalkyl group, a (C3-C14) heterocycle, a (C6-C14) aryl group or a (C5-C14) heteroaryl group, or R3 is a cell targeting moiety,

X1 and X2 are each, independently, a hydrogen atom, a (C1-C6) alkyl group, and

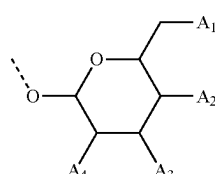
each of Ra, Rb, Rc and Rd is, independently, H or a (C1-C6) alkyl group, or a pharmaceutically acceptable salt thereof.

In some embodiments, the dsRNA comprises one or more compounds of formula (I) wherein Y is a) NR1, R1 is a non-substituted (C1-C20) alkyl group; b) NR1, R1 is a non-substituted (C1-C16) alkyl group, which includes an alkyl group selected from a group comprising methyl, isopropyl, butyl, octyl, and hexadecyl; c) NR1, R1 is a (C3-C8) cycloalkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group; d) NR1, R1 is a cyclohexyl group; e) NR1, R1 is a (C1-C20) alkyl group substituted by a (C6-C14) aryl group; f) NR1, R1 is a methyl group substituted by a phenyl group; g) N—C(=O)—R1, R1 is an optionally substituted (C1-C20) alkyl group; or h) N—C(=O)—R1, R1 is methyl or pentadecyl.

In some embodiments, the dsRNA comprises one or more compounds of formula (I) wherein B is selected from a group consisting of a pyrimidine, a substituted pyrimidine, a purine and a substituted purine, or a pharmaceutically acceptable salt thereof.

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In some embodiments, R3 is of the formula (II):



wherein A1, A2 and A3 are OH,

A4 is OH or $\text{NHC}(=\text{O})\text{—R5}$, wherein R5 is a (C1-C6) alkyl group, optionally substituted by an halogen atom, or a pharmaceutically acceptable salt thereof.

In some embodiments, R3 is N-acetyl-galactosamine, or a pharmaceutically acceptable salt thereof.

In some embodiments, the present dsRNA comprises one or more nucleotides from Tables A and B.

In some embodiments, the present dsRNA comprises 2 to 10 compounds of formula (I), or a pharmaceutically acceptable salt thereof, wherein the compounds are optionally on the sense strand.

In some embodiments, the sense strand of the dsRNA comprises two to five compounds of formula (I) at the 5' end, and/or comprises one to three compounds of formula (I) at the 3' end. In further embodiments, the two to five compounds of formula (I) at the 5' end of the sense strand comprise IgT3 and/or IgT7, optionally comprising three consecutive IgT3 nucleotides; and/or the one to three compounds of formula (I) at the 3' end of the sense strand comprise IT4 or IT3; optionally comprising two consecutive IT4.

In some embodiments, the dsRNA of the present disclosure comprises one or more internucleoside linking groups independently selected from the group consisting of phosphodiester, phosphotriester, phosphorothioate, phosphorodithioate, alkyl-phosphonate and phosphoramidate backbone linking groups, or a pharmaceutically acceptable salt thereof.

In some embodiments, the present dsRNA is a small interfering RNA (siRNA). In some embodiments, the present dsRNA is a short hairpin RNA (shRNA). The present disclosure encompasses a DNA vector encoding such a dsRNA. In some embodiments, the dsRNA is selected from Tables 1-4.

The present disclosure further provides a pharmaceutical composition comprising a dsRNA or DNA vector described herein, and a pharmaceutically acceptable excipient.

Also provided in this disclosure is the present dsRNA, DNA vector, or composition for use in inhibiting ANGPTL8 expression in a human in need thereof, or for use in treating or preventing an ANGPTL8-associated condition in a human in need thereof.

Further provided in this disclosure is a method of inhibiting ANGPTL8 expression, or treating or preventing an ANGPTL8-associated condition, in a mammal (e.g., a human) in need thereof by administering the present dsRNA, DNA vector, or composition to the mammal.

Further provided in this disclosure is the use of the present dsRNA or DNA vector in the manufacture of a medicament for inhibiting ANGPTL8 expression, or treating or preventing an ANGPTL8-associated condition, in a mammal (e.g., a human) in need thereof, as well as articles of manufacture (e.g., kits).

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In some embodiments, the dsRNA inhibits the expression of the ANGPTL8 gene in the liver of a human in the treatment methods. In certain embodiments, the ANGPTL8-associated condition is a lipid metabolism disorder such as hypertriglyceridemia and associated diseases such as pancreatitis.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A, 1B, and 1C are graphs showing RT-qPCR analysis of ANGPTL8 mRNA expression in human Hep3B cell lysates following treatment with 132 test siRNAs as indicated at 0.1 or 1 nM, respectively. Expression of mRNA is represented relative to cells treated with a non-targeting siRNA control. Error bars indicate standard deviation.

FIG. 2 is a graph showing cytotoxic effects of 18 selected test siRNAs in human Hep3B cells. Cells were treated with siRNAs as indicated at 5 or 50 nM before being analyzed for viability (CellTiter-Glo assay) and toxicity (ToxiLight assay). Ratios of the resulting readings are shown relative to results for a non-targeting siRNA control. Error bars indicate standard deviation.

FIG. 3 is a graph of immune stimulation showing the amount of interferon α (IFN α) protein released into the supernatant of human peripheral blood mononuclear cells (PBMCs) isolated from three donors and transfected with selected GalNAc-conjugated siRNAs targeting ANGPTL8 or controls. Protein concentration was determined by ELISA. Error bars indicate standard deviation.

FIG. 4 is a graph showing cytotoxic effects of 12 selected GalNAc-conjugated test siRNAs in human primary hepatocytes following free uptake. Cells were treated with siRNAs as indicated at 1, 5, or 25 μM before being analyzed for viability (CellTiter-Glo assay) and toxicity (ToxiLight assay). Ratios of the resulting readings are shown relative to results for an untreated control and in comparison to toxic positive controls and a non-targeting siRNA control. Error bars indicate standard deviation.

FIGS. 5A-5D are graphs showing RT-qPCR analysis of ANGPTL8 mRNA expression in cynomolgus primary hepatocyte cell lysates following treatment with selected GalNAc-siRNAs at multiple doses as indicated (free uptake). mRNA expression is represented relative to cells treated with a non-targeting siRNA control. Error bars indicate standard deviation.

FIG. 6 is a graph showing serum ANGPTL3 protein levels as an indirect monitoring of ANGPTL8 expression from a bicistronic ANGPTL3-ANGPTL8 construct, over time of mice treated subcutaneously with selected GalNAc-siRNAs at 15 mg/kg at day 0. Treated mice express human ANGPTLs 3 and 8 from a liver specific bicistronic adeno-associated viral vector, where knockdown of mRNA leads to reduction of expression of both proteins. Human ANGPTL3 levels were quantified by ELISA. Error bars indicate standard error of the mean. The curve with symbol “◆” corresponds to siRNA#023-c. The curve with symbol “■” corresponds to siRNA#042-c. The curve with symbol “▲” corresponds to siRNA#043-c. The curve with symbol “▼” corresponds to siRNA#044-c.

FIGS. 7A and 7B are graphs showing RT-qPCR analysis of ANGPTL8 mRNA expression in primary human hepatocytes following treatment of the cells with 2x54 test siRNAs at 1 nM, 10 nM, or 100 nM, respectively. Neg. Ctrl.: negative control (untreated cells and cells treated with a non-targeting siRNA control LV2). Pos. Ctrl.: cells treated with parent constructs siRNA#023-c and siRNA#042-c.

Expression of mRNA is represented relative to cells treated with LV2. Error bars indicate standard deviation.

FIG. 8 is a graph showing immune stimulation indicated by the amount of interferon α 2a (IFN- α 2a) protein released into the supernatant of human peripheral blood mononuclear cells (PBMCs) isolated from three donors and transfected for 24 hours with 100 nM concentration of 24 selected modified GalNAc ANGPTL8 siRNAs together with respective parental sequences (siRNA#023-c and siRNA#042-c) or controls. Protein concentration was determined by ELISA. Error bars indicate standard deviation.

FIG. 9 is a graph showing cytotoxic effects of 24 selected modified GalNAc ANGPTL8 siRNAs together with respective parental sequences (siRNA#023-c and siRNA#042-c) in human Hep3B cells. Cells were transfected with siRNAs as indicated at 5 or 50 nM concentration for 72 hours before being analyzed for viability (CellTiter-Glo assay) and toxicity (ToxiLight assay). Ratios of the resulting readings are shown relative to results for a non-targeting siRNA control. Error bars indicate standard deviation.

FIGS. 10A and 10B are graphs showing serum ANGPTL3 protein levels as a surrogate indicator for ANGPTL8 expression from a bicistronic ANGPTL3-ANGPTL8 AAV8 construct, over time in mice treated subcutaneously with selected GalNAc-siRNAs at 6 mg/kg at day 0. Serum ANGPTL3 levels were measured by ELISA. FIG. 10A: siRNA#023-c and modified variants thereof. FIG. 10B: siRNA#042-c and modified variants thereof. Error bars indicate standard error of the mean.

DETAILED DESCRIPTION OF THE INVENTION

The present disclosure provides novel double-stranded RNAs (dsRNAs) that inhibit expression of an angiopoietin-like protein 8 (ANGPTL8) gene. In some embodiments, the dsRNAs are small interfering RNAs (siRNAs). The dsRNAs can be used to treat conditions such as lipid metabolism disorders (e.g., dyslipidemia, hypertriglyceridemia, and associated diseases such as pancreatitis). Unless otherwise stated, "ANGPTL8" refers to human ANGPTL8 herein. An mRNA sequence of a human ANGPTL8 protein is available under NCBI Reference Sequence No. NM_018687.6 (SEQ ID NO: 529) and its polypeptide sequence is available under NCBI Reference Sequence No. NP_061157.3 (SEQ ID NO: 530). In certain embodiments, the present disclosure refers to cynomolgus ANGPTL8. An mRNA sequence of a cynomolgus ANGPTL8 protein is available under NCBI Reference Sequence No. XM_005588007.1 (SEQ ID NO: 531) and its polypeptide sequence is available under NCBI Reference Sequence No. XP_005588064.1 (SEQ ID NO: 532).

A dsRNA of the present disclosure may have one or more of the following properties: (i) has a half-life of at least 24, 28, 32, 48, 52, 56, 60, or 72 hours in vitro; (ii) does not increase production of interferon α secreted from human primary PBMCs; (iii) has an IC_{50} value of at least 300, 325, 350, 375, 400, 425, 450, 475, 500, 525, 550, 575, or 600 nM for inhibition of human ANGPTL8 expression in vitro; and (iv) reduces protein levels of ANGPTL8 by at least 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% in vivo in C57BL/6 mice expressing human ANGPTL3 and ANGPTL8.

In some embodiments, a dsRNA of the present disclosure has at least one of the following properties: (i) has a half-life of at least 24 hours in vitro; (ii) does not increase production of interferon α secreted from human primary PBMCs, (iii) has an IC_{50} value of at least 475 nM for inhibition of human

ANGPTL8 expression in vitro; and (iv) reduces protein levels of human ANGPTL8 by at least 80% in vivo in C57BL/6 mice expressing human ANGPTL3 and ANGPTL8. In certain embodiments, the dsRNA has all of said properties.

It will be understood by a person skilled in the art that the dsRNAs described herein do not occur in nature ("isolated" dsRNAs).

I. Double-Stranded RNAs

Certain aspects of the present disclosure relate to double-stranded ribonucleic acid (dsRNA) molecules targeting ANGPTL8. As used herein, the term "double-stranded RNA" or "dsRNA" refers to an oligoribonucleotide molecule comprising a duplex structure having two anti-parallel and substantially complementary nucleic acid strands. The two strands forming the duplex structure may be different portions of one larger RNA molecule, or they may be on separate RNA molecules. When the two strands are on separate RNA molecules, the dsRNA structure may function as a small interfering RNA (siRNA). Where the two strands are part of one larger molecule and are connected by an uninterrupted chain of nucleotides between the 3'-end of a first strand and the 5'-end of a second strand, the connecting RNA chain is referred to as a "hairpin loop" and the RNA molecule may be termed "short hairpin RNA," or "shRNA." The RNA strands may have the same or a different number of nucleotides. In addition to the duplex structure, a dsRNA may comprise overhangs of one or more (e.g., 1, 2, or 3) nucleotides.

As used herein, the term "polynucleotide" refers to a polymeric form of nucleotides (e.g., ribonucleotides, deoxyribonucleotides, or modified forms of either type of nucleotide) of at least 10 bases in length. The term includes single and double stranded forms.

A "dsRNA" may include naturally occurring ribonucleotides, and/or chemically modified analogs thereof. As used herein, "dsRNAs" are not limited to those with ribose-containing nucleotides. A dsRNA herein encompasses a double-stranded polynucleotide molecule where the ribose moiety in some or all of its nucleotides has been replaced by another moiety, so long as the resultant double-stranded molecule can inhibit the expression of a target gene by RNA interference. The dsRNA may also include one or more, but not more than 60% (e.g., not more than 50%, 40%, 30%, 20%, or 10%) deoxyribonucleotides or chemically modified analogs thereof.

A dsRNA of the present disclosure comprises a sense strand comprising a sense sequence, and an antisense strand comprising an antisense sequence, wherein the sense strand and the antisense strand are sufficiently complementary to hybridize to form a duplex structure. The term "antisense sequence" refers to a sequence that is substantially or fully complementary, and binds under physiological conditions, to a target RNA sequence in a cell. A "target sequence" refers to a nucleotide sequence on an RNA molecule (e.g., a primary RNA transcript or a messenger RNA transcript) transcribed from a target gene, e.g., an ANGPTL8 gene. The term "sense sequence" refers to a sequence that is substantially or fully complementary to the antisense sequence.

The ANGPTL8-targeting dsRNA of the present disclosure comprises a sense strand comprising a sense sequence and an antisense strand comprising an antisense sequence, wherein the sense and antisense sequences are substantially or fully complementary to each other. Unless otherwise indicated, the term "complementary" refers herein to the ability of a polynucleotide comprising a first contiguous nucleotide sequence, under certain conditions, e.g., physi-

ological conditions, to hybridize to and form a duplex structure with another polynucleotide comprising a second contiguous nucleotide sequence. This may include base-pairing of the two polynucleotides over the entire length of the first or second contiguous nucleotide sequence; in this case, the two nucleotide sequences are considered “fully complementary” to each other. For example, in a case where a dsRNA comprises a first oligonucleotide 21 nucleotides in length and a second oligonucleotide 23 nucleotides in length, and where the two oligonucleotides form 21 contiguous base-pairs, the two oligonucleotides may be referred to as “fully complementary” to each other. Where a first polynucleotide sequence is referred to as “substantially complementary” to a second polynucleotide sequence, the two sequences may base-pair with each other over 80% or more (e.g., 90% or more) of their length of hybridization, with no more than 20% (e.g., no more than 10%) of mismatching base-pairs (e.g., for a duplex of 20 nucleotides, no more than 4 or no more than 2 mismatched base-pairs). Where two oligonucleotides are designed to form a duplex with one or more single-stranded overhangs, such overhangs shall not be regarded as mismatches for the determination of complementarity. Complementarity of two sequences may be based on Watson-Crick base-pairs and/or non-Watson-Crick base-pairs. As used herein, a polynucleotide which is “substantially complementary to at least part of” an mRNA refers to a polynucleotide which is substantially complementary to a contiguous portion of an mRNA of interest (e.g., an mRNA encoding ANGPTL8).

In some embodiments, the ANGPTL8-targeting dsRNA is an siRNA where the sense and antisense strands are not covalently linked to each other. In some embodiments, the sense and antisense strands of the ANGPTL8-targeting dsRNA are covalently linked to each other, e.g., through a hairpin loop (such as in the case of shRNA), or by means other than a hairpin loop (such as by a connecting structure referred to as a “covalent linker”).

I.1 Lengths

In some embodiments, each of the sense sequence (in the sense strand) and the antisense sequence (in the antisense strand) is 9-30 nucleotides in length. For example, each sequence can be any of a range of nucleotide lengths having an upper limit of 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 and an independently selected lower limit of 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20. In some embodiments, the number of nucleotides in each sequence may be 15-25 (i.e., 15 to 25 nucleotides in each sequence), 15-30, 16-29, 17-28, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, or 19-21.

In some embodiments, each sequence is greater than 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length. In some embodiments, each sequence is less than 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, or 31 nucleotides in length. In some embodiments, each sequence is 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length.

In some embodiments, the sense and antisense sequences are each at least 15 and no greater than 25 nucleotides in length. In some embodiments, the sense and antisense sequences are each at least 17 (e.g. at least 19) and no greater than 23 nucleotides in length. For example, the sequences are 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleotides in length.

The sense sequence and antisense sequence may be of the same or different lengths. For example, the antisense

sequence may have 19 nucleotides while the sense sequence may have 17 nucleotides. In another example, the antisense sequence and the sense sequence both have 19 nucleotides.

In some embodiments, the ANGPTL8-targeting dsRNA has sense and antisense strands of the same length or different lengths. For example, the sense strand may be 1, 2, 3, 4, 5, 6, or 7 nucleotides longer than the antisense strand. Alternatively, the sense strand may be 1, 2, 3, 4, 5, 6, or 7 nucleotides shorter than the antisense strand.

In some embodiments, each of the sense strand and the antisense strand is 9-36 nucleotides in length. For example, each strand can be any of a range of nucleotide lengths having an upper limit of 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, or 36 and an independently selected lower limit of 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20. In some embodiments, the number of nucleotides in each strand may be 15-25, 15-30, 16-29, 17-28, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, or 19-21.

In some embodiments, each strand is greater than 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length. In some embodiments, each strand is less than 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, or 37 nucleotides in length. In some embodiments, each strand is 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, or 36 nucleotides in length.

In some embodiments, the sense and antisense strands are each at least 15 and no greater than 25 nucleotides in length. In some embodiments, the sense and antisense strands are each at least 19 and no greater than 23 nucleotides in length. For example, the strands are 19, 20, 21, 22, or 23 nucleotides in length.

In some embodiments, the sense strand may have 21, 22, 23, or 24 nucleotides, including any modified nucleotides, while the antisense strand may have 21 nucleotides, including any modified nucleotides; in certain embodiments, the sense strand may have a sense sequence having 17, 18, or 19 nucleotides, while the antisense strand may have an antisense sequence having 19 nucleotides.

I.2 Overhangs

In some embodiments, a dsRNA of the present disclosure comprises one or more overhangs at the 3'-end, 5'-end, or both ends of one or both of the sense and antisense strands. In some embodiments, the one or more overhangs improve the stability and/or inhibitory activity of the dsRNA.

“Overhang” refers herein to the unpaired nucleotide(s) that protrude from the duplex structure of a dsRNA when a 3' end of a first strand of the dsRNA extends beyond the 5' end of a second strand, or vice versa. “Blunt end” means that there are no unpaired nucleotides at that end of the dsRNA, i.e., no nucleotide overhang. A “blunt-ended” dsRNA is a dsRNA that is double-stranded over its entire length, i.e., no nucleotide overhang at either end of the duplex molecule. Chemical caps or non-nucleotide chemical moieties conjugated to the 3' end and/or the 5' end of a dsRNA are not considered herein in determining whether a dsRNA has an overhang or not.

In some embodiments, an overhang comprises one or more, two or more, three or more, or four or more nucleotides. For example, the overhang may comprise 1, 2, 3, or 4 nucleotides.

In some embodiments, an overhang of the present disclosure comprises one or more nucleotides (e.g., ribonucleotides or deoxyribonucleotides, naturally occurring or chemically modified analogs thereof). In some embodi-

ments, the overhang comprises one or more thymines or chemically modified analogs thereof. In certain embodiments, the overhang comprises one or more thymines.

In some embodiments, the dsRNA comprises an overhang located at the 3'-end of the antisense strand. In some embodiments, the dsRNA comprises a blunt end at the 5'-end of the antisense strand. In some embodiments, the dsRNA comprises an overhang located at the 3'-end of the antisense strand and a blunt end at the 5'-end of the antisense strand. In some embodiments, the dsRNA comprises an overhang located at the 3'-end of the sense strand. In some embodiments, the dsRNA comprises a blunt end at the 5'-end of the sense strand. In some embodiments, the dsRNA comprises an overhang located at the 3'-end of the sense strand and a blunt end at the 5'-end of the sense strand. In some embodiments, the dsRNA comprises overhangs located at the 3'-end of both the sense and antisense strands of the dsRNA.

In some embodiments, the dsRNA comprises an overhang located at the 5'-end of the antisense strand. In some embodiments, the dsRNA comprises a blunt end at the 3'-end of the antisense strand. In some embodiments, the dsRNA comprises an overhang located at the 5'-end of the antisense strand and a blunt end at the 3'-end of the antisense strand. In some embodiments, the dsRNA comprises an overhang located at the 5'-end of the sense strand. In some embodiments, the dsRNA comprises a blunt end at the 3'-end of the sense strand. In some embodiments, the dsRNA comprises an overhang located at the 5'-end of the sense strand and a blunt end at the 3'-end of the sense strand. In some embodiments, the dsRNA comprises overhangs located at both the 5'-end of the sense and antisense strands of the dsRNA.

In some embodiments, the dsRNA comprises an overhang located at the 3'-end of the antisense strand and an overhang at the 5'-end of the antisense strand. In some embodiments, the dsRNA comprises an overhang located at the 3'-end of the sense strand and an overhang at the 5'-end of the sense strand.

In some embodiments, the dsRNA has two blunt ends.

In some embodiments, the overhang is the result of the sense strand being longer than the antisense strand. In some embodiments, the overhang is the result of the antisense strand being longer than the sense strand. In some embodiments, the overhang is the result of sense and antisense strands of the same length being staggered. In some embodiments, the overhang forms a mismatch with the target mRNA. In some embodiments, the overhang is complementary to the target mRNA.

In certain embodiments, a dsRNA of the present disclosure contains a sense strand having the sequence of 5'-CCA-[sense sequence]-invdT, and the antisense strand having the sequence of 5'-[antisense sequence]-dTdT-3', where the trinucleotide CCA may be modified (e.g., 2'-O-Methyl-C and 2'-O-Methyl-A).

I.3 Target and dsRNA Sequences

The antisense strand of a dsRNA of the present disclosure comprises an antisense sequence that may be substantially or fully complementary to a target sequence of 12-30 nucleotides in length in an ANGPTL8 RNA (e.g., an mRNA). For example, the target sequence can be any of a range of nucleotide lengths having an upper limit of 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 and an independently selected lower limit of 12, 13, 14, 15, 16, 17, 18, or 19. In some embodiments, the number of nucleotides in the target sequence may be 15-25, 15-30, 16-29, 17-28, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, or 19-21.

In some embodiments, the target sequence is greater than 12, 13, 14, 15, 16, 17, 18, 19, or 20 nucleotides in length. In some embodiments, the target sequence is less than 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length. In some embodiments, the target sequence is 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 nucleotides in length. In certain embodiments, the target sequence is at least 15 and no greater than 25 nucleotides in length; for example, at least 19 and no greater than 23 nucleotides in length, or 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, or 25 nucleotides in length.

The target sequence may be in the 5' noncoding region, the coding region, or the 3' noncoding region of the ANGPTL8 mRNA transcript. The target sequence may also be located at the junction of the non-coding and coding regions.

In some embodiments, the dsRNA antisense strand comprises an antisense sequence having one or more mismatch (e.g., one, two, three, or four mismatches) to the target sequence. In certain embodiments, the antisense sequence is fully complementary to the corresponding portion in the human ANGPTL8 mRNA sequence and is fully complementary or substantially complementary (e.g., comprises at least one or two mismatches) to the corresponding portion in a cynomolgus ANGPTL8 mRNA sequence. One advantage of such dsRNAs is to allow pre-clinical in vivo studies of the dsRNAs in non-human primates such as cynomolgus monkeys. In certain embodiments, the dsRNA sense strand comprises a sense sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identical to the target sequence (e.g., in human or cynomolgus ANGPTL8 mRNA).

In some embodiments, the target sequence in a human ANGPTL8 mRNA sequence (SEQ ID NO: 529) has the start and end nucleotide positions at or around (e.g., within 3 nucleotides of) the following nucleotides: 211 and 229, 315 and 333, 455 and 473, 456 and 474, 457 and 475, 458 and 476, 459 and 477, 573 and 591, 648 and 666, 843 and 861, 844 and 862, 851 and 869, 455 and 474, 455 and 475, 455 and 476, 455 and 477, 456 and 475, 456 and 476, 456 and 477, 457 and 476, or 457 and 477, respectively. In certain embodiments, the target sequence corresponds to nucleotide positions 315-333, 457-475, 458-476, or 459-477 of the human ANGPTL8 mRNA sequence, where the start and end positions may vary within 3 nucleotides of the numbered positions. In some embodiments, the target sequence is a sequence listed in Table 1 as a sense sequence, or a sequence that includes at least 80% nucleotides (e.g., at least 90%) of the listed sequence.

In some embodiments, a dsRNA of the present disclosure comprises a sense strand comprising a sense sequence shown in Table 1. For example, the sense strand comprises a sequence selected from SEQ ID NOs: 6, 23, 40-44, 67, 100, 123, 124, and 131, or a sequence having at least 15, 16, 17, or 18 contiguous nucleotides derived from said selected sequence. In certain embodiments, the sense strand comprises a sequence selected from SEQ ID NOs: 23 and 42-44.

In some embodiments, a dsRNA of the present disclosure comprises an antisense strand comprising an antisense sequence shown in Table 1. In some embodiments, the antisense strand comprises a sequence selected from SEQ ID NOs: 138, 155, 172-176, 199, 232, 255, 256, and 263, or a sequence having at least 15, 16, 17, or 18 contiguous nucleotides derived from said selected sequence. In certain embodiments, the antisense strand comprises a sequence selected from SEQ ID NOs: 155 and 174-176.

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In some embodiments, a dsRNA of the present disclosure comprises a sense strand comprising a sense sequence shown in Table 1 and an antisense strand comprising an antisense sequence shown in Table 1. In some embodiments, the sense and antisense strands respectively comprise the sequences of:

SEQ ID NOs: 6 and 138;
 SEQ ID NOs: 23 and 155;
 SEQ ID NOs: 40 and 172;
 SEQ ID NOs: 41 and 173;
 SEQ ID NOs: 42 and 174;
 SEQ ID NOs: 43 and 175;
 SEQ ID NOs: 44 and 176;
 SEQ ID NOs: 67 and 199;
 SEQ ID NOs: 100 and 232;
 SEQ ID NOs: 123 and 255;
 SEQ ID NOs: 124 and 256; or
 SEQ ID NOs: 131 and 263.

In certain embodiments, the sense and antisense strands respectively comprise the sequences of:

SEQ ID NOs: 23 and 155;
 SEQ ID NOs: 42 and 174;
 SEQ ID NOs: 43 and 175; or
 SEQ ID NOs: 44 and 176.

In some embodiments, the antisense sequence is fully complementary to a sequence selected from SEQ ID NOs: 6, 23, 40-44, 67, 100, 123, 124, and 131. In some embodiments, the antisense sequence is substantially complementary to a sequence selected from SEQ ID NOs: 6, 23, 40-44,

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67, 100, 123, 124, and 131, wherein the antisense sequence comprises at least one mismatch (e.g., one, two, three, or four mismatches) to the selected sequence.

In some embodiments, the antisense sequence is fully complementary to a sequence selected from SEQ ID NOs: 23 and 42-44. In some embodiments, the antisense sequence is substantially complementary to a sequence selected from SEQ ID NOs: 23 and 42-44, wherein the antisense sequence comprises at least one mismatch (e.g., one, two, three, or four mismatches) to the selected sequence.

In some embodiments, the antisense sequence of the ANGPTL8-targeting dsRNA comprises one or more mismatches to the target sequence (for example, due to allelic differences among individuals in a general population). For example, the antisense sequence comprises one or more mismatches (e.g., one, two, three, or four mismatches) to the target sequence. In some embodiments, the one or more mismatches are not located in the center of the region of complementarity. In some embodiments, the one or more mismatches are located within five, four, three, two, or one nucleotide of the 5' and/or 3' ends of the region of complementarity. For example, for a dsRNA containing a 19 nucleotide antisense sequence, in some embodiments the antisense sequence may not contain any mismatch within the central 9 nucleotides of the region of complementarity between it and its target sequence in the ANGPTL8 mRNA.

Table 1 below lists the sense and antisense sequences of exemplary siRNA constructs (CNST). The start (ST) and end (ED) nucleotide positions in NM_018687.6 (SEQ ID NO: 529) are indicated. "SEQ" denotes SEQ ID NOs.

TABLE 1

Sequences of Exemplary siRNA Constructs						
CNST#	ST	ED	Sense Sequence (5'-3')	SEQ	Antisense Sequence (5'-3')	SEQ
001	5	23	CUUAGACCCUCAGUCAUGC	1	GCAUGACUGAGGGUCUAAG	133
002	99	117	CAGAACUGGCACAGCAUGA	2	UCAUGCUGUGCCAGUUCUG	134
003	100	118	AGAACUGGCACAGCAUGAG	3	CUCAUGCUGUGCCAGUUCU	135
004	197	215	CUGACAAAGGCCAGGAACA	4	UGUUCUGGCCUUUGUCAG	136
005	198	216	UGACAAAGGCCAGGAACAG	5	CUGUUCUGGCCUUUGUCA	137
006	211	229	GAACAGCCUGGGUCUCUAU	6	AUAGAGACCCAGGCUGUUC	138
007	212	230	AACAGCCUGGGUCUCUAUG	7	CAUAGAGACCCAGGCUGUU	139
008	222	240	GUCUCUAUGGCCGCACAAU	8	AUUGUGCGGCCAUAGAGAC	140
009	288	306	AACUUCGGGCAAGCCUGUU	9	AACAGGCUUGCCGAAGUU	141
010	297	315	CAAGCCUGUUGGAGACUCA	10	UGAGUCUCCAACAGGCUUG	142
011	298	316	AAGCCUGUUGGAGACUCAG	11	CUGAGUCUCCAACAGGCUU	143
012	299	317	AGCCUGUUGGAGACUCAGA	12	UCUGAGUCUCCAACAGGCU	144
013	300	318	GCCUGUUGGAGACUCAGAU	13	AUCUGAGUCUCCAACAGGC	145
014	301	319	CCUGUUGGAGACUCAGAUG	14	CAUCUGAGUCUCCAACAGG	146
015	302	320	CUGUUGGAGACUCAGAUGG	15	CCAUCUGAGUCUCCAACAG	147
016	303	321	UGUUGGAGACUCAGAUGGA	16	UCCAUCUGAGUCUCCAACA	148
017	304	322	GUUGGAGACUCAGAUGGAG	17	CUCCAUCUGAGUCUCCAAC	149
018	305	323	UUGGAGACUCAGAUGGAGG	18	CCUCCAUCUGAGUCUCCAA	150
019	310	328	GACUCAGAUGGAGGAGGAU	19	AUCCUCCUCCAUCUGAGUC	151

TABLE 1-continued

Sequences of Exemplary siRNA Constructs						
CNST#	ST	ED	Sense Sequence (5'-3')	SEQ	Antisense Sequence (5'-3')	SEQ
020	312	330	CUCAGAUGGAGGAGGAUUAU	20	AUAUCCUCCUCCAUCUGAG	152
021	313	331	UCAGAUGGAGGAGGAUUAU	21	AAUAUCCUCCUCCAUCUGA	153
022	314	332	CAGAUGGAGGAGGAUAUUC	22	GAAUAUCCUCCUCCAUCUG	154
023	315	333	AGAUGGAGGAGGAUAUUCU	23	AGAAUAUCCUCCUCCAUCU	155
024	316	334	GAUGGAGGAGGAUAUUCUG	24	CAGAAUAUCCUCCUCCAUC	156
025	317	335	AUGGAGGAGGAUAUUCUGC	25	GCAGAAUAUCCUCCUCCAUC	157
026	318	336	UGGAGGAGGAUAUUCUGCA	26	UGCAGAAUAUCCUCCUCCA	158
027	319	337	GGAGGAGGAUAUUCUGCAG	27	CUGCAGAAUAUCCUCCUCC	159
028	320	338	GAGGAGGAUAUUCUGCAGC	28	GCUGCAGAAUAUCCUCCUC	160
029	321	339	AGGAGGAUAUUCUGCAGCU	29	AGCUGCAGAAUAUCCUCCU	161
030	328	346	UAUUCUGCAGCUGCAGGCA	30	UGCCUGCAGCUGCAGAAUA	162
031	416	434	CUAGAAGUCCAGCUGAGGA	31	UCCUCAGCUGGACUUCUAG	163
032	417	435	UAGAAGUCCAGCUGAGGAG	32	CUCCUCAGCUGGACUUCUA	164
033	448	466	CCUGCCUACCGAGAAUUU	33	AAAUUCUCGGUAGGCAGGG	165
034	449	467	CCUGCCUACCGAGAAUUUG	34	CAAAUUCUCGGUAGGCAGG	166
035	450	468	CUGCCUACCGAGAAUUUGA	35	UCAAUUUCUCGGUAGGCAG	167
036	451	469	UGCCUACCGAGAAUUUGAG	36	CUCAAAUUCUCGGUAGGCA	168
037	452	470	GCCUACCGAGAAUUUGAGG	37	CCUCAAAUUCUCGGUAGGC	169
038	453	471	CCUACCGAGAAUUUGAGGU	38	ACCUCAAAUUCUCGGUAGG	170
039	454	472	CUACCGAGAAUUUGAGGUC	39	GACCUCAAAUUCUCGGUAG	171
040	455	473	UACCGAGAAUUUGAGGUCU	40	AGACCUCAAAUUCUCGGUA	172
041	456	474	ACCGAGAAUUUGAGGUCUU	41	AAGACCUCAAAUUCUCGGU	173
042	457	475	CCGAGAAUUUGAGGUCUUA	42	UAAGACCUCAAAUUCUCGG	174
043	458	476	CGAGAAUUUGAGGUCUUA	43	UUAAGACCUCAAAUUCUCG	175
044	459	477	GAGAAUUUGAGGUCUUA	44	UUUAAGACCUCAAAUUCUC	176
045	460	478	AGAAUUUGAGGUCUUAAG	45	CUUUAAGACCUCAAAUUCU	177
046	461	479	GAAUUUGAGGUCUUAAGG	46	CCUUUAAGACCUCAAAUUC	178
047	462	480	AAUUUGAGGUCUUAAGGC	47	GCCUUUAAGACCUCAAAUUC	179
048	463	481	AUUUGAGGUCUUAAGGCU	48	AGCCUUUAAGACCUCAAAU	180
049	464	482	UUUGAGGUCUUAAGGCUC	49	GAGCCUUUAAGACCUCAAA	181
050	465	483	UUGAGGUCUUAAGGCUCA	50	UGAGCCUUUAAGACCUCAA	182
051	466	484	UGAGGUCUUAAGGCUCAC	51	GUGAGCCUUUAAGACCUCA	183
052	467	485	GAGGUCUUAAGGCUCACG	52	CGUGAGCCUUUAAGACCUC	184
053	468	486	AGGUCUUAAGGCUCACGC	53	GCGUGAGCCUUUAAGACCU	185
054	469	487	GGUCUUAAGGCUCACGCU	54	AGCGUGAGCCUUUAAGACC	186
055	470	488	GUCUUAAGGCUCACGCUG	55	CAGCGUGAGCCUUUAAGAC	187
056	471	489	UCUUAAGGCUCACGCUGA	56	UCAGCGUGAGCCUUUAAGA	188

TABLE 1-continued

Sequences of Exemplary siRNA Constructs						
CNST#	ST	ED	Sense Sequence (5'-3')	SEQ	Antisense Sequence (5'-3')	SEQ
057	472	490	CUUAAAGGCUCACGCUGAC	57	GUCAGCGUGAGCCUUUAAG	189
058	473	491	UUAAGGCUCACGCUGACA	58	UGUCAGCGUGAGCCUUUAA	190
059	474	492	UAAAGGCUCACGCUGACAA	59	UUGUCAGCGUGAGCCUUUA	191
060	475	493	AAAGGCUCACGCUGACAAG	60	CUUGUCAGCGUGAGCCUUU	192
061	486	504	CUGACAAGCAGAGCCACAUC	61	AUGUGGCUCUGCUUGUCAG	193
062	487	505	UGACAAGCAGAGCCACAUC	62	GAUGUGGCUCUGCUUGUCA	194
063	489	507	ACAAGCAGAGCCACAUCCU	63	AGGAUGUGGCUCUGCUUGU	195
064	490	508	CAAGCAGAGCCACAUCCUA	64	UAGGAUGUGGCUCUGCUUG	196
065	491	509	AAGCAGAGCCACAUCCUAU	65	AUAGGAUGUGGCUCUGCUU	197
066	492	510	AGCAGAGCCACAUCCUAUG	66	CAUAGGAUGUGGCUCUGCU	198
067	573	591	GACAGAUCCAGGAGAGACU	67	AGUCUCUCCUGGAUCUGUC	199
068	574	592	ACAGAUCCAGGAGAGACUC	68	GAGUCUCUCCUGGAUCUGU	200
069	576	594	AGAUCAGGAGAGACUCCA	69	UGGAGUCUCUCCUGGAUCU	201
070	610	628	AGCCUGAAUCUGCCUGGAU	70	AUCCAGGCAGAUUCAGGCU	202
071	613	631	CUGAAUCUGCCUGGAUGGA	71	UCCAUCAGGCAGAUUCAG	203
072	614	632	UGAAUCUGCCUGGAUGGAA	72	UUCAUCAGGCAGAUUCA	204
073	615	633	GAAUCUGCCUGGAUGGAAC	73	GUUCAUCAGGCAGAUUC	205
074	616	634	AAUCUGCCUGGAUGGAACU	74	AGUCCAUCAGGCAGAUU	206
075	617	635	AUCUGCCUGGAUGGAACUG	75	CAGUCCAUCAGGCAGAU	207
076	618	636	UCUGCCUGGAUGGAACUGA	76	UCAGUCCAUCAGGCAGA	208
077	624	642	UGGAUGGAACUGAGGACCA	77	UGGUCCUCAGUCCAUCCA	209
078	626	644	GAUGGAACUGAGGACCAAU	78	AUUGGUCCUCAGUCCAUC	210
079	627	645	AUGGAACUGAGGACCAAUC	79	GAUUGGUCCUCAGUCCA	211
080	628	646	UGGAACUGAGGACCAAUCA	80	UGAUUGGUCCUCAGUCCA	212
081	629	647	GGAACUGAGGACCAAUCAU	81	AUGAUUGGUCCUCAGUCC	213
082	630	648	GAACUGAGGACCAAUCAUG	82	CAUGAUUGGUCCUCAGUUC	214
083	631	649	AACUGAGGACCAAUCAUGC	83	GCAUGAUUGGUCCUCAGUU	215
084	632	650	ACUGAGGACCAAUCAUGCU	84	AGCAUGAUUGGUCCUCAGU	216
085	633	651	CUGAGGACCAAUCAUGCUG	85	CAGCAUGAUUGGUCCUCAG	217
086	634	652	UGAGGACCAAUCAUGCUGC	86	GCAGCAUGAUUGGUCCUCA	218
087	635	653	GAGGACCAAUCAUGCUGCA	87	UGCAGCAUGAUUGGUCCUC	219
088	636	654	AGGACCAAUCAUGCUGCAA	88	UUGCAGCAUGAUUGGUCCU	220
089	637	655	GGACCAAUCAUGCUGCAAG	89	CUUGCAGCAUGAUUGGUCC	221
090	638	656	GACCAAUCAUGCUGCAAGG	90	CCUUGCAGCAUGAUUGGUC	222
091	639	657	ACCAAUCAUGCUGCAAGGA	91	UCCUUGCAGCAUGAUUGGU	223
092	640	658	CCAAUCAUGCUGCAAGGAA	92	UUCUUGCAGCAUGAUUGG	224
093	641	659	CAAUCAUGCUGCAAGGAAC	93	GUUCCUUGCAGCAUGAUUG	225
094	642	660	AAUCAUGCUGCAAGGAACA	94	UGUCCUUGCAGCAUGAUU	226

TABLE 1-continued

Sequences of Exemplary siRNA Constructs						
CNST#	ST	ED	Sense Sequence (5'-3')	SEQ	Antisense Sequence (5'-3')	SEQ
095	643	661	AUCAUGCUGCAAGGAACAC	95	GUGUUCUUGCAGCAUGAU	227
096	644	662	UCAUGCUGCAAGGAACACU	96	AGUGUUCUUGCAGCAUGA	228
097	645	663	CAUGCUGCAAGGAACACUU	97	AAGUGUUCUUGCAGCAUG	229
098	646	664	AUGCUGCAAGGAACACUUC	98	GAAGUGUUCUUGCAGCAU	230
099	647	665	UGCUGCAAGGAACACUUC	99	GGAAGUGUUCUUGCAGCA	231
100	648	666	GCUGCAAGGAACACUCCA	100	UGGAAGUGUUCUUGCAGC	232
101	698	716	UGCCUGUUCACUGGGAUCA	101	UGAUCCAGUGAACAGGCA	233
102	702	720	UGUUCACUGGGAUCAGCCA	102	UGGCUGAUCCAGUGAACA	234
103	734	752	CACUUCUGAGCACAGAGCA	103	UGCUCUGUGCUCAGAAGUG	235
104	735	753	ACUUCUGAGCACAGAGCAG	104	CUGCUCUGUGCUCAGAAGU	236
105	740	758	UGAGCACAGAGCAGAGACA	105	UGUCUCUGCUCUGUGCUCA	237
106	742	760	AGCACAGAGCAGAGACAGA	106	UCUGUCUCUGCUCUGUGCU	238
107	770	788	GGAcAAAGGcAGAGGAuGU	107	ACAUCCUCUGCCUUUGUCC	239
108	771	789	GACAAAGGCAGAGGAUGUA	108	UACAUCCUCUGCCUUUGUC	240
109	772	790	ACAAAGGCAGAGGAUGUAG	109	CUACAUCCUCUGCCUUUGU	241
110	773	791	CAAAGGCAGAGGAUGUAGC	110	GCUACAUCCUCUGCCUUUG	242
111	774	792	AAAGGCAGAGGAUGUAGCC	111	GGCUACAUCCUCUGCCUUU	243
112	779	797	CAGAGGAUGUAGCCCCAUU	112	AAUGGGGCUACAUCCUCUG	244
113	780	798	AGAGGAUGUAGCCCCAUUG	113	CAAUGGGGCUACAUCCUCU	245
114	820	838	UGUACCCUUUCAUGCCUAC	114	GUAGGCAUGAAAGGGUACA	246
115	821	839	GUACCCUUUCAUGCCUACA	115	UGUAGGCAUGAAAGGGUAC	247
116	822	840	UACCCUUUCAUGCCUACAC	116	GUGUAGGCAUGAAAGGGUA	248
117	823	841	ACCCUUUCAUGCCUACACA	117	UGUGUAGGCAUGAAAGGGU	249
118	824	842	CCCUUUCAUGCCUACACAC	118	GUGUGUAGGCAUGAAAGGG	250
119	825	843	CCUUUCAUGCCUACACACC	119	GGUGUGUAGGCAUGAAAGG	251
120	826	844	CUUUCAUGCCUACACACCC	120	GGGUGUGUAGGCAUGAAAG	252
121	827	845	UUUCAUGCCUACACACCCC	121	GGGGUGUGUAGGCAUGAAA	253
122	842	860	CCCCAUUAAAGCAGAGU	122	ACUCUGCUUUAUGAGGGG	254
123	843	861	CCCUCAUUAAGCAGAGUC	123	GACUCUGCUUUAUGAGGG	255
124	844	862	CCUCAUUAAGCAGAGUCG	124	CGACUCUGCUUUAUGAGG	256
125	845	863	CUCAUUAAGCAGAGUCGU	125	ACGACUCUGCUUUAUGAG	257
126	846	864	UCAUUAAGCAGAGUCGUG	126	CACGACUCUGCUUUAUGA	258
127	847	865	CAUUAAGCAGAGUCGUGG	127	CCACGACUCUGCUUUAUG	259
128	848	866	AUUAAGCAGAGUCGUGGC	128	GCCACGACUCUGCUUUAU	260
129	849	867	UUAAGCAGAGUCGUGGCA	129	UGCCACGACUCUGCUUUA	261

TABLE 1-continued

Sequences of Exemplary siRNA Constructs						
CNST#	ST	ED	Sense Sequence (5'-3')	SEQ	Antisense Sequence (5'-3')	SEQ
130	850	868	UAAAGCAGAGUCGUGGCAU	130	AUGCCACGACUCUGCUUUA	262
131	851	869	AAAGCAGAGUCGUGGCAUC	131	GAUGCCACGACUCUGCUUU	263
132	852	870	AAGCAGAGUCGUGGCAUCU	132	AGAUGCCACGACUCUGCUU	264

I.4 Nucleotide Modifications

A dsRNA of the present disclosure may comprise one or more modifications, e.g., to enhance cellular uptake, affinity for the target sequence, inhibitory activity, and/or stability. Modifications may include any modification known in the art, including, for example, end modifications, base modifications, sugar modifications/replacements, and backbone modifications. End modifications may include, for example, 5' end modifications (e.g., phosphorylation, conjugation, and inverted linkages) and 3' end modifications (e.g., conjugation, DNA nucleotides, and inverted linkages). Base modifications may include, e.g., replacement with stabilizing bases, destabilizing bases or bases that base-pair with an expanded repertoire of partners, removal of bases (abasic modifications of nucleotides), or conjugated bases. Sugar modifications or replacements may include, e.g., modifications at the 2' or 4' position of the sugar moiety, or replacement of the sugar moiety. Backbone modifications may include, for example, modification or replacement of the phosphodiester linkages, e.g., with one or more phosphorothioates, phosphorodithioates, phosphotriesters, methyl and other alkyl phosphonates, phosphinates, and phosphoramidates.

As used herein, the term "nucleotide" includes naturally occurring or modified nucleotide, or a surrogate replacement moiety. A modified nucleotide is a non-naturally occurring nucleotide and is also referred to herein as a "nucleotide analog." One of ordinary skill in the art would understand that guanine, cytosine, adenine, uracil, or thymine in a nucleotide may be replaced by other moieties without substantially altering the base-pairing properties of the modified nucleotide. For example, a nucleotide comprising inosine as its base may base-pair with nucleotides containing adenine, cytosine, or uracil. Hence, nucleotides containing uracil, guanine, or adenine may be replaced in the nucleotide sequences of the present disclosure by a nucleotide containing, for example, inosine. Sequences comprising such replacement moieties are included as embodiments of the present disclosure. A modified nucleotide may also be a nucleotide whose ribose moiety is replaced with a non-ribose moiety.

The dsRNAs of the present disclosure may include one or more modified nucleotides known in the art, including, without limitation, 2'-O-methyl modified nucleotides, 2'-fluoro modified nucleotides, 2'-deoxy modified nucleotides, 2'-O-methoxyethyl modified nucleotides, modified nucleotides comprising alternate internucleotide linkages such as thiophosphates and phosphorothioates, phosphotriester modified nucleotides, modified nucleotides terminally linked to a cholesterol derivative or lipophilic moiety, peptide nucleic acids (PNAs; see, e.g., Nielsen et al., Science (1991) 254:1497-1500), constrained ethyl (cEt) modified nucleotides, inverted deoxy modified nucleotides, inverted dideoxy modified nucleotides, locked nucleic acid modified

nucleotides, abasic modifications of nucleotides, 2'-amino modified nucleotides, 2'-alkyl modified nucleotides, morpholino-modified nucleotides, phosphoramidate modified nucleotides, modified nucleotides comprising modifications at other sites of the sugar or base of an oligonucleotide, and non-natural base-containing modified nucleotides. In some embodiments, at least one of the one or more modified nucleotides is a 2'-O-methyl nucleotide, a 5'-phosphorothioate nucleotide, or a terminal nucleotide linked to a cholesterol derivative, lipophilic or other targeting moiety. The incorporation of 2'-O-methyl, 2'-O-ethyl, 2'-O-propyl, 2'-O-alkyl, 2'-O-aminoalkyl, or 2'-deoxy-2'-fluoro (i.e., 2'-fluoro) groups in nucleosides of an oligonucleotide may confer enhanced hybridization properties and/or enhanced nuclease stability to the oligonucleotide. Further, oligonucleotides containing phosphorothioate backbones (e.g., phosphorothioate linkage between two neighboring nucleotides at one or more positions of the dsRNA) may have enhanced nuclease stability. In some embodiments, the dsRNA may contain nucleotides with a modified ribose, such as locked nucleic acid (LNA) units.

In some embodiments, a dsRNA of the present disclosure comprises one or more 2'-O-methyl nucleotides and one or more 2'-fluoro nucleotides. In some embodiments, the dsRNA comprises two or more 2'-O-methyl nucleotides and two or more 2'-fluoro nucleotides. In some embodiments, the dsRNA comprises two or more 2'-O-methyl nucleotides (OMe) and two or more 2'-fluoro nucleotides (F) in an alternating pattern, e.g., the pattern OMe-F-OMe-F or the pattern F-OMe-F-OMe. In some embodiments, the dsRNA comprises up to 10 contiguous nucleotides that are each a 2'-O-methyl nucleotide. In some embodiments, the dsRNA comprises up to 10 contiguous nucleotides that are each a 2'-fluoro nucleotide. In some embodiments, the dsRNA comprises two or more 2'-fluoro nucleotides at the 5'- or 3'-end of the antisense strand.

In some embodiments, a dsRNA of the present disclosure comprises one or more phosphorothioate groups. In some embodiments, a dsRNA of the present disclosure comprises two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or 10 or more phosphorothioate groups. In some embodiments, the dsRNA does not comprise any phosphorothioate group.

In some embodiments, the dsRNA comprises one or more phosphotriester groups. In some embodiments, the dsRNA comprises two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or 10 or more phosphotriester groups. In some embodiments, the dsRNA does not comprise any phosphotriester group.

In some embodiments, the dsRNA comprises a modified ribonucleoside such as a deoxyribonucleoside, including, for example, deoxyribonucleoside overhang(s), and one or more deoxyribonucleosides within the double-stranded portion of

a dsRNA. However, it is self-evident that under no circumstances is a double-stranded DNA molecule encompassed by the term “dsRNA.”

In some embodiments, the dsRNA comprises two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or 10 or more different modified nucleotides described herein. In some embodiments, the dsRNA comprises up to two contiguous modified nucleotides, up to three contiguous modified nucleotides, up to four contiguous modified nucleotides, up to five contiguous modified nucleotides, up to six contiguous modified nucleotides, up to seven contiguous modified nucleotides, up to eight contiguous modified nucleotides, up to nine contiguous modified nucleotides, or up to 10 contiguous modified nucleotides. In some embodiments, the contiguous modified nucleotides are the same modified nucleotide. In some embodiments, the contiguous modified nucleotides are two or more, three or more, four or more, five or more, six or more, seven or more, eight or more, nine or more, or ten or more different modified nucleotides.

Table 2 below lists the sequences of exemplary siRNA constructs (CNST) with modified nucleotides. The start (ST) and end (ED) nucleotide positions in NM_018687.6 (SEQ ID NO: 529) are indicated. Abbreviations are as follows: SEQ=SEQ ID NO; mX=2'-O-Me nucleotide; fX=2'-F nucleotide; dX=DNA nucleotide; invdX=inverted dX; PO=phosphodiester linkage; and Hy=hydroxyl group. In these constructs, the sequences of their sense strands and antisense strands correspond to the sense and antisense sequences of the constructs in Table 1 with the same construct numbers, but for the inclusion of (1) the modified nucleotides mX and fX, (2) “Hy” at the 5' and 3' ends of both strands, (3) mC-mC-mA at the 5' end of the sense strand nucleotide sequence, (4) invdT at the 3' end of the sense strand nucleotide sequence, and (5) dT-dT at the 3' end of the antisense strand nucleotide sequence. In these constructs, a base-pair of nucleotides may be modified differently in some embodiments, e.g., one nucleotide in the base-pair is a 2'-O-Me ribonucleotide and the other is a 2'-F nucleotide. In some embodiments, the antisense strand comprises two 2'-F nucleotides at its 5' end.

TABLE 2

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
001	5	23	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mU-PO-fU-PO-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-invdt-Hy	265	Hy-fG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mG-PO-fU-PO-mC-PO-fU-PO-mA-PO-fA-PO-mG-PO-dT-PO-dT-Hy	397
002	99	117	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fG-PO-mA-PO-fA-PO-mC-PO-fU-PO-mG-PO-fG-PO-mC-PO-fA-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-invdt-Hy	266	Hy-fU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mU-PO-fG-PO-mC-PO-fC-PO-mA-PO-fG-PO-mU-PO-fU-PO-mC-PO-fU-PO-mG-PO-dT-PO-dT-Hy	398
003	100	118	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fA-PO-mA-PO-fC-PO-mU-PO-fG-PO-mG-PO-fC-PO-mA-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fG-PO-invdt-Hy	267	Hy-fC-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-mG-PO-fU-PO-mG-PO-fC-PO-mC-PO-fA-PO-mG-PO-fU-PO-mU-PO-fC-PO-mU-PO-dT-PO-dT-Hy	399
004	197	215	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mU-PO-fG-PO-mA-PO-fC-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fC-PO-mC-PO-fA-PO-mG-PO-fG-PO-mA-PO-fA-PO-mC-PO-fA-PO-invdt-Hy	268	Hy-fU-PO-fG-PO-mU-PO-fU-PO-mC-PO-fC-PO-mU-PO-fG-PO-mG-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mG-PO-fU-PO-mC-PO-fA-PO-mG-PO-dT-PO-dT-Hy	400
005	198	216	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mC-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mC-PO-fC-PO-mA-PO-fG-PO-mG-PO-fA-PO-mA-PO-fC-PO-mA-PO-fG-PO-invdt-Hy	269	Hy-fC-PO-fU-PO-mG-PO-fU-PO-mU-PO-fC-PO-mC-PO-fU-PO-mG-PO-fG-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fG-PO-mU-PO-fC-PO-mA-PO-dT-PO-dT-Hy	401
006	211	229	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fA-PO-mC-PO-fA-PO-mG-PO-fC-PO-mC-PO-fU-PO-mG-PO-fG-PO-mG-PO-fU-PO-mC-PO-fU-PO-mC-PO-fU-PO-mA-PO-fU-PO-invdt-Hy	270	Hy-fA-PO-fU-PO-mA-PO-fG-PO-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fC-PO-mA-PO-fG-PO-mG-PO-fC-PO-mU-PO-fG-PO-mU-PO-fU-PO-mC-PO-dT-PO-dT-Hy	402

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
007	212	230	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fC-PO-mA-PO-fG-PO-mC-PO-fC-PO-mU-PO-fG-PO-mG-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mU-PO-fA-PO-mU-PO-fG-PO-invdt-Hy	271	Hy-fC-PO-fA-PO-mU-PO-fA-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-fC-PO-mC-PO-fA-PO-mG-PO-fG-PO-mC-PO-fU-PO-mG-PO-fU-PO-mU-PO-dT-PO-dT-Hy	403
008	222	240	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mU-PO-fA-PO-mU-PO-fG-PO-mG-PO-fC-PO-mC-PO-fG-PO-mC-PO-fA-PO-mC-PO-fA-PO-mA-PO-fU-PO-invdt-Hy	272	Hy-fA-PO-fU-PO-mU-PO-fG-PO-mU-PO-fG-PO-mC-PO-fG-PO-mG-PO-fC-PO-mC-PO-fA-PO-mU-PO-fA-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-dT-PO-dT-Hy	404
009	288	306	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fC-PO-mU-PO-fU-PO-mC-PO-fG-PO-mG-PO-fG-PO-mC-PO-fA-PO-mA-PO-fG-PO-mC-PO-fC-PO-mU-PO-fG-PO-mU-PO-fU-PO-invdt-Hy	273	Hy-fA-PO-fA-PO-mC-PO-fA-PO-mG-PO-fG-PO-mC-PO-fU-PO-mU-PO-fG-PO-mC-PO-fC-PO-mC-PO-fG-PO-mA-PO-fA-PO-mG-PO-fU-PO-mU-PO-dT-PO-dT-Hy	405
010	297	315	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fA-PO-mG-PO-fC-PO-mC-PO-fU-PO-mG-PO-fU-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-fU-PO-mC-PO-fA-PO-invdt-Hy	274	Hy-fU-PO-fG-PO-mA-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mA-PO-fC-PO-mA-PO-fG-PO-mG-PO-fC-PO-mU-PO-fU-PO-mG-PO-dT-PO-dT-Hy	406
011	298	316	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fG-PO-mC-PO-fC-PO-mU-PO-fG-PO-mU-PO-fU-PO-mG-PO-fG-PO-mA-PO-fG-PO-mA-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-invdt-Hy	275	Hy-fC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fU-PO-mC-PO-fU-PO-mC-PO-fC-PO-mA-PO-fA-PO-mC-PO-fA-PO-mG-PO-fG-PO-mC-PO-fU-PO-mU-PO-dT-PO-dT-Hy	407
012	299	317	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fC-PO-mC-PO-fU-PO-mG-PO-fU-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-fA-PO-invdt-Hy	276	Hy-fU-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mA-PO-fC-PO-mA-PO-fG-PO-mG-PO-fC-PO-mU-PO-dT-PO-dT-Hy	408
013	300	318	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mC-PO-fC-PO-mU-PO-fG-PO-mU-PO-fU-PO-mG-PO-fG-PO-mA-PO-fG-PO-mA-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-mA-PO-fU-PO-invdt-Hy	277	Hy-fA-PO-fU-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fU-PO-mC-PO-fU-PO-mC-PO-fC-PO-mA-PO-fA-PO-mC-PO-fA-PO-mG-PO-fG-PO-mC-PO-dT-PO-dT-Hy	409
014	301	319	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fU-PO-mG-PO-fU-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-fA-PO-mU-PO-fG-PO-invdt-Hy	278	Hy-fC-PO-fA-PO-mU-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mA-PO-fC-PO-mA-PO-fG-PO-mG-PO-dT-PO-dT-Hy	410
015	302	320	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mU-PO-fG-PO-mU-PO-fU-PO-mG-PO-fG-PO-mA-PO-fG-PO-mA-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-mA-PO-fU-PO-mG-PO-fG-PO-invdt-Hy	279	Hy-fC-PO-fC-PO-mA-PO-fU-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fU-PO-mC-PO-fU-PO-mC-PO-fC-PO-mA-PO-fA-PO-mC-PO-fA-PO-mG-PO-dT-PO-dT-Hy	411
016	303	321	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fU-PO-mU-PO-fG-PO-mG-PO-fA-PO-	280	Hy-fU-PO-fC-PO-mC-PO-fA-PO-mU-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-	412

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
			mG-PO-fA-PO-mC-PO-fU- PO-mC-PO-fA-PO-mG-PO- fA-PO-mU-PO-fG-PO-mG- PO-fA-PO-invdt-Hy		mU-PO-fC-PO-mU-PO-fC- PO-mC-PO-fA-PO-mA-PO- fC-PO-mA-PO-dT-PO-dT- Hy	
017	304	322	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mU-PO-fU-PO-mG- PO-fG-PO-mA-PO-fG-PO- mA-PO-fC-PO-mU-PO-fC- PO-mA-PO-fG-PO-mA-PO- fU-PO-mG-PO-fG-PO-mA- PO-fG-PO-invdt-Hy	281	Hy-fC-PO-fU-PO-mC-PO- fC-PO-mA-PO-fU-PO-mC- PO-fU-PO-mG-PO-fA-PO- mG-PO-fU-PO-mC-PO-fU- PO-mC-PO-fC-PO-mA-PO- fA-PO-mC-PO-dT-PO-dT- Hy	413
018	305	323	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mU-PO-fG-PO-mG- PO-fA-PO-mG-PO-fA-PO- mC-PO-fU-PO-mC-PO-fA- PO-mG-PO-fA-PO-mU-PO- fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-invdt-Hy	282	Hy-fC-PO-fC-PO-mU-PO- fC-PO-mC-PO-fA-PO-mU- PO-fC-PO-mU-PO-fG-PO- mA-PO-fG-PO-mU-PO-fC- PO-mU-PO-fC-PO-mC-PO- fA-PO-mA-PO-dT-PO-dT- Hy	414
019	310	328	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mA-PO-fC-PO-mU- PO-fC-PO-mA-PO-fG-PO- mA-PO-fU-PO-mG-PO-fG- PO-mA-PO-fG-PO-mG-PO- fA-PO-mG-PO-fG-PO-mA- PO-fU-PO-invdt-Hy	283	Hy-fA-PO-fU-PO-mC-PO- fC-PO-mU-PO-fC-PO-mC- PO-fU-PO-mC-PO-fC-PO- mA-PO-fU-PO-mC-PO-fU- PO-mG-PO-fA-PO-mG-PO- fU-PO-mC-PO-dT-PO-dT- Hy	415
020	312	330	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mU-PO-fC-PO-mA- PO-fG-PO-mA-PO-fU-PO- mG-PO-fG-PO-mA-PO-fG- PO-mG-PO-fA-PO-mG-PO- fG-PO-mA-PO-fU-PO-mA- PO-fU-PO-invdt-Hy	284	Hy-fA-PO-fU-PO-mA-PO- fU-PO-mC-PO-fC-PO-mU- PO-fC-PO-mC-PO-fU-PO- mC-PO-fC-PO-mA-PO-fU- PO-mC-PO-fU-PO-mG-PO- fA-PO-mG-PO-dT-PO-dT- Hy	416
021	313	331	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mC-PO-fA-PO-mG- PO-fA-PO-mU-PO-fG-PO- mG-PO-fA-PO-mG-PO-fG- PO-mA-PO-fG-PO-mG-PO- fA-PO-mU-PO-fA-PO-mU- PO-fU-PO-invdt-Hy	285	Hy-fA-PO-fA-PO-mU-PO- fA-PO-mU-PO-fC-PO-mC- PO-fU-PO-mC-PO-fC-PO- mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PO- fG-PO-mA-PO-dT-PO-dT- Hy	417
022	314	332	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mA-PO-fG-PO-mA- PO-fU-PO-mG-PO-fG-PO- mA-PO-fG-PO-mG-PO-fA- PO-mG-PO-fG-PO-mA-PO- fU-PO-mA-PO-fU-PO-mU- PO-fC-PO-invdt-Hy	286	Hy-fG-PO-fA-PO-mA-PO- fU-PO-mA-PO-fU-PO-mC- PO-fC-PO-mU-PO-fC-PO- mC-PO-fU-PO-mC-PO-fC- PO-mA-PO-fU-PO-mC-PO- fU-PO-mG-PO-dT-PO-dT- Hy	418
023	315	333	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mG-PO-fA-PO-mU- PO-fG-PO-mG-PO-fA-PO- mG-PO-fG-PO-mA-PO-fG- PO-mG-PO-fA-PO-mU-PO- fA-PO-mU-PO-fU-PO-mC- PO-fU-PO-invdt-Hy	287	Hy-fA-PO-fG-PO-mA-PO- fA-PO-mU-PO-fA-PO-mU- PO-fC-PO-mC-PO-fU-PO- mC-PO-fC-PO-mU-PO-fC- PO-mC-PO-fA-PO-mU-PO- fC-PO-mU-PO-dT-PO-dT- Hy	419
024	316	334	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mA-PO-fU-PO-mG- PO-fG-PO-mA-PO-fG-PO- mG-PO-fA-PO-mG-PO-fG- PO-mA-PO-fU-PO-mA-PO- fU-PO-mU-PO-fC-PO-mU- PO-fG-PO-invdt-Hy	288	Hy-fC-PO-fA-PO-mG-PO- fA-PO-mA-PO-fU-PO-mA- PO-fU-PO-mC-PO-fC-PO- mU-PO-fC-PO-mC-PO-fU- PO-mC-PO-fC-PO-mA-PO- fU-PO-mC-PO-dT-PO-dT- Hy	420

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
025	317	335	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-invdt-Hy	289	Hy-fG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mU-PO-dT-PO-dT-Hy	421
026	318	336	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fU-PO-mA-PO-fU-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-invdt-Hy	290	Hy-fU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mA-PO-fU-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mC-PO-fC-PO-mA-PO-dT-PO-dT-Hy	422
027	319	337	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-invdt-Hy	291	Hy-fC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mC-PO-dT-PO-dT-Hy	423
028	320	338	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fU-PO-mA-PO-fU-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-invdt-Hy	292	Hy-fG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mA-PO-fU-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-dT-PO-dT-Hy	424
029	321	339	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fU-PO-invdt-Hy	293	Hy-fA-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fC-PO-mU-PO-dT-PO-dT-Hy	425
030	328	346	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mA-PO-fU-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-invdt-Hy	294	Hy-fU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mA-PO-fU-PO-mA-PO-dT-PO-dT-Hy	426
031	416	434	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mU-PO-fA-PO-mG-PO-fA-PO-mA-PO-fG-PO-mU-PO-fC-PO-mC-PO-fA-PO-mG-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-invdt-Hy	295	Hy-fU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-fC-PO-mU-PO-fG-PO-mG-PO-fA-PO-mC-PO-fU-PO-mU-PO-fC-PO-mU-PO-fA-PO-mG-PO-dT-PO-dT-Hy	427
032	417	435	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mA-PO-fG-PO-mA-PO-fA-PO-mG-PO-fU-PO-mC-PO-fC-PO-mA-PO-fG-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fG-PO-mA-PO-fG-PO-invdt-Hy	296	Hy-fC-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-mC-PO-fU-PO-mG-PO-fG-PO-mA-PO-fC-PO-mU-PO-fU-PO-mC-PO-fU-PO-mA-PO-dT-PO-dT-Hy	428
033	448	466	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fC-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fA-PO-mC-PO-fC-PO-mG-PO-fA-PO-mG-PO-fA-PO-mA-PO-fU-PO-mU-PO-fU-PO-invdt-Hy	297	Hy-fA-PO-fA-PO-mA-PO-fU-PO-mU-PO-fC-PO-mU-PO-fC-PO-mG-PO-fG-PO-mU-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-mG-PO-fG-PO-mG-PO-dT-PO-dT-Hy	429
034	449	467	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-	298	Hy-fC-PO-fA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-	430

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
			mA-PO-fC-PO-mC-PO-fG- PO-mA-PO-fG-PO-mA-PO- fA-PO-mU-PO-fU-PO-mU- PO-fG-PO-invdt-Hy		mG-PO-fU-PO-mA-PO-fG- PO-mG-PO-fC-PO-mA-PO- fG-PO-mG-PO-dT-PO-dT- Hy	
035	450	468	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mU-PO-fG-PO-mC- PO-fC-PO-mU-PO-fA-PO- mC-PO-fC-PO-mG-PO-fA- PO-mG-PO-fA-PO-mA-PO- fU-PO-mU-PO-fU-PO-mG- PO-fA-PO-invdt-Hy	299	Hy-fU-PO-fC-PO-mA-PO- fA-PO-mA-PO-fU-PO-mU- PO-fC-PO-mU-PO-fC-PO- mG-PO-fG-PO-mU-PO-fA- PO-mG-PO-fG-PO-mC-PO- fA-PO-mG-PO-dT-PO-dT- Hy	431
036	451	469	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mG-PO-fC-PO-mC- PO-fU-PO-mA-PO-fC-PO- mC-PO-fG-PO-mA-PO-fG- PO-mA-PO-fA-PO-mU-PO- fU-PO-mU-PO-fG-PO-mA- PO-fG-PO-invdt-Hy	300	Hy-fC-PO-fU-PO-mC-PO- fA-PO-mA-PO-fA-PO-mU- PO-fU-PO-mC-PO-fU-PO- mC-PO-fG-PO-mG-PO-fU- PO-mA-PO-fG-PO-mG-PO- fC-PO-mA-PO-dT-PO-dT- Hy	432
037	452	470	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mC-PO-fC-PO-mU- PO-fA-PO-mC-PO-fC-PO- mG-PO-fA-PO-mG-PO-fA- PO-mA-PO-fU-PO-mU-PO- fU-PO-mG-PO-fA-PO-mG- PO-fG-PO-invdt-Hy	301	Hy-fC-PO-fC-PO-mU-PO- fC-PO-mA-PO-fA-PO-mA- PO-fU-PO-mU-PO-fC-PO- mU-PO-fC-PO-mG-PO-fG- PO-mU-PO-fA-PO-mG-PO- fG-PO-mC-PO-dT-PO-dT- Hy	433
038	453	471	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mC-PO-fU-PO-mA- PO-fC-PO-mC-PO-fG-PO- mA-PO-fG-PO-mA-PO-fA- PO-mU-PO-fU-PO-mU-PO- fG-PO-mA-PO-fG-PO-mG- PO-fU-PO-invdt-Hy	302	Hy-fA-PO-fC-PO-mC-PO- fU-PO-mC-PO-fA-PO-mA- PO-fA-PO-mU-PO-fU-PO- mC-PO-fU-PO-mC-PO-fG- PO-mG-PO-fU-PO-mA-PO- fG-PO-mG-PO-dT-PO-dT- Hy	434
039	454	472	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mU-PO-fA-PO-mC- PO-fC-PO-mG-PO-fA-PO- mG-PO-fA-PO-mA-PO-fU- PO-mU-PO-fU-PO-mG-PO- fA-PO-mG-PO-fG-PO-mU- PO-fC-PO-invdt-Hy	303	Hy-fG-PO-fA-PO-mC-PO- fC-PO-mU-PO-fC-PO-mA- PO-fA-PO-mA-PO-fU-PO- mU-PO-fC-PO-mU-PO-fC- PO-mG-PO-fG-PO-mU-PO- fA-PO-mG-PO-dT-PO-dT- Hy	435
040	455	473	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mA-PO-fC-PO-mC- PO-fG-PO-mA-PO-fG-PO- mA-PO-fA-PO-mU-PO-fU- PO-mU-PO-fG-PO-mA-PO- fG-PO-mG-PO-fU-PO-mC- PO-fU-PO-invdt-Hy	304	Hy-fA-PO-fG-PO-mA-PO- fC-PO-mC-PO-fU-PO-mC- PO-fA-PO-mA-PO-fA-PO- mU-PO-fU-PO-mC-PO-fU- PO-mC-PO-fG-PO-mG-PO- fU-PO-mA-PO-dT-PO-dT- Hy	436
041	456	474	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mC-PO-fC-PO-mG- PO-fA-PO-mG-PO-fA-PO- mA-PO-fU-PO-mU-PO-fU- PO-mG-PO-fA-PO-mG-PO- fG-PO-mU-PO-fC-PO-mU- PO-fU-PO-invdt-Hy	305	Hy-fA-PO-fA-PO-mG-PO- fA-PO-mC-PO-fC-PO-mU- PO-fC-PO-mA-PO-fA-PO- mA-PO-fU-PO-mU-PO-fC- PO-mU-PO-fC-PO-mG-PO- fG-PO-mU-PO-dT-PO-dT- Hy	437
042	457	475	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mC-PO-fG-PO-mA- PO-fG-PO-mA-PO-fA-PO- mU-PO-fU-PO-mU-PO-fG- PO-mA-PO-fG-PO-mG-PO- fU-PO-mC-PO-fU-PO-mU- PO-fA-PO-invdt-Hy	306	Hy-fU-PO-fA-PO-mA-PO- fG-PO-mA-PO-fC-PO-mC- PO-fU-PO-mC-PO-fA-PO- mA-PO-fA-PO-mU-PO-fU- PO-mC-PO-fU-PO-mC-PO- fG-PO-mG-PO-dT-PO-dT- Hy	438

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
043	458	476	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mG-PO-fA-PO-mG-PO-fA-PO-mA-PO-fU-PO-mU-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mU-PO-fC-PO-mU-PO-fU-PO-mA-PO-fA-PO-invdt-Hy	307	Hy-fU-PO-fU-PO-mA-PO-fA-PO-mG-PO-fA-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fA-PO-mA-PO-fU-PO-mU-PO-fC-PO-mU-PO-fC-PO-mG-PO-dT-PO-dT-Hy	439
044	459	477	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fU-PO-mC-PO-fU-PO-mU-PO-fA-PO-mA-PO-fA-PO-invdt-Hy	308	Hy-fU-PO-fU-PO-mU-PO-fA-PO-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-dT-PO-dT-Hy	440
045	460	478	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fA-PO-mA-PO-fU-PO-mU-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mU-PO-fC-PO-mU-PO-fU-PO-mA-PO-fA-PO-mA-PO-fG-PO-invdt-Hy	309	Hy-fC-PO-fU-PO-mU-PO-fU-PO-mA-PO-fA-PO-mG-PO-fA-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fA-PO-mA-PO-fU-PO-mU-PO-fC-PO-mU-PO-dT-PO-dT-Hy	441
046	461	479	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fA-PO-mU-PO-fU-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fU-PO-mC-PO-fU-PO-mU-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-invdt-Hy	310	Hy-fC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fA-PO-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-dT-PO-dT-Hy	442
047	462	480	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fU-PO-mU-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mU-PO-fC-PO-mU-PO-fU-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fC-PO-invdt-Hy	311	Hy-fG-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mA-PO-fA-PO-mG-PO-fA-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fA-PO-mA-PO-fU-PO-mU-PO-dT-PO-dT-Hy	443
048	463	481	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mU-PO-fU-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fU-PO-mC-PO-fU-PO-mU-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mC-PO-fU-PO-invdt-Hy	312	Hy-fA-PO-fG-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fA-PO-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-fA-PO-mU-PO-dT-PO-dT-Hy	444
049	464	482	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mU-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mU-PO-fC-PO-mU-PO-fU-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fC-PO-mU-PO-fC-PO-invdt-Hy	313	Hy-fG-PO-fA-PO-mG-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mU-PO-fA-PO-mG-PO-fA-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fA-PO-mA-PO-dT-PO-dT-Hy	445
050	465	483	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fU-PO-mC-PO-fU-PO-mU-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mC-PO-fU-PO-mC-PO-fA-PO-invdt-Hy	314	Hy-fU-PO-fG-PO-mA-PO-fG-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fA-PO-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-dT-PO-dT-Hy	446
051	466	484	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mU-PO-fC-PO-mU-PO-fU-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fC-PO-mU-PO-fC-PO-mA-PO-fC-PO-invdt-Hy	315	Hy-fG-PO-fU-PO-mG-PO-fA-PO-mG-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mA-PO-fA-PO-mG-PO-fA-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-dT-PO-dT-Hy	447
052	467	485	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mG-PO-fU-PO-mC-PO-fU-PO-	316	Hy-fC-PO-fG-PO-mU-PO-fG-PO-mA-PO-fG-PO-mC-PO-fC-PO-mU-PO-fU-PO-	448

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
			mU-PO-fA-PO-mA-PO-fA- PO-mG-PO-fG-PO-mC-PO- fU-PO-mC-PO-fA-PO-mC- PO-fG-PO-invdt-Hy		mU-PO-fA-PO-mA-PO-fG- PO-mA-PO-fC-PO-mC-PO- fU-PO-mC-PO-dT-PO-dT- Hy	
053	468	486	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mG-PO-fG-PO-mU- PO-fC-PO-mU-PO-fU-PO- mA-PO-fA-PO-mA-PO-fG- PO-mG-PO-fC-PO-mU-PO- fC-PO-mA-PO-fC-PO-mG- PO-fC-PO-invdt-Hy	317	Hy-fG-PO-fC-PO-mG-PO- fU-PO-mG-PO-fA-PO-mG- PO-fC-PO-mC-PO-fU-PO- mU-PO-fU-PO-mA-PO-fA- PO-mG-PO-fA-PO-mC-PO- fC-PO-mU-PO-dT-PO-dT- Hy	449
054	469	487	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mG-PO-fU-PO-mC- PO-fU-PO-mU-PO-fA-PO- mA-PO-fA-PO-mG-PO-fG- PO-mC-PO-fU-PO-mC-PO- fA-PO-mC-PO-fG-PO-mC- PO-fU-PO-invdt-Hy	318	Hy-fA-PO-fG-PO-mC-PO- fG-PO-mU-PO-fG-PO-mA- PO-fG-PO-mC-PO-fC-PO- mU-PO-fU-PO-mU-PO-fA- PO-mA-PO-fG-PO-mA-PO- fC-PO-mC-PO-dT-PO-dT- Hy	450
055	470	488	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mU-PO-fC-PO-mU- PO-fU-PO-mA-PO-fA-PO- mA-PO-fG-PO-mG-PO-fC- PO-mU-PO-fC-PO-mA-PO- fC-PO-mG-PO-fC-PO-mU- PO-fG-PO-invdt-Hy	319	Hy-fC-PO-fA-PO-mG-PO- fC-PO-mG-PO-fU-PO-mG- PO-fA-PO-mG-PO-fC-PO- mC-PO-fU-PO-mU-PO-fU- PO-mA-PO-fA-PO-mG-PO- fA-PO-mC-PO-dT-PO-dT- Hy	451
056	471	489	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mC-PO-fU-PO-mU- PO-fA-PO-mA-PO-fA-PO- mG-PO-fG-PO-mC-PO-fU- PO-mC-PO-fA-PO-mC-PO- fG-PO-mC-PO-fU-PO-mG- PO-fA-PO-invdt-Hy	320	Hy-fU-PO-fC-PO-mA-PO- fG-PO-mC-PO-fG-PO-mU- PO-fG-PO-mA-PO-fG-PO- mC-PO-fC-PO-mU-PO-fU- PO-mU-PO-fA-PO-mA-PO- fG-PO-mA-PO-dT-PO-dT- Hy	452
057	472	490	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mU-PO-fU-PO-mA- PO-fA-PO-mA-PO-fG-PO- mG-PO-fC-PO-mU-PO-fC- PO-mA-PO-fC-PO-mG-PO- fC-PO-mU-PO-fG-PO-mA- PO-fC-PO-invdt-Hy	321	Hy-fG-PO-fU-PO-mC-PO- fA-PO-mG-PO-fC-PO-mG- PO-fU-PO-mG-PO-fA-PO- mG-PO-fC-PO-mC-PO-fU- PO-mU-PO-fU-PO-mA-PO- fA-PO-mG-PO-dT-PO-dT- Hy	453
058	473	491	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mU-PO-fA-PO-mA- PO-fA-PO-mG-PO-fG-PO- mC-PO-fU-PO-mC-PO-fA- PO-mC-PO-fG-PO-mC-PO- fU-PO-mG-PO-fA-PO-mC- PO-fA-PO-invdt-Hy	322	Hy-fU-PO-fG-PO-mU-PO- fC-PO-mA-PO-fG-PO-mC- PO-fG-PO-mU-PO-fG-PO- mA-PO-fG-PO-mC-PO-fC- PO-mU-PO-fU-PO-mU-PO- fA-PO-mA-PO-dT-PO-dT- Hy	454
059	474	492	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mA-PO-fA-PO-mA- PO-fG-PO-mG-PO-fC-PO- mU-PO-fC-PO-mA-PO-fC- PO-mG-PO-fC-PO-mU-PO- fG-PO-mA-PO-fC-PO-mA- PO-fA-PO-invdt-Hy	323	Hy-fU-PO-fU-PO-mG-PO- fU-PO-mC-PO-fA-PO-mG- PO-fC-PO-mG-PO-fU-PO- mG-PO-fA-PO-mG-PO-fC- PO-mC-PO-fU-PO-mU-PO- fU-PO-mA-PO-dT-PO-dT- Hy	455
060	475	493	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mA-PO-fA-PO-mG- PO-fG-PO-mC-PO-fU-PO- mC-PO-fA-PO-mC-PO-fG- PO-mC-PO-fU-PO-mG-PO- fA-PO-mC-PO-fA-PO-mA- PO-fG-PO-invdt-Hy	324	Hy-fC-PO-fU-PO-mU-PO- fG-PO-mU-PO-fC-PO-mA- PO-fG-PO-mC-PO-fG-PO- mU-PO-fG-PO-mA-PO-fG- PO-mC-PO-fC-PO-mU-PO- fU-PO-mU-PO-dT-PO-dT- Hy	456

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
061	486	504	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mU-PO-fG-PO-mA-PO-fC-PO-mA-PO-fA-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mC-PO-fC-PO-mA-PO-fC-PO-mA-PO-fU-PO-invdt-Hy	325	Hy-fA-PO-fU-PO-mG-PO-fU-PO-mG-PO-fG-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-mU-PO-fU-PO-mG-PO-fU-PO-mC-PO-fA-PO-mG-PO-dT-PO-dT-Hy	457
062	487	505	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mC-PO-fA-PO-mA-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fC-PO-mC-PO-fA-PO-mC-PO-fA-PO-mU-PO-fC-PO-invdt-Hy	326	Hy-fG-PO-fA-PO-mU-PO-fG-PO-mU-PO-fG-PO-mG-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fU-PO-mU-PO-fG-PO-mU-PO-fC-PO-mA-PO-dT-PO-dT-Hy	458
063	489	507	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mC-PO-fA-PO-mA-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fC-PO-mC-PO-fA-PO-mC-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-invdt-Hy	327	Hy-fA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fG-PO-mU-PO-fG-PO-mG-PO-fC-PO-mU-PO-fC-PO-mU-PO-fC-PO-mC-PO-fU-PO-mU-PO-fG-PO-mU-PO-dT-PO-dT-Hy	459
064	490	508	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fA-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mC-PO-fC-PO-mA-PO-fC-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fA-PO-invdt-Hy	328	Hy-fU-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fU-PO-mG-PO-fG-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-mU-PO-fU-PO-mG-PO-dT-PO-dT-Hy	460
065	491	509	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fC-PO-mC-PO-fA-PO-mC-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-mA-PO-fU-PO-invdt-Hy	329	Hy-fA-PO-fU-PO-mA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fG-PO-mU-PO-fG-PO-mG-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fU-PO-mU-PO-dT-PO-dT-Hy	461
066	492	510	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mC-PO-fC-PO-mA-PO-fC-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fA-PO-mU-PO-fG-PO-invdt-Hy	330	Hy-fC-PO-fA-PO-mU-PO-fA-PO-mG-PO-fG-PO-mA-PO-fU-PO-mG-PO-fU-PO-mG-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mC-PO-fC-PO-mU-PO-dT-PO-dT-Hy	462
067	573	591	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fC-PO-mA-PO-fG-PO-mA-PO-fU-PO-mC-PO-fC-PO-mA-PO-fG-PO-mG-PO-fA-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-fU-PO-invdt-Hy	331	Hy-fA-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mU-PO-fC-PO-mC-PO-fU-PO-mG-PO-fG-PO-mA-PO-fU-PO-mC-PO-fU-PO-mG-PO-fU-PO-mC-PO-dT-PO-dT-Hy	463
068	574	592	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mC-PO-fA-PO-mG-PO-fA-PO-mU-PO-fC-PO-mC-PO-fA-PO-mG-PO-fG-PO-mA-PO-fG-PO-mA-PO-fG-PO-mA-PO-fC-PO-mU-PO-fC-PO-mU-PO-fC-PO-invdt-Hy	332	Hy-fG-PO-fA-PO-mG-PO-fU-PO-mC-PO-fU-PO-mC-PO-fU-PO-mC-PO-fC-PO-mU-PO-fG-PO-mG-PO-fA-PO-mU-PO-fC-PO-mU-PO-fG-PO-mU-PO-dT-PO-dT-Hy	464
069	576	594	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fA-PO-mU-PO-fC-PO-mC-PO-fA-PO-mG-PO-fG-PO-mA-PO-fG-PO-mA-PO-fC-PO-mU-PO-fC-PO-mC-PO-fA-PO-invdt-Hy	333	Hy-fU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fU-PO-mC-PO-fU-PO-mC-PO-fU-PO-mC-PO-fC-PO-mU-PO-fG-PO-mG-PO-fA-PO-mU-PO-fC-PO-mU-PO-dT-PO-dT-Hy	465
070	610	628	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fC-PO-mC-PO-fU-PO-mG-PO-fA-PO-	334	Hy-fA-PO-fU-PO-mC-PO-fC-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-fG-PO-	466

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
			mA-PO-fU-PO-mC-PO-fU- PO-mG-PO-fC-PO-mC-PO- fU-PO-mG-PO-fG-PO-mA- PO-fU-PO-invdt-Hy		mA-PO-fU-PO-mU-PO-fC- PO-mA-PO-fG-PO-mG-PO- fC-PO-mU-PO-dT-PO-dT- Hy	
071	613	631	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mU-PO-fG-PO-mA- PO-fA-PO-mU-PO-fC-PO- mU-PO-fG-PO-mC-PO-fC- PO-mU-PO-fG-PO-mG-PO- fA-PO-mU-PO-fG-PO-mG- PO-fA-PO-invdt-Hy	335	Hy-fU-PO-fC-PO-mC-PO- fA-PO-mU-PO-fC-PO-mC- PO-fA-PO-mG-PO-fG-PO- mC-PO-fA-PO-mG-PO-fA- PO-mU-PO-fU-PO-mC-PO- fA-PO-mG-PO-dT-PO-dT- Hy	467
072	614	632	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mG-PO-fA-PO-mA- PO-fU-PO-mC-PO-fU-PO- mG-PO-fC-PO-mC-PO-fU- PO-mG-PO-fG-PO-mA-PO- fU-PO-mG-PO-fG-PO-mA- PO-fA-PO-invdt-Hy	336	Hy-fU-PO-fU-PO-mC-PO- fC-PO-mA-PO-fU-PO-mC- PO-fC-PO-mA-PO-fG-PO- mG-PO-fC-PO-mA-PO-fG- PO-mA-PO-fU-PO-mU-PO- fC-PO-mA-PO-dT-PO-dT- Hy	468
073	615	633	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mA-PO-fA-PO-mU- PO-fC-PO-mU-PO-fG-PO- mC-PO-fC-PO-mU-PO-fG- PO-mG-PO-fA-PO-mU-PO- fG-PO-mG-PO-fA-PO-mA- PO-fC-PO-invdt-Hy	337	Hy-fG-PO-fU-PO-mU-PO- fC-PO-mC-PO-fA-PO-mU- PO-fC-PO-mC-PO-fA-PO- mG-PO-fG-PO-mC-PO-fA- PO-mG-PO-fA-PO-mU-PO- fU-PO-mC-PO-dT-PO-dT- Hy	469
074	616	634	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mA-PO-fU-PO-mC- PO-fU-PO-mG-PO-fC-PO- mC-PO-fU-PO-mG-PO-fG- PO-mA-PO-fU-PO-mG-PO- fG-PO-mA-PO-fA-PO-mC- PO-fU-PO-invdt-Hy	338	Hy-fA-PO-fG-PO-mU-PO- fU-PO-mC-PO-fC-PO-mA- PO-fU-PO-mC-PO-fC-PO- mA-PO-fG-PO-mG-PO-fC- PO-mA-PO-fG-PO-mA-PO- fU-PO-mU-PO-dT-PO-dT- Hy	470
075	617	635	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mU-PO-fC-PO-mU- PO-fG-PO-mC-PO-fC-PO- mU-PO-fG-PO-mG-PO-fA- PO-mU-PO-fG-PO-mG-PO- fA-PO-mA-PO-fC-PO-mU- PO-fG-PO-invdt-Hy	339	Hy-fC-PO-fA-PO-mG-PO- fU-PO-mU-PO-fC-PO-mC- PO-fA-PO-mU-PO-fC-PO- mC-PO-fA-PO-mG-PO-fG- PO-mC-PO-fA-PO-mG-PO- fA-PO-mU-PO-dT-PO-dT- Hy	471
076	618	636	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mC-PO-fU-PO-mG- PO-fC-PO-mC-PO-fU-PO- mG-PO-fG-PO-mA-PO-fU- PO-mG-PO-fG-PO-mA-PO- fA-PO-mC-PO-fU-PO-mG- PO-fA-PO-invdt-Hy	340	Hy-fU-PO-fC-PO-mA-PO- fG-PO-mU-PO-fU-PO-mC- PO-fC-PO-mA-PO-fU-PO- mC-PO-fC-PO-mA-PO-fG- PO-mG-PO-fC-PO-mA-PO- fG-PO-mA-PO-dT-PO-dT- Hy	472
077	624	642	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mG-PO-fG-PO-mA- PO-fU-PO-mG-PO-fG-PO- mA-PO-fA-PO-mC-PO-fU- PO-mG-PO-fA-PO-mG-PO- fG-PO-mA-PO-fC-PO-mC- PO-fA-PO-invdt-Hy	341	Hy-fU-PO-fG-PO-mG-PO- fU-PO-mC-PO-fC-PO-mU- PO-fC-PO-mA-PO-fG-PO- mU-PO-fU-PO-mC-PO-fC- PO-mA-PO-fU-PO-mC-PO- fC-PO-mA-PO-dT-PO-dT- Hy	473
078	626	644	Hy-mC-PO-mC-PO-mA-PO- fG-PO-mA-PO-fU-PO-mG- PO-fG-PO-mA-PO-fA-PO- mC-PO-fU-PO-mG-PO-fA- PO-mG-PO-fG-PO-mA-PO- fC-PO-mC-PO-fA-PO-mA- PO-fU-PO-invdt-Hy	342	Hy-fA-PO-fU-PO-mU-PO- fG-PO-mG-PO-fU-PO-mC- PO-fC-PO-mU-PO-fC-PO- mA-PO-fG-PO-mU-PO-fU- PO-mC-PO-fC-PO-mA-PO- fU-PO-mC-PO-dT-PO-dT- Hy	474

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
079	627	645	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mU-PO-fG-PO-mG-PO-fA-PO-mA-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mC-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-invdt-Hy	343	Hy-fG-PO-fA-PO-mU-PO-fU-PO-mG-PO-fG-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-fU-PO-mU-PO-fC-PO-mC-PO-fA-PO-mU-PO-dT-PO-dT-Hy	475
080	628	646	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fG-PO-mA-PO-fA-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fC-PO-mC-PO-fA-PO-mA-PO-fU-PO-mC-PO-fA-PO-invdt-Hy	344	Hy-fU-PO-fG-PO-mA-PO-fU-PO-mU-PO-fG-PO-mG-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-mU-PO-fU-PO-mC-PO-fC-PO-mA-PO-dT-PO-dT-Hy	476
081	629	647	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mG-PO-fA-PO-mA-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mC-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-invdt-Hy	345	Hy-fA-PO-fU-PO-mG-PO-fA-PO-mU-PO-fU-PO-mG-PO-fG-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-fU-PO-mU-PO-fC-PO-mC-PO-dT-PO-dT-Hy	477
082	630	648	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fA-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fC-PO-mC-PO-fA-PO-mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-invdt-Hy	346	Hy-fC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fU-PO-mU-PO-fG-PO-mG-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-mU-PO-fU-PO-mC-PO-dT-PO-dT-Hy	478
083	631	649	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mC-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-invdt-Hy	347	Hy-fG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mU-PO-fU-PO-mG-PO-fG-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-fU-PO-mU-PO-dT-PO-dT-Hy	479
084	632	650	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fC-PO-mC-PO-fA-PO-mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-invdt-Hy	348	Hy-fA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fU-PO-mU-PO-fG-PO-mG-PO-fU-PO-mC-PO-fC-PO-mA-PO-fG-PO-mU-PO-dT-PO-dT-Hy	480
085	633	651	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mC-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-invdt-Hy	349	Hy-fC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mU-PO-fU-PO-mG-PO-fG-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-dT-PO-dT-Hy	481
086	634	652	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fC-PO-mC-PO-fA-PO-mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-invdt-Hy	350	Hy-fG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fU-PO-mU-PO-fG-PO-mG-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-dT-PO-dT-Hy	482
087	635	653	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mC-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-invdt-Hy	351	Hy-fU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mU-PO-fU-PO-mG-PO-fG-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-dT-PO-dT-Hy	483
088	636	654	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fG-PO-mA-PO-fC-PO-mC-PO-fA-PO-	352	Hy-fU-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-	484

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
			mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fA-PO-invdt-Hy		mA-PO-fU-PO-mU-PO-fG-PO-mG-PO-fU-PO-mC-PO-fC-PO-mU-PO-dT-PO-dT-Hy	
089	637	655	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mG-PO-fA-PO-mC-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mA-PO-fG-PO-invdt-Hy	353	Hy-fC-PO-fU-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mU-PO-fU-PO-mG-PO-fG-PO-mU-PO-fC-PO-mC-PO-dT-PO-dT-Hy	485
090	638	656	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fA-PO-mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fA-PO-mG-PO-fG-PO-invdt-Hy	354	Hy-fC-PO-fC-PO-mU-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fU-PO-mU-PO-fG-PO-mG-PO-fU-PO-mC-PO-dT-PO-dT-Hy	486
091	639	657	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mC-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mA-PO-fG-PO-mG-PO-fA-PO-invdt-Hy	355	Hy-fU-PO-fC-PO-mC-PO-fU-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mU-PO-fU-PO-mG-PO-fG-PO-mU-PO-dT-PO-dT-Hy	487
092	640	658	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fA-PO-mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fA-PO-mG-PO-fG-PO-mA-PO-fA-PO-invdt-Hy	356	Hy-fU-PO-fU-PO-mC-PO-fC-PO-mU-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fU-PO-mU-PO-fG-PO-mG-PO-dT-PO-dT-Hy	488
093	641	659	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mA-PO-fG-PO-mG-PO-fA-PO-mA-PO-fC-PO-invdt-Hy	357	Hy-fG-PO-fU-PO-mU-PO-fC-PO-mC-PO-fU-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mU-PO-fU-PO-mG-PO-dT-PO-dT-Hy	489
094	642	660	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fC-PO-mA-PO-fA-PO-mG-PO-fG-PO-mA-PO-fA-PO-mC-PO-fA-PO-invdt-Hy	358	Hy-fU-PO-fG-PO-mU-PO-fU-PO-mC-PO-fC-PO-mU-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fU-PO-mU-PO-dT-PO-dT-Hy	490
095	643	661	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mA-PO-fG-PO-mG-PO-fA-PO-mA-PO-fC-PO-invdt-Hy	359	Hy-fG-PO-fU-PO-mG-PO-fU-PO-mU-PO-fC-PO-mC-PO-fU-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mU-PO-dT-PO-dT-Hy	491
096	644	662	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fA-PO-mG-PO-fG-PO-mA-PO-fA-PO-mC-PO-fU-PO-invdt-Hy	360	Hy-fA-PO-fG-PO-mU-PO-fG-PO-mU-PO-fU-PO-mC-PO-fC-PO-mU-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-dT-PO-dT-Hy	492

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
097	645	663	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mA-PO-fG-PO-mG-PO-fA-PO-mA-PO-fC-PO-mA-PO-fC-PO-mU-PO-fU-PO-invdt-Hy	361	Hy-fA-PO-fA-PO-mG-PO-fU-PO-mG-PO-fU-PO-mU-PO-fC-PO-mC-PO-fU-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-dT-PO-dT-Hy	493
098	646	664	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mU-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fA-PO-mG-PO-fG-PO-mA-PO-fA-PO-mC-PO-fA-PO-mC-PO-fU-PO-mU-PO-fC-PO-invdt-Hy	362	Hy-fG-PO-fA-PO-mA-PO-fG-PO-mU-PO-fG-PO-mU-PO-fU-PO-mC-PO-fC-PO-mU-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-fA-PO-mU-PO-dT-PO-dT-Hy	494
099	647	665	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mU-PO-fG-PO-mC-PO-fA-PO-mA-PO-fG-PO-mG-PO-fA-PO-mA-PO-fC-PO-mA-PO-fC-PO-mU-PO-fU-PO-mC-PO-fC-PO-invdt-Hy	363	Hy-fG-PO-fG-PO-mA-PO-fA-PO-mG-PO-fU-PO-mG-PO-fU-PO-mU-PO-fC-PO-mC-PO-fU-PO-mU-PO-fG-PO-mC-PO-fA-PO-mG-PO-fC-PO-mA-PO-dT-PO-dT-Hy	495
100	648	666	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mC-PO-fU-PO-mG-PO-fC-PO-mA-PO-fA-PO-mG-PO-fG-PO-mA-PO-fA-PO-mC-PO-fA-PO-mC-PO-fU-PO-mU-PO-fC-PO-mC-PO-fA-PO-invdt-Hy	364	Hy-fU-PO-fG-PO-mG-PO-fA-PO-mA-PO-fG-PO-mU-PO-fG-PO-mU-PO-fU-PO-mC-PO-fC-PO-mU-PO-fU-PO-mG-PO-fC-PO-mA-PO-fG-PO-mC-PO-dT-PO-dT-Hy	496
101	698	716	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mG-PO-fU-PO-mU-PO-fC-PO-mA-PO-fC-PO-mU-PO-fG-PO-mG-PO-fG-PO-mA-PO-fU-PO-mC-PO-fA-PO-invdt-Hy	365	Hy-fU-PO-fG-PO-mA-PO-fU-PO-mC-PO-fC-PO-mC-PO-fA-PO-mG-PO-fU-PO-mG-PO-fA-PO-mA-PO-fC-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-dT-PO-dT-Hy	497
102	702	720	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fU-PO-mU-PO-fC-PO-mA-PO-fC-PO-mU-PO-fG-PO-mG-PO-fG-PO-mA-PO-fU-PO-mC-PO-fA-PO-mG-PO-fC-PO-mC-PO-fA-PO-invdt-Hy	366	Hy-fU-PO-fG-PO-mG-PO-fC-PO-mU-PO-fG-PO-mA-PO-fU-PO-mC-PO-fC-PO-mC-PO-fA-PO-mG-PO-fU-PO-mG-PO-fA-PO-mA-PO-fC-PO-mA-PO-dT-PO-dT-Hy	498
103	734	752	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fC-PO-mU-PO-fU-PO-mC-PO-fU-PO-mG-PO-fA-PO-mG-PO-fC-PO-mA-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mC-PO-fA-PO-invdt-Hy	367	Hy-fU-PO-fG-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fU-PO-mG-PO-fC-PO-mU-PO-fC-PO-mA-PO-fG-PO-mA-PO-fA-PO-mG-PO-fU-PO-mG-PO-dT-PO-dT-Hy	499
104	735	753	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mC-PO-fU-PO-mU-PO-fC-PO-mU-PO-fG-PO-mA-PO-fG-PO-mC-PO-fA-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fC-PO-mA-PO-fG-PO-invdt-Hy	368	Hy-fC-PO-fU-PO-mG-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mU-PO-fG-PO-mC-PO-fU-PO-mC-PO-fA-PO-mG-PO-fA-PO-mA-PO-fG-PO-mU-PO-dT-PO-dT-Hy	500
105	740	758	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mG-PO-fC-PO-mA-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-fA-PO-invdt-Hy	369	Hy-fU-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fU-PO-mG-PO-fC-PO-mU-PO-fC-PO-mA-PO-dT-PO-dT-Hy	501
106	742	760	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fC-PO-mA-PO-fC-PO-mA-PO-fG-PO-	370	Hy-fU-PO-fC-PO-mU-PO-fG-PO-mU-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-	502

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
			mA-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fA-PO-mC-PO-fA-PO-mG-PO-fA-PO-invdt-Hy		mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fU-PO-mG-PO-fC-PO-mU-PO-dT-PO-dT-Hy	
107	770	788	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mG-PO-fA-PO-mC-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fU-PO-mG-PO-fU-PO-invdt-Hy	371	Hy-fA-PO-fC-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fG-PO-mU-PO-fC-PO-mC-PO-dT-PO-dT-Hy	503
108	771	789	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mA-PO-fC-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fG-PO-mU-PO-fA-PO-invdt-Hy	372	Hy-fU-PO-fA-PO-mC-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mG-PO-fU-PO-mC-PO-dT-PO-dT-Hy	504
109	772	790	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mC-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fU-PO-mG-PO-fU-PO-mA-PO-fG-PO-invdt-Hy	373	Hy-fC-PO-fU-PO-mA-PO-fC-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fG-PO-mU-PO-dT-PO-dT-Hy	505
110	773	791	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fG-PO-mU-PO-fA-PO-mG-PO-fC-PO-invdt-Hy	374	Hy-fG-PO-fC-PO-mU-PO-fA-PO-mC-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mG-PO-dT-PO-dT-Hy	506
111	774	792	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fU-PO-mG-PO-fU-PO-mA-PO-fG-PO-mC-PO-fC-PO-invdt-Hy	375	Hy-fG-PO-fG-PO-mC-PO-fU-PO-mA-PO-fC-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-dT-PO-dT-Hy	507
112	779	797	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mG-PO-fA-PO-mU-PO-fG-PO-mU-PO-fA-PO-mG-PO-fC-PO-mC-PO-fC-PO-mC-PO-fA-PO-mU-PO-fU-PO-invdt-Hy	376	Hy-fA-PO-fA-PO-mU-PO-fG-PO-mG-PO-fG-PO-mG-PO-fC-PO-mU-PO-fA-PO-mC-PO-fA-PO-mU-PO-fC-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-dT-PO-dT-Hy	508
113	780	798	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mG-PO-fA-PO-mG-PO-fG-PO-mA-PO-fU-PO-mG-PO-fU-PO-mA-PO-fG-PO-mC-PO-fC-PO-mC-PO-fC-PO-mA-PO-fU-PO-mU-PO-fG-PO-invdt-Hy	377	Hy-fC-PO-fA-PO-mA-PO-fU-PO-mG-PO-fG-PO-mG-PO-fG-PO-mC-PO-fU-PO-mA-PO-fC-PO-mA-PO-fU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mU-PO-dT-PO-dT-Hy	509
114	820	838	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mG-PO-fU-PO-mA-PO-fC-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mA-PO-fC-PO-invdt-Hy	378	Hy-fG-PO-fU-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mG-PO-fU-PO-mA-PO-fC-PO-mA-PO-dT-PO-dT-Hy	510

TABLE 2-continued

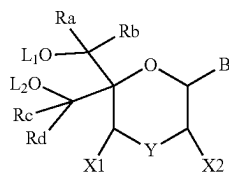
Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
115	821	839	Hy-mC-PO-mC-PO-mA-PO-fG-PO-mU-PO-fA-PO-mC-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fA-PO-mC-PO-fA-PO-invdt-Hy	379	Hy-fU-PO-fG-PO-mU-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fG-PO-mU-PO-fA-PO-mC-PO-dT-PO-dT-Hy	511
116	822	840	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mA-PO-fC-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mA-PO-fC-PO-mA-PO-fC-PO-invdt-Hy	380	Hy-fG-PO-fU-PO-mG-PO-fU-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mG-PO-fU-PO-mA-PO-dT-PO-dT-Hy	512
117	823	841	Hy-mC-PO-mC-PO-mA-PO-fA-PO-mC-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fA-PO-mC-PO-fA-PO-mC-PO-fA-PO-invdt-Hy	381	Hy-fU-PO-fG-PO-mU-PO-fG-PO-mU-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-fG-PO-mU-PO-dT-PO-dT-Hy	513
118	824	842	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fC-PO-mU-PO-fU-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mA-PO-fC-PO-mA-PO-fC-PO-mA-PO-fC-PO-invdt-Hy	382	Hy-fG-PO-fU-PO-mG-PO-fU-PO-mG-PO-fU-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mA-PO-fA-PO-mG-PO-fG-PO-mG-PO-dT-PO-dT-Hy	514
119	825	843	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fU-PO-mU-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fA-PO-mC-PO-fA-PO-mC-PO-fA-PO-mC-PO-fC-PO-invdt-Hy	383	Hy-fG-PO-fG-PO-mU-PO-fG-PO-mU-PO-fG-PO-mU-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fA-PO-mA-PO-fG-PO-mG-PO-dT-PO-dT-Hy	515
120	826	844	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mU-PO-fU-PO-mU-PO-fC-PO-mA-PO-fU-PO-mG-PO-fC-PO-mC-PO-fU-PO-mA-PO-fC-PO-mA-PO-fC-PO-mC-PO-fC-PO-invdt-Hy	384	Hy-fG-PO-fG-PO-mG-PO-fU-PO-mG-PO-fU-PO-mG-PO-fU-PO-mA-PO-fG-PO-mG-PO-fC-PO-mA-PO-fU-PO-mG-PO-fA-PO-mA-PO-fA-PO-mG-PO-dT-PO-dT-Hy	516
121	827	845	Hy-mC-PO-mC-PO-mA-PO-fU-PO-mU-PO-fU-PO-mC-PO-fA-PO-mU-PO-fG-PO-mC-PO-fC-PO-mU-PO-fA-PO-mC-PO-fA-PO-mC-PO-fA-PO-mC-PO-fC-PO-mC-PO-fC-PO-invdt-Hy	385	Hy-fG-PO-fG-PO-mG-PO-fG-PO-mU-PO-fG-PO-mU-PO-fG-PO-mU-PO-fA-PO-mG-PO-fG-PO-mC-PO-fA-PO-mU-PO-fG-PO-mA-PO-fA-PO-mA-PO-dT-PO-dT-Hy	517
122	842	860	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mU-PO-fU-PO-mA-PO-fA-PO-mA-PO-fG-PO-mC-PO-fA-PO-mG-PO-fA-PO-mG-PO-fU-PO-invdt-Hy	386	Hy-fA-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fU-PO-mU-PO-fU-PO-mA-PO-fA-PO-mU-PO-fG-PO-mA-PO-fG-PO-mG-PO-fG-PO-mG-PO-dT-PO-dT-Hy	518
123	843	861	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fC-PO-mU-PO-fC-PO-mA-PO-fU-PO-mU-PO-fA-PO-mA-PO-fA-PO-mG-PO-fC-PO-mA-PO-fG-PO-mA-PO-fG-PO-mU-PO-fC-PO-invdt-Hy	387	Hy-fG-PO-fA-PO-mC-PO-fU-PO-mC-PO-fU-PO-mG-PO-fC-PO-mU-PO-fU-PO-mU-PO-fA-PO-mA-PO-fU-PO-mG-PO-fA-PO-mG-PO-fG-PO-mG-PO-dT-PO-dT-Hy	519
124	844	862	Hy-mC-PO-mC-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mU-PO-fU-PO-	388	Hy-fC-PO-fG-PO-mA-PO-fC-PO-mU-PO-fC-PO-mU-PO-fG-PO-mC-PO-fU-PO-	520

TABLE 2-continued

Sequences of Exemplary Modified siRNA Constructs						
CNST #	ST	ED	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
			mA-PO-fA-PO-mA-PO-fG- PO-mC-PO-fA-PO-mG-PO- fA-PO-mG-PO-fU-PO-mC- PO-fG-PO-invdt-Hy		mU-PO-fU-PO-mA-PO-fA- PO-mU-PO-fG-PO-mA-PO- fG-PO-mG-PO-dT-PO-dT- Hy	
125	845	863	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mU-PO-fC-PO-mA- PO-fU-PO-mU-PO-fA-PO- mA-PO-fA-PO-mG-PO-fC- PO-mA-PO-fG-PO-mA-PO- fG-PO-mU-PO-fC-PO-mG- PO-fU-PO-invdt-Hy	389	Hy-fA-PO-fC-PO-mG-PO- fA-PO-mC-PO-fU-PO-mC- PO-fU-PO-mG-PO-fC-PO- mU-PO-fU-PO-mU-PO-fA- PO-mA-PO-fU-PO-mG-PO- fA-PO-mG-PO-dT-PO-dT- Hy	521
126	846	864	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mC-PO-fA-PO-mU- PO-fU-PO-mA-PO-fA-PO- mA-PO-fG-PO-mC-PO-fA- PO-mG-PO-fA-PO-mG-PO- fU-PO-mC-PO-fG-PO-mU- PO-fG-PO-invdt-Hy	390	Hy-fC-PO-fA-PO-mC-PO- fG-PO-mA-PO-fC-PO-mU- PO-fC-PO-mU-PO-fG-PO- mC-PO-fU-PO-mU-PO-fU- PO-mA-PO-fA-PO-mU-PO- fG-PO-mA-PO-dT-PO-dT- Hy	522
127	847	865	Hy-mC-PO-mC-PO-mA-PO- fC-PO-mA-PO-fU-PO-mU- PO-fA-PO-mA-PO-fA-PO- mG-PO-fC-PO-mA-PO-fG- PO-mA-PO-fG-PO-mU-PO- fC-PO-mG-PO-fU-PO-mG- PO-fG-PO-invdt-Hy	391	Hy-fC-PO-fC-PO-mA-PO- fC-PO-mG-PO-fA-PO-mC- PO-fU-PO-mC-PO-fU-PO- mG-PO-fC-PO-mU-PO-fU- PO-mU-PO-fA-PO-mA-PO- fU-PO-mG-PO-dT-PO-dT- Hy	523
128	848	866	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mU-PO-fU-PO-mA- PO-fA-PO-mA-PO-fG-PO- mC-PO-fA-PO-mG-PO-fA- PO-mG-PO-fU-PO-mC-PO- fG-PO-mU-PO-fG-PO-mG- PO-fC-PO-invdt-Hy	392	Hy-fG-PO-fC-PO-mC-PO- fA-PO-mC-PO-fG-PO-mA- PO-fC-PO-mU-PO-fC-PO- mU-PO-fG-PO-mC-PO-fU- PO-mU-PO-fU-PO-mA-PO- fA-PO-mU-PO-dT-PO-dT- Hy	524
129	849	867	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mU-PO-fA-PO-mA- PO-fA-PO-mG-PO-fC-PO- mA-PO-fG-PO-mA-PO-fG- PO-mU-PO-fC-PO-mG-PO- fU-PO-mG-PO-fG-PO-mC- PO-fA-PO-invdt-Hy	393	Hy-fU-PO-fG-PO-mC-PO- fC-PO-mA-PO-fC-PO-mG- PO-fA-PO-mC-PO-fU-PO- mC-PO-fU-PO-mG-PO-fC- PO-mU-PO-fU-PO-mU-PO- fA-PO-mA-PO-dT-PO-dT- Hy	525
130	850	868	Hy-mC-PO-mC-PO-mA-PO- fU-PO-mA-PO-fA-PO-mA- PO-fG-PO-mC-PO-fA-PO- mG-PO-fA-PO-mG-PO-fU- PO-mC-PO-fG-PO-mU-PO- fG-PO-mG-PO-fC-PO-mA- PO-fU-PO-invdt-Hy	394	Hy-fA-PO-fU-PO-mG-PO- fC-PO-mC-PO-fA-PO-mC- PO-fG-PO-mA-PO-fC-PO- mU-PO-fC-PO-mU-PO-fG- PO-mC-PO-fU-PO-mU-PO- fU-PO-mA-PO-dT-PO-dT- Hy	526
131	851	869	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mA-PO-fA-PO-mG- PO-fC-PO-mA-PO-fG-PO- mA-PO-fG-PO-mU-PO-fC- PO-mG-PO-fU-PO-mG-PO- fG-PO-mC-PO-fA-PO-mU- PO-fC-PO-invdt-Hy	395	Hy-fG-PO-fA-PO-mU-PO- fG-PO-mC-PO-fC-PO-mA- PO-fC-PO-mG-PO-fA-PO- mC-PO-fU-PO-mC-PO-fU- PO-mG-PO-fC-PO-mU-PO- fU-PO-mU-PO-dT-PO-dT- Hy	527
132	852	870	Hy-mC-PO-mC-PO-mA-PO- fA-PO-mA-PO-fG-PO-mC- PO-fA-PO-mG-PO-fA-PO- mG-PO-fU-PO-mC-PO-fG- PO-mU-PO-fG-PO-mG-PO- fC-PO-mA-PO-fU-PO-mC- PO-fU-PO-invdt-Hy	396	Hy-fA-PO-fG-PO-mA-PO- fU-PO-mG-PO-fC-PO-mC- PO-fA-PO-mC-PO-fG-PO- mA-PO-fC-PO-mU-PO-fC- PO-mU-PO-fG-PO-mC-PO- fU-PO-mU-PO-dT-PO-dT- Hy	528

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In some embodiments, the dsRNA comprises one or more modified nucleotides described in PCT Publication WO 2019/170731, the disclosure of which is incorporated herein in its entirety. In such modified nucleotides, the ribose ring has been replaced by a six-membered heterocyclic ring. Such a modified nucleotide has the structure of formula (I):



wherein:

B is a heterocyclic nucleobase;

one of L1 and L2 is an internucleoside linking group linking the compound of formula (I) to a polynucleotide and the other of L1 and L2 is H, a protecting group, a phosphorus moiety or an internucleoside linking group linking the compound of formula (I) to a polynucleotide,

Y is O, NH, NR1 or N—C(=O)—R1, wherein R1 is:

a (C1-C20) alkyl group, optionally substituted by one or more groups selected from a halogen atom, a (C1-C6) alkyl group, a (C3-C8) cycloalkyl group, a (C3-C14) heterocycle, a (C6-C14) aryl group, a (C5-C14) heteroaryl group, —O—Z1, —N(Z1)(Z2), —S—Z1, —CN, —C(=J)—O—Z1, —O—C(=J)—Z1, —C(=J)—N(Z1)(Z2), and —N(Z1)—C(=J)—Z2,

wherein

J is O or S,

each of Z1 and Z2 is, independently, H, a (C1-C6) alkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group, a (C3-C8) cycloalkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group,

a group [C(=O)]_m-R2-(O—CH₂—CH₂)_p-R3, wherein m is an integer meaning 0 or 1,

p is an integer ranging from 0 to 10,

R2 is a (C1-C20) alkylene group optionally substituted by a (C1-C6) alkyl group, —O—Z3, —N(Z3)(Z4), —S—Z3, —CN, —C(=K)—O—Z3, —O—C(=K)—Z3, —C(=K)—N(Z3)(Z4), or —N(Z3)—C(=K)—Z4, wherein

K is O or S,

each of Z3 and Z4 is, independently, H, a (C1-C6) alkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group, and R3 is selected from the group consisting of a hydrogen atom, a (C1-C6) alkyl group, a (C1-C6) alkoxy group, a (C3-C8) cycloalkyl group, a (C3-C14) heterocycle, a (C6-C14) aryl group or a (C5-C14) heteroaryl group, or R3 is a cell targeting moiety,

X1 and X2 are each, independently, a hydrogen atom, a (C1-C6) alkyl group, and

each of Ra, Rb, Rc and Rd is, independently, H or a (C1-C6) alkyl group, or is a pharmaceutically acceptable salt thereof.

In some embodiments, Y is NR1, R1 is a non-substituted (C1-C20) alkyl group, and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

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In some embodiments, Y is NR1, R1 is a non-substituted (C1-C16) alkyl group, which includes an alkyl group selected from a group comprising methyl, isopropyl, butyl, octyl, hexadecyl, and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

In some embodiments, Y is NR1, R1 is a (C3-C8) cycloalkyl group, optionally substituted by one or more groups selected from a halogen atom and a (C1-C6) alkyl group, and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

In some embodiments, Y is NR1, R1 is a cyclohexyl group, and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

In some embodiments, Y is NR1, R1 is a (C1-C20) alkyl group substituted by a (C6-C14) aryl group and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

In some embodiments, Y is NR1, R1 is a methyl group substituted by a phenyl group, and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

In some embodiments, Y is N—C(=O)—R1, R1 is an optionally substituted (C1-C20) alkyl group, and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

In some embodiments, Y is N—C(=O)—R1, R1 is selected from a group comprising methyl and pentadecyl and L1, L2, Ra, Rb, Rc, Rd, X1, X2, R2, R3 and B have the same meaning as defined for the general formula (I), or a pharmaceutically acceptable salt thereof.

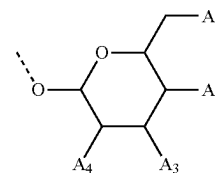
In some embodiments, B is selected from a group comprising a pyrimidine, a substituted pyrimidine, a purine and a substituted purine, or a pharmaceutically acceptable salt thereof.

In some embodiments, the internucleoside linking group in the dsRNA is independently selected from the group consisting of phosphodiester, phosphotriester, phosphorothioate, phosphorodithioate, alkyl-phosphonate and phosphoramidate backbone linking groups, or a pharmaceutically acceptable salt thereof.

In some embodiments, the dsRNA comprises from 2 to 10 compounds of formula (I), or a pharmaceutically acceptable salt thereof.

In further embodiments, the dsRNA comprises one or more targeted nucleotides or a pharmaceutically acceptable salt thereof.

In some embodiments, R3 is of the formula (II):



wherein A1, A2 and A3 are OH, A4 is OH or NHC(=O)—R5, wherein R5 is a (C1-C6) alkyl group, optionally substituted by a halogen atom.

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In some embodiments, R3 is N-acetyl-galactosamine.

The precursors that can be used to make modified siRNAs having nucleotides of formula (I) are exemplified in Tables A and B below. Table A shows examples of phosphoramidite nucleotide analogs for oligonucleotide synthesis; Table B shows some solid support of nucleotide analogs for oligonucleotide synthesis. In the (2S,6R) diastereomeric series, the phosphoramidites as nucleotide precursors are abbreviated with a “pre-l”, the nucleotide analogs are abbreviated

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with an “l”, followed by the nucleobase and a number, which specifies the group Y in formula (I). To distinguish both stereochemistries, the analogues (2R,6R)-diastereoisomers are indicated with an additional “b.” For solid supports, the abbreviation “CPG-l” is used with the additional information as described above. Targeted nucleotide precursors, targeted nucleotide analogs and solid supports are abbreviated as described above, but with an “lg” instead of the “l.”

TABLE A

No	Structure	Precursor name	Name in oligo-sequence	Stereochemistry
1		pre-1T3	1T3	(2S,6R)
2		pre-1U3	1U3	(2S,6R)
3		pre-1G3	1G3	(2S,6R)
4		pre-1A3	1A3	(2S,6R)

TABLE A-continued

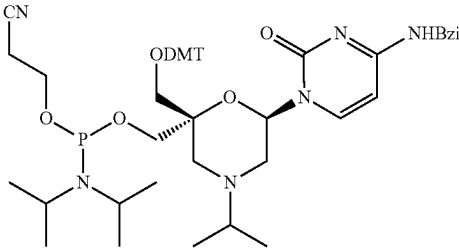
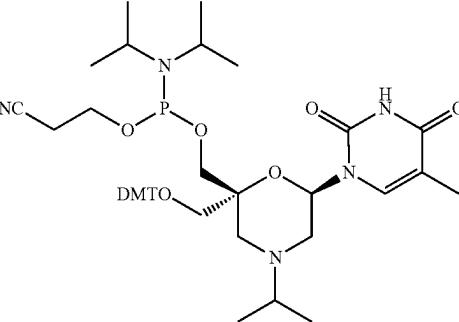
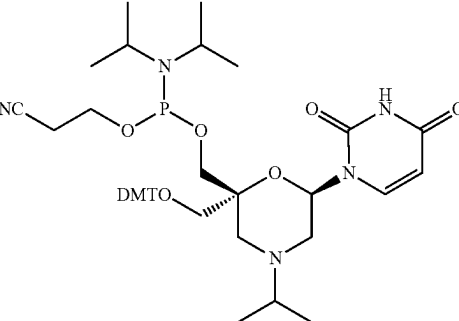
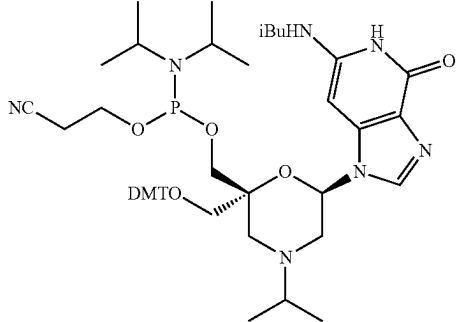
No	Structure	Name in		
		Precursor name	oligo- sequence	Stereochemistry
5		pre-1C3	1C3	(2S,6R)
6		pre-1T3b	1T3b	(2R,6R)
7		pre-1U3b	1U3b	(2R,6R)
8		pre-1G3b	1G3b	(2R,6R)

TABLE A-continued

No	Structure	Precursor name	Name in oligo-sequence	Stereochemistry
9		pre-1A3b	1A3b	(2R,6R)
10		pre-1C3b	1C3b	(2R,6R)
11		pre-1T2	1T2	(2S,6R)
12		pre-1T6	1T6	(2S,6R)
13		pre-1T7	1T7	(2S,6R)

TABLE A-continued

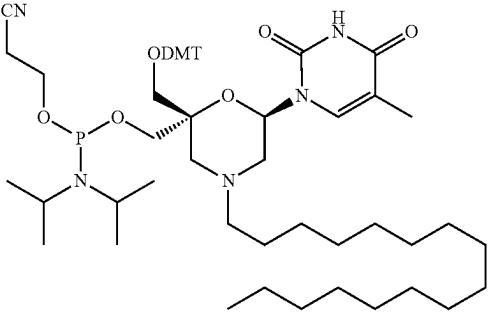
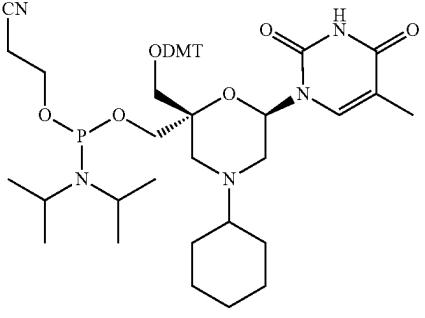
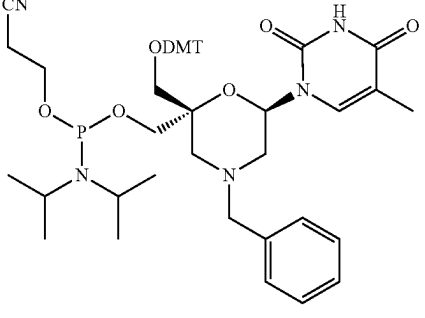
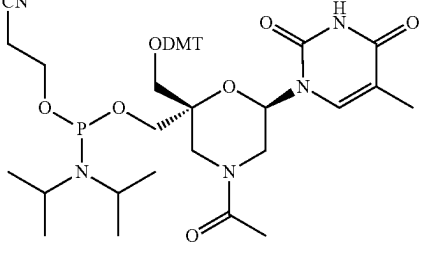
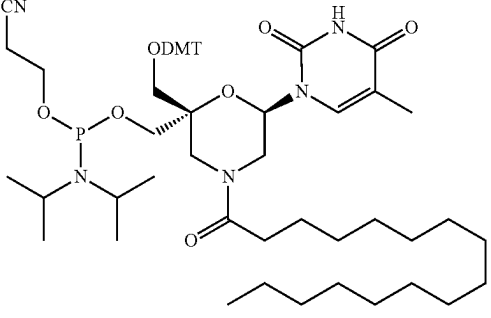
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14		pre-1T8	1T8	(2S,6R)
15		pre-1T4	1T4	(2S,6R)
16		pre-1T5	1T5	(2S,6R)
17		pre-1T9	1T9	(2S,6R)
18		pre-1T10	1T10	(2S,6R)

TABLE A-continued

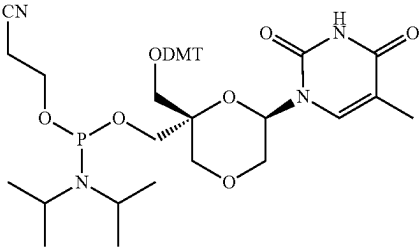
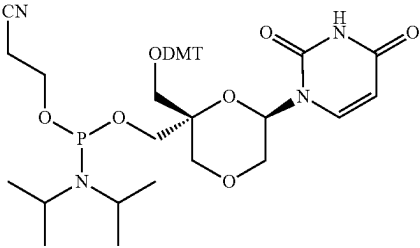
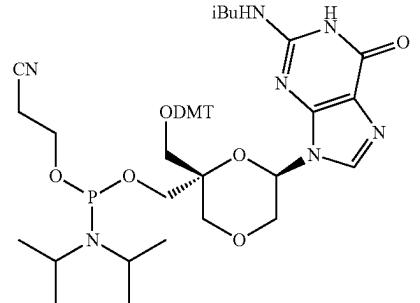
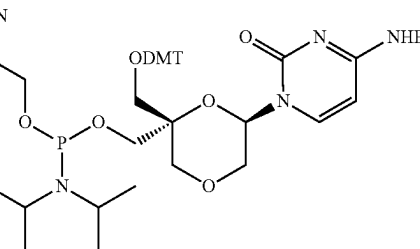
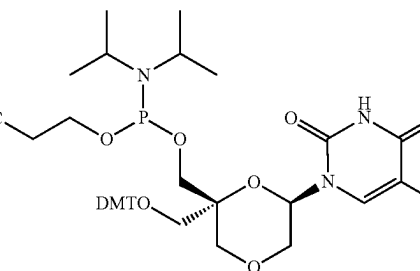
No	Structure	Name in		
		Precursor name	oligo- sequence	Stereochemistry
19		pre-1T1	1T1	(2S,6R)
20		pre-1U1	1U1	(2S,6R)
21		pre-1G1	1G1	(2S,6R)
22		pre-1C1	1C1	(2S,6R)
23		pre-1T1b	1T1b	(2R,6R)

TABLE A-continued

No	Structure	Precursor name	Name in oligo- sequence	Stereochemistry
24		pre-1U1b	1U1b	(2R,6R)
25		pre-1C1b	1C1b	(2R,6R)
26		pre-1gT9	1gT9	(2S,6R)
27		pre-1gT8	1gT8	(2S,6R)

TABLE A-continued

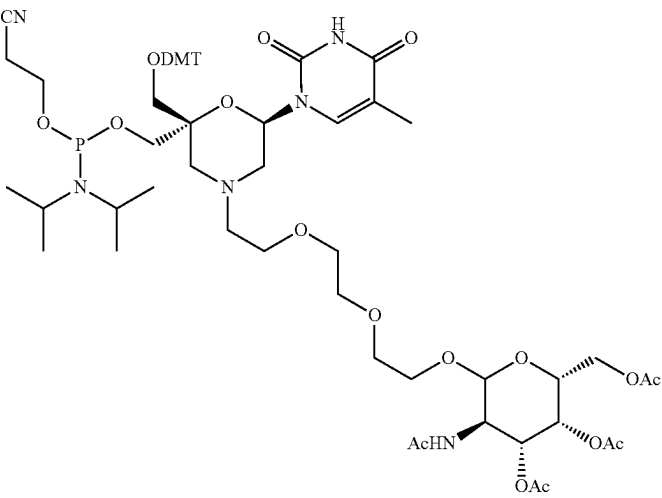
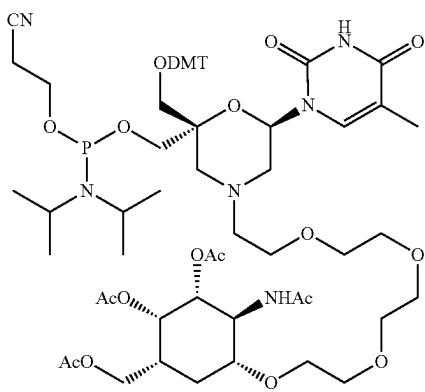
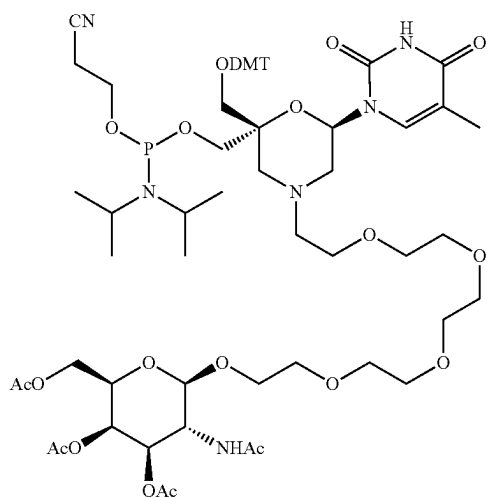
No	Structure	Name in		
		Precursor name	oligo- sequence	Stereochemistry
28		pre-1gT7	1gT7	(2S,6R)
29		pre-1gT6	1gT6	(2S,6R)
30		pre-1gT5	1gT5	(2S,6R)

TABLE A-continued

No	Structure	Precursor name	Name in oligo-sequence	Stereochemistry
31		pre-1gT3	1gT3	(2S,6R)
32		pre-1gT4	1gT4	(2S,6R)
33		pre-1gT12	1gT12	(2S,6R)
34		pre-1gT11	1gT11	(2S,6R)

TABLE A-continued

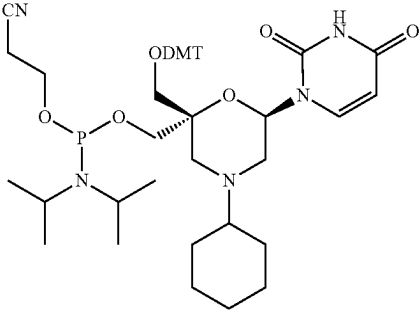
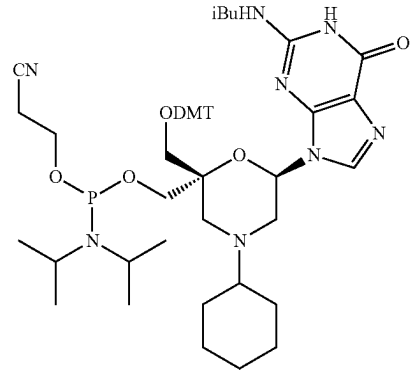
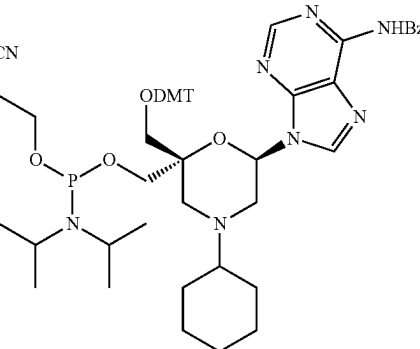
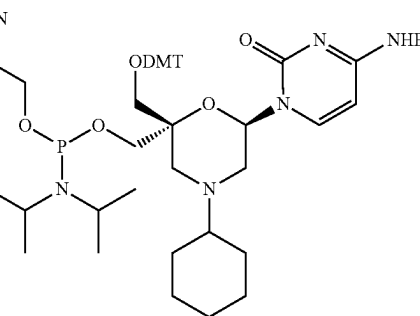
No	Structure	Name in		
		Precursor name	oligo- sequence	Stereochemistry
38		pre-1U4	1U4	(2S,6R)
39		pre-1G4	1G4	(2S,6R)
40		pre-1A4	1A4	(2S,6R)
41		pre-1C4	1C4	(2S,6R)

TABLE A-continued

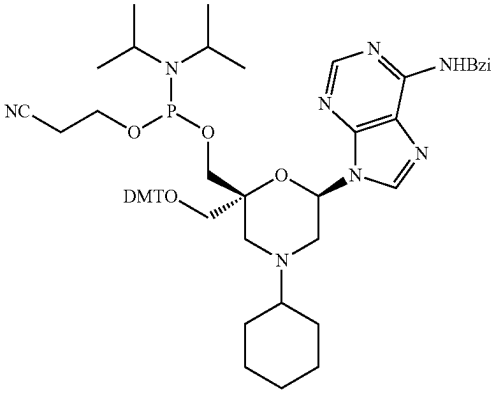
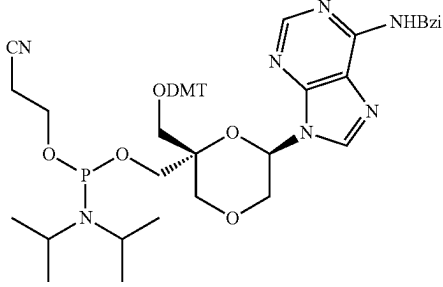
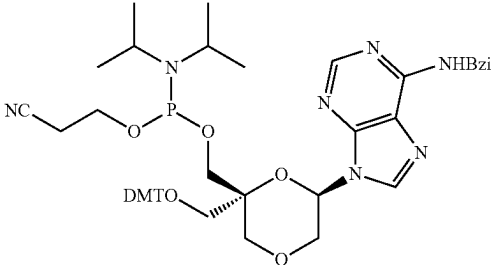
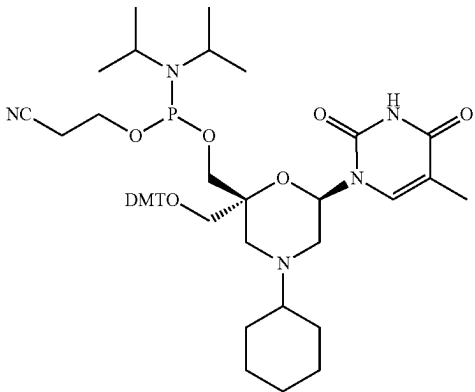
No	Structure	Name in		
		Precursor name	oligo- sequence	Stereochemistry
42		pre-1A4b	1A4b	(2R,6R)
43		pre-1A1	1A1	(2S,6R)
44		pre-1A1b	1A1b	(2R,6R)
45		pre-1T4b	1T4b	(2R,6R)

TABLE A-continued

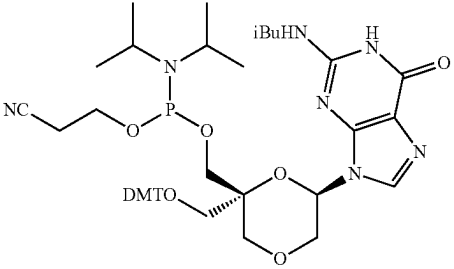
No	Structure	Precursor name	Name in oligo-sequence	Stereochemistry
46		pre-1G1b	1G1b	(2R,6R)

TABLE B

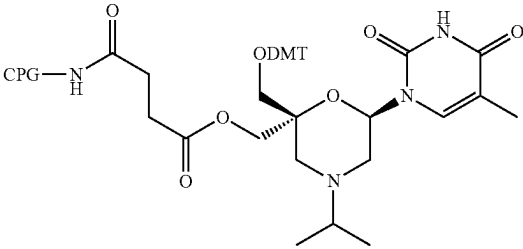
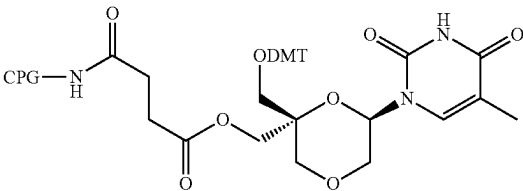
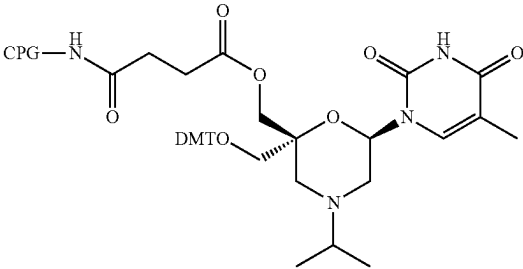
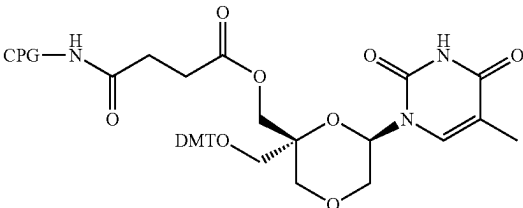
No	Structure	Name	Name in oligo-sequence	Stereochemistry
1		cpg-1T3	1T3	(2S,6R)
2		cpg-1T1	1T1	(2S,6R)
3		cpg-1T3b	1T3b	(2R,6R)
4		cpg-1T1b	1T1b	(2R,6R)

TABLE B-continued

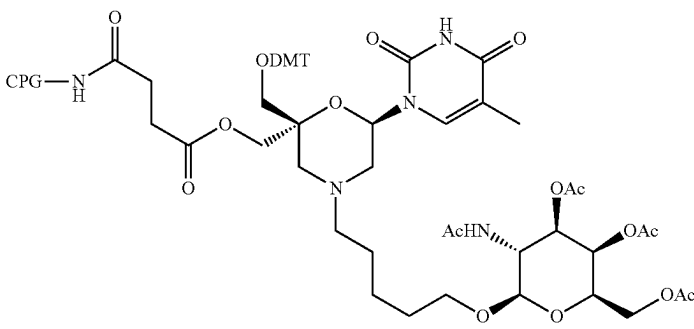
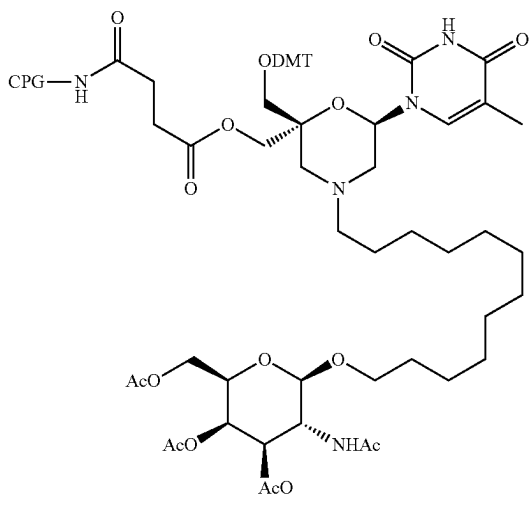
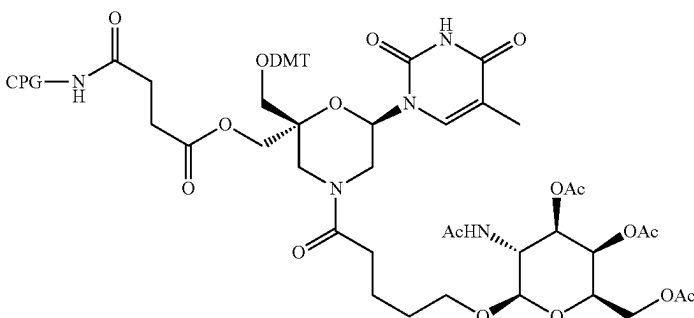
No	Structure	Name	Name in oligo-sequence	Stereochemistry
5		cpg-1gT3	1gT3	(2S,6R)
6		cpg-1gT4	1gT4	(2S,6R)
7		cpg-1gT1	1gT1	(2S,6R)

TABLE B-continued

No	Structure	Name	Name in oligo-sequence	Stereochemistry
8		cpg-1gT2	1gT2	(2S,6R)
9		cpg-1gT5	1gT5	(2S,6R)

The modified nucleotides of formula (I) may be incorporated at the 5', 3', or both ends of the sense strand and/or antisense strand of the dsRNA. By way of example, one or more (e.g., 1, 2, 3, 4, or 5 or more) modified nucleotides may be incorporated at the 5' end of the sense strand of the dsRNA. In some embodiments, one or more (e.g., 1, 2, 3, or more) modified nucleotides are positioned in the 5' end of the sense strand, where the modified nucleotides do not complement the antisense sequence but may be optionally paired with an equal or smaller number of complementary nucleotides at the corresponding 3' end of the antisense strand.

In some embodiments, the dsRNA may comprise a sense strand having a sense sequence of 17, 18, or 19 nucleotides in length, where three to five nucleotides of formula (I) (e.g., three consecutive 1gT3 or 1gT7 with or without additional nucleotides of formula (I)) are placed in the 5' end of the sense sequence, making the sense strand 20, 21, or 22 nucleotides in length. In such embodiments, the sense strand may additionally comprise two consecutive nucleotides of formula (I) (e.g., 1T4 or 1T3) at the 3' of the sense sequence, making the sense strand 22, 23, or 24 nucleotides in length.

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The dsRNA may comprise an antisense sequence of 19 nucleotides in length, where the antisense sequence may additionally be linked to 2 modified nucleotides or deoxyribonucleotides (e.g., dT) at its 3' end, making the antisense strand 21 nucleotides in length. In further embodiments, the sense strand of the dsRNA contains only naturally occurring internucleotide bonds (phosphodiester bond), where the antisense strand may optionally contain non-naturally occurring internucleotide bonds. For example, the antisense strand may contain phosphoro-thioate bonds in the backbone near or at its 5' and/or 3' ends. In some embodiments, the dsRNA may contain nucleotides with a modified ribose, such as locked nucleic acid (LNA) units.

In some embodiments, the use of modified nucleotides of formula (I) circumvents the need for other RNA modifications such as the use of non-naturally occurring internucleotide bonds, thereby simplifying the chemical synthesis of dsRNAs. Moreover, the modified nucleotides of formula (I) can be readily made to contain cell targeted moieties such as GalNAc derivatives (which include GalNAc itself), enhancing the delivery efficiency of dsRNAs incorporating such nucleotides. Further, it has been shown that dsRNAs incor-

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porating modified nucleotides of formula (I), e.g., at the sense strand, significantly improve the stability and therapeutic potency of the dsRNAs.

Tables 3 and 4 below list the sequences of exemplary modified GalNAc-siRNA constructs derived from constructs

siRNA#023 and siRNA#042 listed in Table 2. In some embodiments, the dsRNA may contain nucleotides with a modified ribose, such as locked nucleic acid (LNA) units (abbreviated with an “l”, followed by the corresponding nucleobase, in tables below).

TABLE 3

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#023				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
023-c-01	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	535	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	589
023-c-02	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	536	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	590
023-c-03	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	537	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-fC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	591
023-c-04	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	538	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-fU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	592
023-c-05	Hy-lgT3-PO-lgT3-PO-lgT3- PO-lA-PO-lG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	539	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	593
023-c-06	Hy-lgT3-PO-lgT3-PO-lgT3- PO-lA-PO-lG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	540	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	594
023-c-07	Hy-lgT3-PO-lgT3-PO-lgT3- PO-lA-PO-lG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	541	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-fC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	595
023-c-08	Hy-lgT3-PO-lgT3-PO-lgT3- PO-lA-PO-lG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	542	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-fU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	596
023-c-09	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	543	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	597
023-c-10	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mU-PS-mC-PS-mU-Hy	544	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PS-mA-PS-mA-Hy	598

TABLE 3-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#023				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
	PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy		PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	
023-c-11	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	545	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	599
023-c-12	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	546	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	600
023-c-13	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	547	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-fC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	601
023-c-14	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	548	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	602
023-c-15	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	549	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	603
023-c-16	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	550	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	604
023-c-17	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	551	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	605
023-c-18	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PS-mC-PS-mU-Hy	552	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-fC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	606
023-c-19	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	553	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	607
023-c-20	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	554	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	608

TABLE 3-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#023				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
023-c-21	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	555	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-fC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	609
023-c-22	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	556	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-fU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	610
023-c-23	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	557	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	611
023-c-24	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	558	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	612
023-c-25	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	559	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-fC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	613
023-c-26	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	560	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-fU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	614
023-c-27	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	561	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-•Hy	615
023-c-28	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	562	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	616
023-c-29	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	563	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	617
023-c-30	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	564	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	618
023-c-31	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG-	565	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-fC- PO-mU-PO-fC-PO-mC-PO-fA-	619

TABLE 3-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#023				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
	PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy		PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	
023-c-32	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	566	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	620
023-c-33	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	567	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	621
023-c-34	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	568	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	622
023-c-35	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	569	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	623
023-c-36	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-1T4-PO-1T4-Hy	570	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-fC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	624
	Hy-1gT3-PO-1gT3-PO-1gT3-	571	Hy-mA-PS-fG-PS-mA-PO-fA-	625
023-c-37	PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-1T4- PO-1T4-Hy		PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	
023-c-38	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-1T4- PO-1T4-Hy	572	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	626
023-c-39	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-1T4- PO-1T4-Hy	573	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-fC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	627
023-c-40	Hy-1gT3-PO-1gT3-PO-1gT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-1T4- PO-1T4-Hy	574	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-fU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	628
023-c-41	Hy-1gT3-PO-1gT3-PO-1gT3- PO-1A-PO-1G-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG-	575	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA-	629

TABLE 3-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#023				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
	PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy		PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	
023-c-42	Hy-lgT3-PO-lgT3-PO-lgT3- PO-lA-PO-lG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	576	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	630
023-c-43	Hy-lgT3-PO-lgT3-PO-lgT3- PO-lA-PO-lG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	577	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-fC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	631
023-c-44	Hy-lgT3-PO-lgT3-PO-lgT3- PO-lA-PO-lG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-fG- PO-fG-PO-mA-PO-mG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	578	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-fU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	632
023-c-45	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	579	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	633
023-c-46	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	580	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-fA-PO-mU-PO-mC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	634
023-c-47	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	581	Hy-mA-PS-fG-PS-mA-PO-fA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	635
023-c-48	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	582	Hy-mA-PS-fG-PS-mA-PO-mA- PO-mU-PO-mA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-mC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-mC-PO-mU-PS-mA- PS-mA-Hy	636
023-c-49	Hy-lgT3-PO-lgT3-PO-lgT3- PO-mA-PO-mG-PO-mA-PO-mU- PO-fG-PO-mG-PO-fA-PO-mG- PO-fG-PO-fA-PO-fG-PO-mG- PO-mA-PO-mU-PO-mA-PO-mU- PO-mU-PO-mC-PO-mU-PO-lT4- PO-lT4-Hy	583	Hy-fA-PS-fG-PS-mA-PO-fA- PO-mU-PO-fA-PO-mU-PO-fC- PO-mC-PO-mU-PO-mC-PO-fC- PO-mU-PO-fC-PO-mC-PO-fA- PO-mU-PO-fC-PO-mU-PS-dT- PS-dT-Hy	637

TABLE 3-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#023				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
023-c-50	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lA-PO-lG-PO-mA-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-fA-PO-fG-PO-mG-PO-mA-PO-mU-PO-mA-PO-mU-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	584	Hy-mA-PS-fG-PS-mA-PO-fA-PO-mU-PO-mA-PO-mU-PO-mC-PO-mC-PO-mU-PO-mC-PO-mC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mU-PO-mC-PO-mU-PS-mA-PS-mA-Hy	638
023-c-51	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lA-PO-lG-PO-mA-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-fA-PO-fG-PO-mG-PO-mA-PO-mU-PO-mA-PO-mU-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	585	Hy-mA-PS-fG-PS-mA-PO-mA-PO-mU-PO-fA-PO-mU-PO-mC-PO-mC-PO-mU-PO-mC-PO-mC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mU-PO-mC-PO-mU-PS-mA-PS-mA-Hy	639
023-c-52	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lA-PO-lG-PO-mA-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-fA-PO-fG-PO-mG-PO-mA-PO-mU-PO-mA-PO-mU-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	586	Hy-mA-PS-fG-PS-mA-PO-fA-PO-mU-PO-mA-PO-mU-PO-fC-PO-mC-PO-mU-PO-mC-PO-mC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mU-PO-mC-PO-mU-PS-mA-PS-mA-Hy	640
023-c-53	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lA-PO-lG-PO-mA-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-fA-PO-fG-PO-mG-PO-mA-PO-mU-PO-mA-PO-mU-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	587	Hy-mA-PS-fG-PS-mA-PO-mA-PO-mU-PO-mA-PO-mU-PO-fC-PO-mC-PO-mU-PO-mC-PO-mC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mU-PO-mC-PO-mU-PS-mA-PS-mA-Hy	641
023-c-54	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lA-PO-lG-PO-mA-PO-mU-PO-fG-PO-mG-PO-fA-PO-mG-PO-fG-PO-fA-PO-fG-PO-mG-PO-mA-PO-mU-PO-mA-PO-mU-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	588	Hy-fA-PS-fG-PS-mA-PO-fA-PO-mU-PO-fA-PO-mU-PO-fC-PO-mC-PO-mU-PO-mC-PO-fC-PO-mU-PO-fC-PO-mC-PO-fA-PO-mU-PO-fC-PO-mU-PS-dT-PS-dT-Hy	642

TABLE 4

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#042				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
042-c-01	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	643	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	697
042-c-02	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	644	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	698
042-c-03	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	645	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-fC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	699
042-c-04	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-	646	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-	700

TABLE 4-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#042				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
	PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy		mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	
042-c-05	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	647	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	701
042-c-06	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	648	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	702
042-c-07	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	649	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-fC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	703
042-c-08	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	650	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	704
042-c-09	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	651	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	705
042-c-10	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	652	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	706
042-c-11	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	653	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	707
042-c-12	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	654	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	708
042-c-13	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	655	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	709
042-c-14	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	656	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	710

TABLE 4-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#042				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
042-c-15	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	657	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	711
042-c-16	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	658	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	712
042-c-17	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	659	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	713
042-c-18	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PS-mU-PS-mA-Hy	660	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	714
042-c-19	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	661	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	715
042-c-20	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	662	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	716
042-c-21	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	663	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-fC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	717
042-c-22	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	664	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	718
042-c-23	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	665	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	719
042-c-24	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	666	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	720

TABLE 4-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#042				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
042-c-25	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	667	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-fC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	721
042-c-26	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	668	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	722
042-c-27	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	669	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	723
042-c-28	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	670	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	724
042-c-29	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	671	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	725
042-c-30	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	672	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	726
042-c-31	Hy-lgT3-PO-lgT3-PO-lgT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	673	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	727
042-c-32	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	674	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	728
042-c-33	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	675	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	729
042-c-34	Hy-lgT3-PO-lgT3-PO-lgT3-PO-lC-PO-lC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-lT4-PO-lT4-Hy	676	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	730

TABLE 4-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#042				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
042-c-35	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-1T4-PO-1T4-Hy	677	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	731
042-c-36	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-1T4-PO-1T4-Hy Hy-1gT3-PO-1gT3-PO-1gT3-	678	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy Hy-mU-PS-fA-PS-mA-PO-	732
042-c-37	PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy Hy-1gT3-PO-1gT3-PO-1gT3-	679	fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy Hy-mU-PS-fA-PS-mA-PO-	733
042-c-38	PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	680	mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	734
042-c-39	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	681	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-fC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	735
042-c-40	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	682	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	736
042-c-41	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	683	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	737
042-c-42	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	684	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	738
042-c-43	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	685	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-fC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	739
042-c-44	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-fU-PO-fU-PO-mU-PO-mG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-	686	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-fA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-	740

TABLE 4-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#042				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
	PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy		fG-PO-mG-PS-dT-PS-dT-Hy	
042-c-45	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	687	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	741
042-c-46	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	688	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	742
042-c-47	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	689	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	743
042-c-48	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	690	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	744
042-c-49	Hy-1gT3-PO-1gT3-PO-1gT3-PO-mC-PO-mC-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	691	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	745
042-c-50	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	692	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	746
042-c-51	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	693	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-fC-PO-mC-PO-mU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	747
042-c-52	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	694	Hy-mU-PS-fA-PS-mA-PO-fG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	748
042-c-53	Hy-1gT3-PO-1gT3-PO-1gT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	695	Hy-mU-PS-fA-PS-mA-PO-mG-PO-mA-PO-mC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-mA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-mG-PO-mG-PS-mA-PS-mA-Hy	749

TABLE 4-continued

Sequences of Exemplary Modified GalNAc-siRNA Constructs from siRNA#042				
CNST #	Sense strand sequence (5'-3')	SEQ	Antisense strand sequence (5'-3')	SEQ
042-c-54	Hy-lgT3-PO-lgT3-PO-lgT3-PO-1C-PO-1C-PO-mG-PO-mA-PO-fG-PO-mA-PO-fA-PO-mU-PO-fU-PO-fU-PO-fG-PO-mA-PO-mG-PO-mG-PO-mU-PO-mC-PO-mU-PO-mU-PO-mA-PO-1T4-PO-1T4-Hy	696	Hy-fU-PS-fA-PS-mA-PO-fG-PO-mA-PO-fC-PO-mC-PO-fU-PO-mC-PO-mA-PO-mA-PO-fA-PO-mU-PO-fU-PO-mC-PO-fU-PO-mC-PO-fG-PO-mG-PS-dT-PS-dT-Hy	750

While the exemplary siRNAs shown in Tables 2-4 include nucleotide modifications, siRNAs having the same or substantially the same sequences but different numbers, patterns, and/or types of modifications, are also contemplated.

In some embodiments, a dsRNA comprises a sense strand shown in Table 1 with the addition of nucleotides (or modified versions thereof) at either or both of its termini. For example, the dsRNA comprises a sense strand shown in Table 1 with the addition of a 5' CCA and/or a 3' invdT. In some embodiments, a dsRNA comprises an antisense strand shown in Table 1 with the addition of nucleotides (or modified versions thereof) at either or both of its termini. For example, the dsRNA comprises an antisense strand shown in Table 1 with the addition of a 3' dTdT. In certain embodiments, a dsRNA comprises a pair of sense and antisense strands as shown in Table 1, with the addition of a 5' CCA and a 3' invdT to the sense strand and with the addition of a 3' dTdT to the antisense strand. In certain embodiments, a dsRNA comprises a pair of sense and antisense strands as shown in Table 2, with the addition of a 5' lgT3-lgT3-lgT3 and a 3' IT4-IT4 to the sense strand.

In some embodiments, a dsRNA of the present disclosure comprises a sense sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identical in sequence to a sense sequence shown in Table 1. In some embodiments, a dsRNA of the present disclosure comprises an antisense sequence that is at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identical in sequence to an antisense sequence shown in Table 1. In some embodiments, a dsRNA of the present disclosure comprises sense and antisense sequences that are at least 70%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% identical in sequence to sense and antisense sequences, respectively, shown in Table 1. In certain embodiments, the dsRNA comprises sense and antisense strands having the sequences shown in Table 2. In certain embodiments, the dsRNA comprises sense and antisense strands having the sequences shown in Table 3.

The "percentage identity" between two nucleotide sequences is determined by comparing the two optimally-aligned sequences in which the nucleic acid sequence to compare can have additions or deletions compared to the reference sequence for optimal alignment between the two sequences. "Percentage identity" is calculated by determining the number of positions at which the nucleotide residue is identical between the two sequences, preferably between the two complete sequences, dividing the number of identical positions by the total number of positions in the alignment window and multiplying the result by 100 to obtain the percentage identity between the two sequences. For purposes herein, when determining "percentage identity" between two nucleotide sequences, modifications to the nucleotides are not considered. For example, a sequence of

5' -mC-fU-mA-fG-3' is considered having 100% sequence identity as a sequence of 5'-CUAG-3'.

1.5 dsRNA Conjugates

The present dsRNAs may be covalently or noncovalently linked to one or more ligands or moieties. Examples of such ligands and moieties may be found, e.g., in Jeong et al., *Bioconjugate Chem.* (2009) 20:5-14 and Sebestyén et al., *Methods Mol. Biol.* (2015) 1218:163-86. In some embodiments, the dsRNA is conjugated/attached to one or more ligands via a linker. Any linker known in the art may be used, including, for example, multivalent (e.g., bivalent, trivalent, or tetravalent) branched linkers. The linker may be cleavable or non-cleavable. Conjugating a ligand to a dsRNA may alter its distribution, enhance its cellular absorption and/or targeting to a particular tissue and/or uptake by one or more specific cell types (e.g., liver cells), and/or enhance the lifetime or half-life of the dsRNA. In some embodiments, a hydrophobic ligand is conjugated to the dsRNA to facilitate direct permeation of the cellular membrane and/or uptake across cells (e.g., liver cells). For ANGPTL8-targeting dsRNAs (e.g., siRNAs), the target tissue may be the liver, including parenchymal cells of the liver (e.g., hepatocytes).

In some embodiments, the dsRNA of the present disclosure is conjugated to a cell-targeting ligand. A cell-targeting ligand refers to a molecular moiety that facilitates delivery of the dsRNA to the target cell, which encompasses (i) increased specificity of the dsRNA to bind to cells expressing the selected target receptors (e.g., target proteins); (ii) increased uptake of the dsRNA by the target cells; and (iii) increased ability of the dsRNA to be appropriately processed once it has entered into a target cell, such as increased intracellular release of an siRNA, e.g., by facilitating the translocation of the siRNA from transport vesicles into the cytoplasm. The ligand may be, for example, a protein (e.g., a glycoprotein), a peptide, a lipid, a carbohydrate, or a molecule having a specific affinity for a co-ligand.

Specific examples of ligands include, without limitation, an antibody or antigen-binding fragment thereof that binds to a specific receptor on a liver cell, thyrotropin, melano-tropin, surfactant protein A, mucin carbohydrate, multivalent lactose, multivalent galactose, multivalent mannose, multivalent fucose, N-acetylgalactosamine, N-acetylglucosamine, transferrin, bisphosphonate, a steroid, bile acid, lipopolysaccharide, a recombinant or synthetic molecule such as a synthetic polymer, polyamino acids, an alpha helical peptide, polyglutamate, polyaspartate, lectins, and cofactors. In some embodiments, the ligand is one or more dyes, crosslinkers, polycyclic aromatic hydrocarbons, peptide conjugates (e.g., antennapedia peptide, Tat peptide), polyethylene glycol (PEG), enzymes, haptens, transport/absorption facilitators, synthetic ribonucleases (e.g., imidazole, bisimidazole, histamine, or imidazole clusters), human serum albumin (HSA), or LDL.

In some embodiments, the dsRNA is conjugated to one or more cholesterol derivatives or lipophilic moieties such as cholesterol or a cholesterol derivative; cholic acid; a vitamin (such as folate, vitamin A, vitamin E (tocopherol), biotin, or pyridoxal); bile or fatty acid conjugates, including both saturated and non-saturated (such as lauroyl (C12), myristoyl (C14), palmitoyl (C16), stearoyl (C18) and docosanyl (C22), lithocholic acid and/or lithocholic acid oleylamine conjugate (lithocholic-oleyl, C43)); polymeric backbones or scaffolds (such as PEG, triethylene glycol (TEG), hexaethylene glycol (HEG), poly(lactic-co-glycolic acid) (PLGA), poly(lactide-co-glycolide) (PLG), hydrodynamic polymers); steroids (such as dihydrotestosterone); terpene (such as triterpene); cationic lipids or peptides; and/or a lipid or lipid-based molecule. Such a lipid or lipid-based molecule may bind a serum protein, e.g., human serum albumin (HSA). A lipid-based ligand may be used to modulate (e.g., control) the binding of the conjugate to a target tissue. For example, a lipid or lipid-based ligand that binds to HSA more strongly will be less likely to be targeted to the kidney and therefore less likely to be cleared from the body.

In some embodiments, the cell-targeting moiety or ligand is a N-acetylgalactosamine (GalNAc) derivative. In some embodiments, the dsRNA is attached to one or more (e.g., two, three, four, or more) GalNAc derivatives. The attachment may be via one or more linkers (e.g., two, three, four, or more linkers). In some embodiments, a linker described herein is a multivalent (e.g., bivalent, trivalent, or tetravalent) branched linker. In some embodiments, the dsRNA is attached to two or more GalNAc derivatives via a bivalent branched linker. In some embodiments, the dsRNA is attached to three or more GalNAc derivatives via a trivalent branched linker. In some embodiments, the dsRNA is attached to three or more GalNAc derivatives with or without linkers. In some embodiments, the dsRNA is attached to four or more GalNAc derivatives via four separate linkers. In some embodiments, the dsRNA is attached to four or more GalNAc derivatives via a tetravalent branched linker. In some embodiments, the one or more GalNAc derivatives is attached to the 3'-end of the sense strand, the 3'-end of the antisense strand, the 5'-end of the sense strand, and/or the 5'-end of the antisense strand of the dsRNA. In some embodiments, one or more (e.g., two or three) modified nucleotides of formula (I) are placed at the 5' end and/or 3' end of the sense strand of the dsRNA, where those modified nucleotides each contain a GalNAc derivative moiety (see, e.g., the GalNAc-containing nucleotides shown in Tables A and B). Exemplary and non-limiting conjugates and linkers are described, e.g., in Biessen et al., *Bioconjugate Chem.* (2002) 13(2):295-302; Cedillo et al., *Molecules* (2017) 22(8):E1356; Grijalvo et al., *Genes* (2018) 9(2):E74; Huang et al., *Molecular Therapy: Nucleic Acids* (2017) 6:116-32; Nair et al., *J. Am. Chem. Soc.* (2014) 136:16958-61; Ostergaard et al., *Bioconjugate Chem.* (2015) 26:1451-5; Springer et al., *Nucleic Acid Therapeutics* (2018) 28(3):109-18; and U.S. Pat. Nos. 8,106,022, 9,127,276, and 8,927,705. GalNAc conjugation can be readily performed by methods well known in the art (e.g., as described in the above documents).

II. Methods of Making dsRNAs

A dsRNA of the present disclosure may be synthesized by any method known in the art. For example, a dsRNA may be synthesized by use of an automated synthesizer, by in vitro transcription and purification (e.g., using commercially available in vitro RNA synthesis kits), by transcription and purification from cells (e.g., cells comprising an expression cassette/vector encoding the dsRNA), and the like. In some

embodiments, the sense and antisense strands of the dsRNA are synthesized separately and then annealed to form the dsRNA. In some embodiments, the dsRNA comprising modified nucleotides of formula (I) and optionally conjugated to a cell targeting moiety (e.g., GalNAc) may be prepared according to the disclosure of PCT Publication WO 2019/170731.

Ligand-conjugated dsRNAs and ligand molecules bearing sequence-specific linked nucleosides of the present disclosure may be assembled by any method known in the art, including, for example, assembly on a suitable polynucleotide synthesizer utilizing standard nucleotide or nucleoside precursors, or nucleotide or nucleoside conjugate precursors that already bear the linking moiety, ligand-nucleotide, or nucleoside-conjugated precursors that already bear the ligand molecule, or non-nucleoside ligand-bearing building blocks.

Ligand-conjugated dsRNAs of the present disclosure may be synthesized by any method known in the art, including, for example, by the use of a dsRNA bearing a pendant reactive functionality such as that derived from the attachment of a linking molecule onto the dsRNA. In some embodiments, this reactive oligonucleotide may be reacted directly with commercially-available ligands, ligands that are synthesized bearing any of a variety of protecting groups, or ligands that have a linking moiety attached thereto. In some embodiments, the methods facilitate the synthesis of ligand-conjugated dsRNA by the use of nucleoside monomers that have been appropriately conjugated with ligands and that may further be attached to a solid support material. In some embodiments, a dsRNA bearing an alkyl ligand attached to the 3'-end of the dsRNA is prepared by first covalently attaching a monomer building block to a controlled-pore-glass support via a long-chain aminoalkyl group; then, nucleotides are bonded via standard solid-phase synthesis techniques to the monomer building-block bound to the solid support. The monomer building-block may be a nucleoside or other organic compound that is compatible with solid-phase synthesis.

In some embodiments, functionalized nucleoside sequences of the present disclosure possessing an amino group at the 5'-terminus are prepared using a polynucleotide synthesizer, and then reacted with an active ester derivative of a selected ligand. Active ester derivatives are well known to one of ordinary skill in the art. The reaction of the amino group and the active ester produces an oligonucleotide in which the selected ligand is attached to the 5'-position through a linking group. The amino group at the 5'-terminus can be prepared utilizing a 5'-amino-modifier C6 reagent. In some embodiments, ligand molecules are conjugated to oligonucleotides at the 5'-position by the use of a ligand-nucleoside phosphoramidite wherein the ligand is linked to the 5'-hydroxy group directly or indirectly via a linker. Such ligand-nucleoside phosphoramidites are typically used at the end of an automated synthesis procedure to provide a ligand-conjugated oligonucleotide bearing the ligand at the 5'-terminus.

In some embodiments, click chemistry is used to synthesize siRNA conjugates. See, e.g., Astakhova et al., *Mol. Pharm.* (2018) 15(8):2892-9; Mercier et al., *Bioconjugate Chem.* (2011) 22(1):108-14.

III. Compositions and Delivery of dsRNAs

Certain aspects of the present disclosure relate to compositions (e.g., pharmaceutical compositions) comprising a dsRNA as described herein. In some embodiments, the composition further comprises a pharmaceutically acceptable excipient. In some embodiments, the composition is

useful for treating a disease or disorder associated with the expression or activity of the ANGPTL8 gene. In some embodiments, the disease or disorder associated with the expression of the ANGPTL8 gene is a lipid metabolism disorder such as hypertriglyceridemia and/or any other condition described herein. Compositions of the present disclosure may be formulated based upon the mode of delivery, including, for example, compositions formulated for delivery to the liver via parenteral administration.

The present dsRNAs can be formulated with a pharmaceutically acceptable excipient. Pharmaceutically acceptable excipients can be liquid or solid, and may be selected with the planned manner of administration in mind so as to provide for the desired bulk, consistency, and other pertinent transport and chemical properties. Any known pharmaceutically acceptable excipient may be used, including, for example, water, saline solution, binding agents (e.g., polyvinylpyrrolidone or hydroxypropyl methylcellulose), fillers (e.g., lactose and other sugars, gelatin, or calcium sulfate), lubricants (e.g., starch, polyethylene glycol, or sodium acetate), disintegrates (e.g., starch or sodium starch glycolate), calcium salts (e.g., calcium sulfate, calcium chloride, calcium phosphate, and hydroxyapatite), and wetting agents (e.g., sodium lauryl sulfate).

The present dsRNAs can be formulated into compositions (e.g., pharmaceutical compositions) containing the dsRNA admixed, encapsulated, conjugated, or otherwise associated with other molecules, molecular structures, or mixtures of nucleic acids. For example, a composition comprising one or more dsRNAs as described herein can contain other therapeutic agents such as other lipid lowering agents (e.g., statins). In some embodiments, the composition (e.g., pharmaceutical composition) further comprises a delivery vehicle as described herein.

A dsRNA of the present disclosure may be delivered directly or indirectly. In some embodiments, the dsRNA is delivered directly by administering a pharmaceutical composition comprising the dsRNA to a subject. In some embodiments, the dsRNA is delivered indirectly by administering one or more vectors described below.

A dsRNA of the present disclosure may be delivered by any method known in the art, including, for example, by adapting a method of delivering a nucleic acid molecule for use with a dsRNA (see, e.g., Akhtar et al., *Trends Cell Biol.* (1992) 2(5):139-44; PCT Publication WO 94/02595), or via additional methods known in the art (see, e.g., Kanasty et al., *Nature Materials* (2013) 12:967-77; Wittrup and Lieberman, *Nature Reviews Genetics* (2015) 16:543-52; Whitehead et al., *Nature Reviews Drug Discovery* (2009) 8:129-38; Gary et al., *J Control Release* (2007) 121(1-2):64-73; Wang et al., *AAPS J.* (2010) 12(4):492-503; Draz et al., *Theranostics* (2014) 4(9):872-92; Wan et al., *Drug Deliv Transl Res.* (2013) 4(1):74-83; Erdmann and Barciszewski (eds.) (2010) "RNA Technologies and Their Applications," Springer-Verlag Berlin Heidelberg, DOI 10.1007/978-3-642-12168-5; Xu and Wang, *Asian Journal of Pharmaceutical Sciences* (2015) 10(1):1-12). For in vivo delivery, dsRNA can be injected into a tissue site or administered systemically (e.g., in nanoparticle form via inhalation). In vivo delivery can also be mediated by a beta-glucan delivery system (see, e.g., Tesz et al., *Biochem J.* (2011) 436(2):351-62). In vitro introduction into a cell includes methods known in the art such as electroporation and lipofection.

In some embodiments, a dsRNA of the present disclosure is delivered by a delivery vehicle comprising the dsRNA. In some embodiments, the delivery vehicle is a liposome, lipoplex, complex, or nanoparticle.

III.1 Liposomal Formulations

Liposomes are unilamellar or multilamellar vesicles which have a membrane formed from a lipophilic material and an aqueous interior. In some embodiments, a liposome is a vesicle composed of amphiphilic lipids arranged in a spherical bilayer or bilayers. The aqueous portion contains the composition to be delivered. Cationic liposomes possess the advantage of being able to fuse to the cell wall. Advantages of liposomes include, e.g., that liposomes obtained from natural phospholipids are biocompatible and biodegradable; that liposomes can incorporate a wide range of water and lipid soluble drugs; and that liposomes can protect encapsulated drugs in their internal compartments from metabolism and degradation (Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245). Important considerations in the preparation of liposome formulations are the lipid surface charge, vesicle size and the aqueous volume of the liposomes. For example, engineered cationic liposomes and sterically stabilized liposomes can be used to deliver the dsRNA. See, e.g., Podesta et al., *Methods Enzymol.* (2009) 464:343-54; U.S. Pat. No. 5,665,710.

III.2 Nucleic Acid-Lipid Particles

In some embodiments, a dsRNA of the present disclosure is fully encapsulated in a lipid formulation, e.g., to form a nucleic acid-lipid particle such as, without limitation, a SPLP, pSPLP, or SNALP. As used herein, the term "SNALP" refers to a stable nucleic acid-lipid particle, including SPLP. As used herein, the term "SPLP" refers to a nucleic acid-lipid particle comprising plasmid DNA encapsulated within a lipid vesicle. Nucleic acid-lipid particles, e.g., SNALPs, typically contain a cationic lipid, a non-cationic lipid, and a lipid that prevents aggregation of the particle (e.g., a PEG-lipid conjugate). SNALPs and SPLPs are useful for systemic applications, as they exhibit extended circulation lifetimes following intravenous (i.v.) injection and accumulate at distal sites (e.g., sites physically separated from the administration site). SPLPs include "pSPLPs," which include an encapsulated condensing agent-nucleic acid complex as set forth in PCT Publication WO 2000/003683.

In some embodiments, dsRNAs when present in nucleic acid-lipid particles are resistant in aqueous solution to degradation with a nuclease. Nucleic acid-lipid particles and their methods of preparation are disclosed in, e.g., U.S. Pat. Nos. 5,976,567; 5,981,501; 6,534,484; 6,586,410; and 6,815,432; and PCT Publication WO 96/40964.

In some embodiments, the nucleic acid-lipid particles comprise a cationic lipid. Any cationic lipid or mixture thereof known in the art may be used. In some embodiments, the nucleic acid-lipid particles comprise a non-cationic lipid. Any non-cationic lipid or mixture thereof known in the art may be used. In some embodiments, the nucleic acid-lipid particle comprises a conjugated lipid (e.g., to prevent aggregation). Any conjugated lipid known in the art may be used.

III.3 Additional Formulations

Factors that are important to consider in order to successfully deliver a dsRNA molecule in vivo include: (1) biological stability of the delivered molecule, (2) preventing nonspecific effects, and (3) accumulation of the delivered molecule in the target tissue. The nonspecific effects of a dsRNA can be minimized by local administration, for example by direct injection or implantation into a tissue or topically administering the preparation. For administering a dsRNA systemically for the treatment of a disease, the dsRNA may be modified or alternatively delivered using a drug delivery system; both methods act to prevent the rapid

degradation of the dsRNA by endo- and exonucleases in vivo. Modification of the RNA or the pharmaceutical excipient may also permit targeting of the dsRNA composition to the target tissue and avoid undesirable off-target effects. As described above, dsRNA molecules may be modified by chemical conjugation to lipophilic groups such as cholesterol to enhance cellular uptake and prevent degradation. In some embodiments, the dsRNA is delivered using drug delivery systems such as a nanoparticle (e.g., a calcium phosphate nanoparticle), a dendrimer, a polymer, liposomes, or a cationic delivery system. Positively charged cationic delivery systems facilitate binding of a dsRNA molecule (negatively charged) and also enhance interactions at the negatively charged cell membrane to permit efficient uptake of a dsRNA by the cell. Cationic lipids, dendrimers, or polymers can either be bound to a dsRNA, or induced to form a vesicle or micelle (See, e.g., Kim et al., *Journal of Controlled Release* (2008) 129(2):107-16) that encases a dsRNA. The formation of vesicles or micelles further prevents degradation of the dsRNA when administered systemically. Methods for making and administering cationic-dsRNA complexes are known in the art. In some embodiments, a dsRNA may form a complex with cyclodextrin for systemic administration.

III.4 Vector-Encoded dsRNAs

A dsRNA of the present disclosure may be delivered to the target cell indirectly by introducing into the target cell a recombinant vector (DNA or RNA vector) encoding the dsRNA. The dsRNA will be expressed from the vector inside the cell, e.g., in the form of shRNA, where the shRNA is subsequently processed into siRNA intracellularly. In some embodiments, the vector is a plasmid, cosmid, or viral vector. In some embodiments, the vector is compatible with expression in prokaryotic cells. In some embodiments, the vector is compatible with expression in *E. coli*. In some embodiments, the vector is compatible with expression in eukaryotic cells. In some embodiments, the vector is compatible with expression in yeast cells. In some embodiments, the vector is compatible with expression in vertebrate cells. Any expression vector capable of encoding dsRNA known in the art may be used, including, for example, vectors derived from adenovirus (AV), adeno-associated virus (AAV), retroviruses (e.g., lentiviruses (LV), Rhabdoviruses, murine leukemia virus, etc.), herpes virus, SV40 virus, polyoma virus, papilloma virus, picornavirus, pox virus (e.g., orthopox or avipox), and the like. The tropism of viral vectors or viral-derived vectors may be modified by pseudotyping the vectors with envelope proteins or other surface antigens from one or more other viruses, or by substituting different viral capsid proteins, as appropriate. For example, lentiviral vectors may be pseudotypes with surface proteins from vesicular stomatitis virus (VSV), rabies, Ebola, Mokola, and the like. AAV vectors may be made to target different cells by engineering the vectors to express different capsid protein serotypes. For example, an AAV vector expressing a serotype 2 capsid on a serotype 2 genome is called AAV 2/2. This serotype 2 capsid gene in the AAV 2/2 vector can be replaced by a serotype 5 capsid gene to produce an AAV 2/5 vector. Techniques for constructing AAV vectors which express different capsid protein serotypes have been described previously (see, e.g., Rabinowitz et al., *J. Virol.* (2002) 76:791-801).

Selection of recombinant vectors, methods for inserting nucleic acid sequences into the vector for expressing a dsRNA, and methods of delivering vectors into one or more cells of interest are known in the art. See, e.g., Domburg, *Gene Therap.* (1995) 2:301-310; Eglitis et al., *Biotechniques*

(1998) 6:608-14; Miller, *Hum Gene Therap.* 1:5-14 (1990); Anderson et al., *Nature* (1998) 392:25-30; Xia et al., *Nat. Biotech.* (2002) 20:1006-10; Robinson et al., *Nat Genet.* (2003) 33:401-6; Samulski et al., *J. Virol.* (1987) 61:3096-101; Fisher et al., *J Virol.* (1996) 70:520-32; Samulski et al., *J Virol.* (1989) 63:3822-6; U.S. Pat. Nos. 5,252,479 and 5,139,941; and PCT Publications WO 94/13788 and WO 93/24641.

Vectors useful for the delivery of a dsRNA as described herein may include regulatory elements (e.g., heterologous promoter, enhancer, etc.) sufficient for expression of the dsRNA in the desired target cell or tissue. In some embodiments, the vector comprises one or more sequences encoding the dsRNA linked to one or more heterologous promoters. Any heterologous promoter known in the art capable of expressing a dsRNA may be used, including, for example, the U6 or H1 RNA pol III promoters, the T7 promoter, and the cytomegalovirus promoter. The one or more heterologous promoters may be an inducible promoter, a repressible promoter, a regulatable promoter, and/or a tissue-specific promoter. Selection of additional promoters is within the abilities of one of ordinary skill in the art. In some embodiments, the regulatory elements are selected to provide constitutive expression. In some embodiments, the regulatory elements are selected to provide regulated/inducible/repressible expression. In some embodiments, the regulatory elements are selected to provide tissue-specific expression. In some embodiments, the regulatory elements and sequence encoding the dsRNA form a transcription unit.

A dsRNA of the present disclosure may be expressed from transcription units inserted into DNA or RNA vectors (see, e.g., Couture et al., *TIG* (1996) 12:5-10; PCT Publications WO 00/22113 and WO 00/22114; and U.S. Pat. No. 6,054,299). Expression may be transient (on the order of hours to weeks) or sustained (weeks to months or longer), depending upon the specific construct used and the target tissue or cell type. These transgenes can be introduced as a linear construct, a circular plasmid, or a viral vector, which can be an integrating or non-integrating vector. The transgene can also be constructed to permit it to be inherited as an extrachromosomal plasmid (Gassmann et al., *PNAS* (1995) 92:1292).

In some embodiments, the sense and antisense strands of a dsRNA are encoded on separate expression vectors. In some embodiments, the sense and antisense strands are expressed on two separate expression vectors that are co-introduced (e.g., by transfection or infection) into the same target cell. In some embodiments, the sense and antisense strands are encoded on the same expression vector. In some embodiments, the sense and antisense strands are transcribed from separate promoters which are located on the same expression vector. In some embodiments, the sense and antisense strands are transcribed from the same promoter on the same expression vector. In some embodiments, the sense and antisense strands are transcribed from the same promoter as an inverted repeat joined by a linker polynucleotide sequence such that the dsRNA has a stem and loop structure.

IV. dsRNA Therapy

Certain aspects of the present disclosure relate to methods for inhibiting the expression of the ANGPTL8 gene in a subject (e.g., a primate subject such as a human) comprising administering a therapeutically effective amount of one or more dsRNAs of the present disclosure, or one or more vectors of the present disclosure, or one or more pharmaceutical compositions of the present disclosure. Certain aspects of the present disclosure relate to methods of treating and/or preventing one or more conditions described herein (e.g., an

ANGPTL8-associated condition such as hypertriglyceridemia) comprising administering one or more dsRNAs of the present disclosure and/or one or more vectors of the present disclosure and/or one or more pharmaceutical compositions comprising one or more dsRNAs as described herein. In some embodiments, downregulating ANGPTL8 expression in a subject alleviates one or more symptoms of a condition described herein (e.g., an ANGPTL8-associated condition such as hypertriglyceridemia) in the subject.

The pharmaceutical composition of the present disclosure may be administered in dosages sufficient to inhibit expression of the ANGPTL8 gene. In some embodiments, a suitable dose of a dsRNA described herein is in the range of 0.001 mg/kg-200 mg/kg body weight of the recipient. In certain embodiments, a suitable dose is in the range of 0.001 mg/kg-50 mg/kg body weight of the recipient, e.g., in the range of 0.001 mg/kg-20 mg/kg body weight of the recipient. Treatment of a subject with a therapeutically effective amount of a pharmaceutical composition can include a single treatment or a series of treatments.

As used herein, the terms "therapeutically effective amount" and "prophylactically effective amount" refer to an amount that provides a therapeutic benefit in the treatment, prevention, or management of pathological processes mediated by ANGPTL8 expression, or an overt symptom of pathological processes mediated by ANGPTL8 expression.

As used herein, the term "ANGPTL8-associated condition" is intended to include any condition in which inhibiting the activity of ANGPTL8 is beneficial. Such a condition may be caused, for example, by excess production of the ANGPTL8 protein, by ANGPTL8 gene mutations that increase ANGPTL8 activity or expression, by abnormal cleavage of the ANGPTL8 protein that increases activity or decreases degradation, and/or by abnormal interactions between ANGPTL8 and other proteins or other endogenous or exogenous substances such that ANGPTL8 activity is increased or degradation is decreased. An ANGPTL8-associated condition may be, e.g., a lipid metabolism disorder.

In some embodiments, a dsRNA described herein is used to treat a subject with a lipid metabolism disorder such as hypertriglyceridemia or any symptoms or conditions associated with hypertriglyceridemia. In certain embodiments, a dsRNA described herein is used to treat a patient with drug-induced hypertriglyceridemia, diuretic-induced hypertriglyceridemia, alcohol-induced hypertriglyceridemia, β -adrenergic blocking agent-induced hypertriglyceridemia, estrogen-induced hypertriglyceridemia, glucocorticoid-induced hypertriglyceridemia, retinoid-induced hypertriglyceridemia, cimetidine-induced hypertriglyceridemia, familial hypertriglyceridemia, acute pancreatitis associated with hypertriglyceridemia, and/or hepatosplenomegaly associated with hypertriglyceridemia.

In some embodiments, a dsRNA described herein is used to treat a subject having one or more conditions selected from: lipidemia (e.g., hyperlipidemia), dyslipidemia (e.g., atherogenic dyslipidemia, diabetic dyslipidemia, or mixed dyslipidemia), hyperlipoproteinemia, hypercholesterolemia, gout associated with hypercholesterolemia, chylomicronemia, lipodystrophy, lipoatrophy, metabolic syndrome, diabetes (Type I or Type II), pre-diabetes, Cushing's syndrome, acromegaly, systemic lupus erythematosus, dysglobulinemia, polycystic ovary syndrome, Addison's disease, glycogen storage disease type I, hypothyroidism, uremia, adriamycin cardiomyopathy, lipoprotein lipase deficiency, lysosomal acid lipase deficiency, xanthomatosis, eruptive xanthoma, and lipemia retinalis.

Additionally or alternatively, a dsRNA described herein may be used to treat a subject with one or more pathological conditions associated with any of the disorders described herein, such as heart and circulatory conditions (e.g., atherosclerosis, angina, hypertension, congestive heart failure, coronary artery disease, restenosis, myocardial infarction, stroke, aneurysm, cerebrovascular diseases, and peripheral vascular diseases), liver disease, kidney disease, nephrotic syndrome, and chronic renal disease (e.g., uremia, nephrotic syndrome, maintenance dialysis, and renal transplantation).

In some embodiments, a dsRNA described herein may be used to treat a subject with one or more conditions associated with any genetic profile (e.g., familial hypertriglyceridemia, familial combined lipidemia, or familial dysbetalipoproteinemia), treatment (e.g., use of thiazide diuretics, oral contraceptives and other estrogens, certain beta-adrenergic blocking drugs, propofol, HIV medications, isotretinoin, or protease inhibitors), or lifestyle (e.g., cigarette smoking, excessive alcohol consumption, high carbohydrate diet, or high fat diet) that results in or results from elevated blood triglycerides or lipids. Triglyceride levels (e.g., serum triglyceride levels) of over 150 mg/dL are considered elevated for risk of cardiovascular conditions. Triglyceride levels (e.g., serum triglyceride levels) of 500 mg/dL or higher are considered elevated for risk of pancreatitis.

In some embodiments, a dsRNA described herein may be used to manage body weight or reduce fat mass in a subject.

In some embodiments, a dsRNA as described herein inhibits expression of the human ANGPTL8 gene, or both human and cynomolgus ANGPTL8 genes. The expression of the ANGPTL8 gene in a subject may be inhibited, or the ANGPTL8 protein levels in the subject may be reduced, by at least about 20%, at least about 25%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 85%, at least about 90%, at least about 95%, at least about 99%, or about 100% after treatment as compared to pretreatment levels. In some embodiments, expression of the ANGPTL8 gene is inhibited, or the ANGPTL8 protein levels in the subject may be reduced, by at least about 2, at least about 5, at least about 10, at least about 15, at least about 20, at least about 25, at least about 50, at least about 75, or at least about 100 fold after treatment as compared to pretreatment levels. In some embodiments, the ANGPTL8 gene is inhibited or the ANGPTL8 protein levels are reduced in the liver of the subject.

In some embodiments, expression of the ANGPTL8 gene is decreased by the dsRNA for about 12 or more, 24 or more, or 36 or more hours. In some embodiments, expression of the ANGPTL8 gene is decreased for an extended duration, e.g., at least about two, three, four, five, or six days or more, e.g., about one week, two weeks, three weeks, or four weeks or longer.

As used herein, the terms "inhibit the expression of" or "inhibiting expression of," insofar as they refer to the ANGPTL8 gene, refer to the at least partial suppression of the expression of the ANGPTL8 gene, as manifested by a reduction in the amount of mRNA transcribed from the ANGPTL8 gene in a first cell or group of cells treated such that the expression of the ANGPTL8 gene is inhibited, as compared to a second cell or group of cells substantially identical to the first cell or group of cells but which has or have not been so treated (control cells). Such inhibition can be assessed, e.g., by Northern analysis, in situ hybridization, B-DNA analysis, expression profiling, transcription of

reporter constructs, and other techniques known in the art. As used herein, the term “inhibiting” is used interchangeably with “reducing,” “silencing,” “downregulating,” “suppressing,” and other similar terms, and include any level of inhibition. The degree of inhibition is usually expressed in terms of $\frac{((\text{mRNA in control cells}) - (\text{mRNA in treated cells}))}{(\text{mRNA in control cells})} \times 100\%$.

Alternatively, the degree of inhibition may be given in terms of a reduction of a parameter that is functionally linked to ANGPTL8 gene transcription, e.g., the amount of protein encoded by the ANGPTL8 gene in a cell (as assessed, e.g., by Western analysis, expression of a reporter protein, ELISA, immunoprecipitation, or other techniques known in the art), or the number of cells displaying a certain phenotype, e.g., apoptosis. In principle, ANGPTL8 gene silencing may be determined in any cell expressing the target, either constitutively or by genomic engineering, and by any appropriate assay. However, when a reference is needed in order to determine whether a given dsRNA inhibits the expression of the ANGPTL8 gene by a certain degree and therefore is encompassed by the present disclosure, the assays provided in the Examples below shall serve as such a reference.

In some embodiments, the effect of inhibiting ANGPTL8 gene expression by any of the methods described herein results in a decrease in triglyceride levels in a subject (e.g., in the blood and/or serum of the subject). In some embodiments, triglyceride levels are decreased to below one of the following levels: 500, 450, 400, 350, 300, 250, 200, 190, 180, 170, 160, 150, 140, 130, 120, 110, 100, 90, 80, 70, 60, or 50 mg/dL.

A subject's triglyceride levels may be determined in any of numerous ways known in the art. In some embodiments, a subject's triglyceride levels are determined using a sample from the subject such as blood, serum, or plasma.

A dsRNA or pharmaceutical composition described herein may be administered by any means known in the art, including, without limitation, oral or parenteral routes, including intravenous, intramuscular, subcutaneous, pulmonary, transdermal, and airway (aerosol) administration. Typically, when treating a patient with hypertriglyceridemia, the dsRNA molecules are administered systemically via parenteral means. In some embodiments, the dsRNAs and/or compositions are administered by subcutaneous administration. In some embodiments, the dsRNAs and/or compositions are administered by intravenous administration. In some embodiments, the dsRNAs and/or compositions are administered by pulmonary administration.

As used herein, in the context of ANGPTL8 expression, the terms “treat,” “treatment” and the like refer to relief from or alleviation of pathological processes mediated by target gene expression. In the context of the present disclosure, insofar as it relates to any of the conditions recited herein, the terms “treat,” “treatment,” and the like refer to relieving or alleviating one or more symptoms associated with said condition. For example, in the context of hypertriglyceridemia, treatment may involve a decrease in serum triglyceride levels. As used herein, to “alleviate” a disease, disorder or condition means reducing the severity and/or occurrence frequency of the symptoms of the disease, disorder, or condition. Further, references herein to “treatment” include references to curative, palliative and prophylactic treatment.

As used herein, the terms “prevent” or “delay progression of” (and grammatical variants thereof), with respect to a condition relate to prophylactic treatment of a condition, e.g., in an individual suspected to have or be at risk for developing the condition. Prevention may include, but is not

limited to, preventing or delaying onset or progression of the condition and/or maintaining one or more symptoms of the disease at a desired or sub-pathological level. For example, in the context of hypertriglyceridemia, prevention may involve maintaining serum triglyceride levels at a desired level in an individual suspected to have or be at risk for developing hypertriglyceridemia.

It is understood that the dsRNAs of the present disclosure may be for use in a treatment as described herein, may be used in a method of treatment as described herein, and/or may be for use in the manufacture of a medicament for a treatment as described herein.

In some embodiments, a dsRNA of the present disclosure is administered in combination with one or more additional therapeutic agents, such as other siRNA therapeutic agents, monoclonal antibodies, and small molecules, to provide a greater improvement to the condition of the patient than administration of the dsRNA alone. In certain embodiments, the additional therapeutic agent provides an anti-inflammatory effect. In certain embodiments, the additional therapeutic agent is an agent that treats hypertriglyceridemia, such as a lipid-lowering agent.

In some embodiments, the additional agent may be one or more of a HMG-CoA reductase inhibitor (e.g., a statin), a fibrate, a bile acid sequestrant, niacin, nicotinic acid, an antiplatelet agent, an angiotensin converting enzyme inhibitor, an angiotensin II receptor antagonist (e.g., losartan potassium), an acylCoA cholesterol acetyltransferase (ACAT) inhibitor, a cholesterol absorption inhibitor, a cholesterol ester transfer protein (CETP) inhibitor, a microsomal triglyceride transfer protein (MTTP) inhibitor, a cholesterol modulator, a bile acid modulator, a peroxisome proliferation activated receptor (PPAR) agonist, an omega-3 fatty acid (e.g., fish oil or flaxseed oil), and insulin or an insulin analog. Particular examples include, without limitation, atorvastatin, pravastatin, simvastatin, lovastatin, fluvastatin, cerivastatin, rosuvastatin, pitavastatin, ezetimibe, bezafibrate, clofibrate, fenofibrate, gemfibrozil, ciprofibrate, cholestyramine, colestipol, colesvelam, and niacin.

In certain embodiments, a dsRNA as described herein may be administered in combination with another therapeutic intervention such as lipid lowering, weight loss, dietary modification, and/or moderate exercise.

Genetic predisposition plays a role in the development of target gene associated diseases, e.g., hypertriglyceridemia. Therefore, a subject in need of treatment with one or more dsRNAs of the present disclosure may be identified by taking a family history, or, for example, screening for one or more genetic markers or variants. Examples of genes involved in hypertriglyceridemia may include, without limitation, LPL, APOB, APOC2, APOA5, APOE, LMF1, GCKR, GPIHBP1, and GPD1. In certain embodiments, a subject in need of treatment with one or more dsRNAs of the present disclosure may be identified by screening for variants of or loss-of-function mutations in any of these genes or any combination thereof.

A healthcare provider, such as a doctor, nurse, or family member, can take a family history before prescribing or administering a dsRNA of the present disclosure. In addition, a test may be performed to determine a genotype or phenotype. For example, a DNA test may be performed on a sample from the subject, e.g., a blood sample, to identify the ANGPTL8 genotype and/or phenotype before the dsRNA is administered to the subject.

V. Kits and Articles of Manufacture

Certain aspects of the present disclosure relate to an article of manufacture or a kit comprising one or more of the

dsRNAs, vectors, or compositions (e.g., pharmaceutical compositions) as described herein useful for the treatment and/or prevention of an ANGPTL8-associated condition (e.g., a lipid metabolism disorder such as hypertriglyceridemia). The article of manufacture or kit may further comprise a container and a label or package insert on or associated with the container. Suitable containers include, for example, bottles, vials, syringes, IV solution bags, etc. The containers may be formed from a variety of materials such as glass or plastic. The container holds a composition which is by itself or combined with another composition effective for treating or preventing the disease and may have a sterile access port (for example the container may be an intravenous solution bag or a vial having a stopper pierceable by a hypodermic injection needle). At least one active agent in the composition is a dsRNA as described herein. The label or package insert indicates that the composition is used for treating an ANGPTL8-associated condition. In some embodiments, the condition is a lipid metabolism disorder such as hypertriglyceridemia and/or another condition described herein.

Moreover, the article of manufacture or kit may comprise (a) a first container with a composition contained therein, wherein the composition comprises a dsRNA as described herein; and (b) a second container with a composition contained therein, wherein the composition comprises a second therapeutic agent (e.g., an additional agent as described herein). The article of manufacture or kit in this aspect of the present disclosure may further comprise a package insert indicating that the compositions can be used to treat a particular disease. Alternatively, or additionally, the article of manufacture or kit may further comprise a second (or third) container comprising a pharmaceutically-acceptable buffer, such as bacteriostatic water for injection (BWI), phosphate-buffered saline, Ringer's solution and dextrose solution. It may further include other materials desirable from a commercial and/or user standpoint, including other buffers, diluents, filters, needles, and syringes.

Unless otherwise defined herein, scientific and technical terms used in connection with the present disclosure shall have the meanings that are commonly understood by those of ordinary skill in the art. Exemplary methods and materials are described below, although methods and materials similar or equivalent to those described herein can also be used in the practice or testing of the present disclosure. In case of conflict, the present specification, including definitions, will control. Generally, nomenclature used in connection with, and techniques of, cell and tissue culture, molecular biology, immunology, microbiology, genetics, analytical chemistry, synthetic organic chemistry, medicinal and pharmaceutical chemistry, and protein and nucleic acid chemistry and hybridization described herein are those well-known and commonly used in the art. Enzymatic reactions and purification techniques are performed according to the manufacturer's specifications, as commonly accomplished in the art or as described herein. Further, unless otherwise required by context, singular terms shall include pluralities and plural terms shall include the singular. Throughout this specification and embodiments, the words "have" and "comprise," or variations such as "has," "having," "comprises," or "comprising," will be understood to imply the inclusion of a stated integer or group of integers but not the exclusion of any other integer or group of integers. All publications and other references mentioned herein are incorporated by reference in their entirety. Although a number of documents are

cited herein, this citation does not constitute an admission that any of these documents forms part of the common general knowledge in the art.

EXAMPLES

In order for the present disclosure to be better understood, the following examples are set forth. These examples are for illustration only and are not to be construed as limiting the scope of the present disclosure in any manner.

Example 1: siRNA Synthesis and Purification

siRNAs, including non-targeting control siRNAs (NT control), were produced using solid phase oligonucleotide synthesis.

Methods

siRNA Production

RNA oligonucleotides were synthesized at a scale of 1 μ mol (in vitro) or 10 μ mol (in vivo) on a ABI 394 DNA/RNA or BioAutomation MerMade 12 synthesizer using commercially available 5'-O-DMT-3'-O-(2-cyanoethyl-N, N-diisopropyl) phosphoramidite monomers (SAFC) of uridine, 4-N-acetylcytidine (C^{Ac}), 6-N-benzoyladenosine (A^{Bz}) and 2-N-isobutyrylguanosine (G^{IBu}) with 2'-OMe or 2'-F modification, and the solid supports 5'-O-DMT-thymidine-CPG and 3'-O-DMT-thymidine-CPG (invdT, Link) following standard protocols for solid phase synthesis and deprotection (Beaucage et al., *Curr. Opin. Drug Discov. Devel.* (2008) 11:203-16; Mueller et al., *Curr. Org. Synth.* (2004) 1:293-307).

Phosphoramidite building blocks were used as 0.1 M solutions in acetonitrile and activated with 5-(bis-3,5-trifluoromethylphenyl)-1H-tetrazole (activator 42, 0.25 M in acetonitrile, Sigma Aldrich). Reaction times of 300 s were used for the phosphoramidite couplings. As capping reagents, acetic anhydride in THF (CapA for ABI, Sigma Aldrich) and N-methylimidazole in THF (CapB for ABI, Sigma Aldrich) were used. As oxidizing reagent, iodine in THF/pyridine/water (0.02 M; oxidizer for ABI, Sigma Aldrich) was used. Deprotection of the DMT-protecting group was done using dichloroacetic acid in DCM (DCA deblock, Sigma Aldrich). Final cleavage from solid support and deprotection (acyl- and cyanoethyl-protecting groups) was achieved with NH_3 (32% aqueous solution/ethanol, v/v 3:1).

The crude oligonucleotides were analyzed by IEX and LC-MS and purified by anion-exchange high-performance liquid chromatography (IEX-HPLC) using a linear gradient of 10-65% buffer B in 30 min. AKTA purifier (Thermo Fisher Scientific DNAPac PA200 semi prep ion exchange column, 8 μ m particles, width 22 mm $\times$ length 250 mm).

Buffer A: 1.50 L H_2O , 2.107 g $NaClO_4$, 438 mg EDTA, 1.818 g TRIS, 540.54 g urea, pH 7.4.

Buffer B: 1.50 L H_2O , 105.34 g $NaClO_4$, 438 mg EDTA, 1.818 g TRIS, 540.54 g urea, pH 7.4.

Isolation of the oligonucleotides was achieved by precipitation, induced by the addition of 4 volumes of ethanol and storing at $-20^\circ C$.

To ensure high fidelity of the data, all single strands were HPLC purified to >85% purity. The purity and identity of the oligonucleotides was confirmed by ion exchange chromatography and LC-MS, respectively.

Duplex Annealing

For the in vitro experiments (100 μ M solutions) and in vivo experiments (10 mg/ml), stock solutions of siRNAs in PBS were prepared by mixing equimolar amounts of

complementary sense and antisense strands in 1×PBS buffer. The solutions were heated to 90° C. for 10 min and allowed to slowly cool to room temperature to complete the annealing process. siRNAs were further characterized by HPLC and were stored frozen until use.

siRNA Sequences

The sequences of each siRNA, and sequences including nucleotide modifications, are shown in Tables 1 and 2, supra.

siRNA Stability in Mouse Serum

Modified siRNAs were tested for nuclease stability in 50% mouse serum. 160 µL of 2.5 µM siRNA in 1×DPBS (Life Technologies, cat. no. 14190-094) and 160 µL mouse serum (Sigma, cat. no. M5905) were incubated at 37° C. for up to 72 h. At each time-point (0 h, 8 h, 24 h, 32 h, 48 h, 56 h, and 72 h), 20 µL of the reaction was taken out and quenched with a stop solution (Tissue & Cell Lysis Solution (Epicentre, cat. no. MTC096H), Proteinase K (Sigma, cat. no. P2308), water) at 65° C. for 30 min. Prior to HPLC analysis on a Waters 2695 Separation Module and a 2487 Dual Absorbance Detector, RNase-free water was added to each sample. The solution was analyzed by HPLC using a DNAPac PA200 analytical column (Thermo Scientific, cat. no. 063000).

Time (min)	Flow (mL/min)	% Buffer A*	% Buffer B**
0	1	75	25
20	1	35	65

Buffer A: 20 mM sodium phosphate (Sigma, Cat. No. 342483), pH 11;

Buffer B: 20 mM sodium phosphate (Sigma, Cat. No. 342483), 1 M sodium bromide (Sigma, Cat. No. 02119), pH 11.

Serum half-lives were estimated for both strands of the siRNA.

Example 2: Identification of siRNAs for Inhibition of Human ANGPTL8 Expression

Methods

Cells and Tissue Culture

Human Hep3B cells were grown at 37° C., 5% CO₂ and 95% relative humidity (RH), and cultivated in EMEM medium (ATCC, cat. no. 30-2003) supplemented with 10% FBS.

siRNA Transfections

For knock-down experiments in Hep3B cells, 20,000 cells/well were used in 96-well plates (Greiner, cat. no. 655180). The cells were transfected with ANGPTL8 siRNAs at 0.1 nM and 1 nM using 0.2 µL/well of Lipofectamine RNAiMAX transfection reagent (Thermo Fisher) according to the manufacturer's protocol in a reverse transfection setup, and incubated for 48 h without medium change. Usually, N=4 technical replicates were carried out per test sample.

mRNA Expression Analysis

48 or 72 hours after siRNA transfection or free siRNA uptake, the cellular RNA was harvested by usage of Promega's SV96 total RNA isolation system (cat. no. Z3500) according to the manufacturer's protocol, including a DNase step during the procedure.

For cDNA synthesis, the Thermo Fisher Reverse Transcriptase kit (cat. no. N8080234) was used. cDNA was synthesized from 30 ng RNA using 1.2 µL 10×RT buffer, 2.64 µL MgCl₂ (25 mM), 2.4 µL dNTPs (10 mM), 0.6 µL

random hexamers (50 µM), 0.6 µL Oligo(dT)16 (SEQ ID NO: 533) (50 µM), 0.24 µL RNase inhibitor (20 U/µL) and 0.3 µL Multiscribe (50 U/µL) in a total volume of 12 µL. Samples were incubated at 25° C. for 10 minutes and 42° C. for 60 minutes. The reaction was stopped by heating to 95° C. for 5 minutes.

Human and cynomolgus ANGPTL8 mRNA levels were quantified by qPCR using the Thermo Fisher TaqMan Universal PCR Master Mix (cat. no. 4305719) and the following TaqMan Gene Expression assays:

Human	Hs00218820_m1
Cynomolgus	Hs00218820_m1* *using customized forward primer 5'-GCCCTGCCTACCA AGAATTG-3' (SEQ ID NO: 534)

PCR was performed in technical duplicates with an ABI Prism 7900 system under the following PCR conditions: 2 minutes at 50° C., 10 minutes at 95° C., 40 cycles with 95° C. for 15 seconds and 1 minute at 60° C. PCR was set up as a simplex PCR detecting the target gene in one reaction and the housekeeping gene (human/cynomolgus RPL37A) for normalization in a parallel reaction. The final volume for the PCR reaction was 12.5 µL in a 1×PCR master mix; RPL37A primers were used at a final concentration of 50 nM and the probe was used at a final concentration of 200 nM. The ΔΔCt method was applied to calculate relative expression levels of the target transcripts. Percentage of target gene expression was calculated by normalization based on the levels of non-targeting siRNA control treated cells.

IC₅₀ Measurements

For IC₅₀ measurements, 20,000 human Hep3B cells in 96-well plates were transfected with Lipofectamine RNAiMAX for 48 hours with the indicated ANGPTL8 siRNAs at 7 concentrations starting from 25 nM using 5-8-fold dilution steps. The half maximal inhibitory concentration (IC₅₀) for each siRNA was calculated by applying a Biostat-Speed statistical calculation tool. Results were obtained using the 4-parameter logistic model according to Ratkovsky and Reedy (*Biometrics* (1986) 42(3):575-82). The adjustment was obtained by non-linear regression using the Levenberg-Marquardt algorithm in SAS v9.1.3 software.

Cytotoxicity

Cytotoxicity was measured 72 hours after 5 nM and 50 nM siRNA transfections of a culture of 10,000 Hep3B cells per well of a 96-well plate by determining the ratio of cellular viability/toxicity in each sample. Cell viability was measured by determination of the intracellular ATP content using the CellTiter-Glo assay (Promega, cat. no. G7570) according to the manufacturer's protocol. Cell toxicity was measured in the supernatant using the ToxiLight assay (Lonza, cat. no. LT07-217) according to the manufacturer's protocol. 10 nM AllStars Hs Cell Death siRNA (Qiagen, cat. no. SI04381048), 25 µM Ketoconazole (Calbiochem, cat. no. 420600) and 1% Triton X-100 (Sigma, cat. no. T9284) were used as toxic positive controls.

Results

In order to identify siRNAs useful in targeting human ANGPTL8, the following criteria were applied for in silico library generation: First, 19 mers from the human ANGPTL8 mRNA sequence as set forth in NM_018687.6 were identified in silico with an overlap of 18 nucleotides. The resulting pool of 862 19 mers was then aligned to the

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ANGPTL8 mRNA sequence of *Macaca fascicularis* (cynomolgus monkey), resulting in the exclusion of all sequences having more than 1 mismatch in *M. fascicularis*. This left 401 siRNA sequences with 0 mismatches, and an additional 294 siRNA sequences with 1 mismatch.

For the remaining sequences with a G/C content of 30-55%, an in silico analysis was then carried out to identify any potential off-target transcripts matching either siRNA strand (sense/antisense) in the human transcriptome (RefSeq RNA version 2015-12-22). Human off-target sequences with RNAseq expression (Illumina Body Atlas) FPKM<0.5 in liver tissue were not considered. All siRNA sequences of interest had either greater than three mismatches to any human transcript expressed in liver other than ANGPTL8, or had two mismatches with four or fewer human genes; sequences that did not meet one of these two criteria were filtered out. After this filtration, 132 potential siRNAs were left (see Table 1).

As described above, the 132 siRNAs were produced with nucleotides having a fixed pattern of 2'-O-methyl and 2'-fluoro groups (see Table 2). To test the ability of these 132 siRNAs to reduce expression of ANGPTL8, human Hep3B cells were transfected with 0.1 nM or 1.0 nM of each siRNA and incubated for 48 hours. After incubation, mRNA expression of ANGPTL8 was measured in each sample and compared to positive and negative controls (FIGS. 1A-1C). 18 siRNAs that showed reduction of mRNA expression by at least 90% at a concentration of 1.0 nM, or by at least 70% at a concentration of 0.1 nM, were selected for further characterization.

IC₅₀ measurements (Table 5) and a cytotoxicity assay (FIG. 2) were carried out for the selected 18 siRNAs in human Hep3B cells. After removal of 3 siRNAs that showed <50% of NT control Viability/Toxicity ratio (at 50 nM), 12 siRNAs were selected based on their IC₅₀ values for conjugation to GalNAc (Table 5):

TABLE 5

Activity and Cytotoxicity of selected siRNAs				
Compound	I _{max}	IC ₅₀ [nM]	Cytotoxicity detected	Selected for GalNAc conjugation
siRNA#006	1.004	1.37E-01		X
siRNA#023	0.977	3.46E-02		X
siRNA#040	0.984	2.62E-02		X
siRNA#041	0.956	3.33E-02		X
siRNA#042	0.976	9.81E-03		X
siRNA#043	0.957	1.68E-02		X
siRNA#044	0.956	4.15E-02		X
siRNA#064	0.969	6.21E-02		
siRNA#067	0.951	1.98E-01		X
siRNA#093	0.968	5.58E-02		
siRNA#094	0.995	5.74E-02	X	
siRNA#097	0.992	3.70E-02	X	
siRNA#100	0.992	2.24E-02		X
siRNA#123	0.982	4.79E-02		X
siRNA#124	0.976	4.79E-02		X
siRNA#126	0.944	1.34E-01		
siRNA#131	0.972	3.94E-02		X
siRNA#132	0.992	1.04E-01	X	

Taken together, these results demonstrate the identification of siRNAs capable of potent inhibition of human ANGPTL8 expression without significant cytotoxicity in human cells.

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Example 3: Identification of Active GalNAc-Conjugated siRNAs for Inhibition of Human and Cynomolgus ANGPTL8 Expression

Methods

GalNAc-siRNAs, including non-targeting control siRNAs (NT control), were generated based on the sequences as indicated (see sequence listings above).

Cell Culture and Assays

Human (BioreclamationIVT, cat. no. M00995-P) and cynomolgus (Primacyt, cat. no. CHCP-I-T) primary hepatocytes were cultured as follows: cryopreserved cells were thawed and plated using a plating and thawing kit (Primacyt, cat. no. PTK-1), and were incubated at 37° C., 5% CO₂ and 95% RH. 6 hours after plating, the medium was changed to maintenance medium (KaLy-Cell, cat. no. KLC-MM) supplemented with 1% FBS.

mRNA expression analysis was performed as described above in Example 1.

IC₅₀ Measurements

For demonstration of dose-activity relationships and IC₅₀ measurements in human and cynomolgus primary hepatocytes under free uptake conditions, 50,000-70,000 cells in 96-well plates were incubated for 72 hours without medium change with the siRNAs at concentrations ranging from 10 μM-0.01 nM using 10-fold dilution steps. The half maximal inhibitory concentration (IC₅₀) for each siRNA was calculated by applying a Biostat-Speed statistical calculation tool. Results were obtained using the 4-parameter logistic model according to Ratkovsky and Reedy (*Biometrics* (1986) 42(3):575-582). The adjustment was obtained by non-linear regression using the Levenberg-Marquardt algorithm in SAS v9.1.3 software.

Results

Following selection of potent siRNAs, we went on to demonstrate whether the selected molecules retain their activity in the context of a GalNAc-conjugate suitable for liver specific siRNA delivery in vivo. We also assessed whether this activity holds up in cells from *M. fascicularis* (cynomolgus monkey), a pre-clinical species.

The results of the IC₅₀ measurements show that all tested siRNA conjugates retain activity when delivered by free uptake to human primary hepatocytes (Table 6), with IC₅₀ values ranging from 8.12-475 nM. Surprisingly, however, the performance ranking following free uptake of GalNAc-siRNA differs significantly from that obtained after transfection assisted uptake of unconjugated siRNA (Table 5). This indicates that siRNAs seem to have inherent properties based on their sequence that makes them differentially suited for application in the context of GalNAc conjugates with regard to resulting knock-down potency.

TABLE 6

I _{max} and IC ₅₀ of selected GalNAc-conjugated siRNAs		
Compound	I _{max}	IC ₅₀ [nM]
siRNA#006-c	0.4301279	4.75E+02
siRNA#023-c	0.8143869	3.03E+01
siRNA#040-c	0.8949594	2.70E+01
siRNA#041-c	0.5851301	8.12E+00
siRNA#042-c	0.946814	1.27E+01
siRNA#043-c	0.9174046	1.64E+01

TABLE 6-continued

I_{max} and IC_{50} of selected GalNAc-conjugated siRNAs		
Compound	I_{max}	IC_{50} [nM]
siRNA#044-c	0.958233	1.29E+01
siRNA#067-c	0.3105127	3.97E+01
siRNA#100-c	0.6889424	1.28E+01
siRNA#123-c	0.51566	7.04E+01
siRNA#124-c	0.5422029	6.48E+01
siRNA#131-c	0.7886907	1.63E+01

Even more surprisingly, the siRNAs show differential dose-activity relationships in cynomolgus hepatocytes that do not correlate with their mismatches to the *M. fascicularis* sequence (FIGS. 5A-D). Contrary to what is expected, the tested siRNAs without mismatch (siRNA#006-c, siRNA#023-c, siRNA#100-c and siRNA#131) overall show an inferior dose-activity relationship compared to the tested siRNAs with one mismatch (particularly siRNA#041-c, siRNA#042-c, siRNA#043-c and siRNA#044-c). This surprising discovery allows the identification of siRNAs that show highest activity in the human target and still allow use in this pre-clinical species.

Example 4: In Vitro and In Vivo Characterization of GalNAc-Conjugated siRNAs for Inhibition of Human ANGPTL8 Expression

Methods

Cells and Tissue Culture

Human Hep3B cells were grown at 37° C., 5% CO₂ and 95% RH, and cultivated in EMEM medium (ATCC, cat. no. 30-2003) supplemented with 10% FBS.

Human (BioreclamationIVT, cat. no. M00995-P) and cynomolgus (Primacyt, cat. no. CHCP-I-T) primary hepatocytes were cultured as follows: cryopreserved cells were thawed and plated using a plating and thawing kit (Primacyt, cat. no. PTK-1), and were incubated at 37° C., 5% CO₂ and 95% RH. 6 hours after plating, the medium was changed to maintenance medium (KaLy-Cell, cat. no. KLC-MM) supplemented with 1% FBS.

Human peripheral blood mononuclear cells (PBMCs) were isolated from approximately 16 mL of blood from three healthy donors that were collected in Vacutainer tubes coated with sodium heparin (BD, Heidelberg Germany) according to the manufacturer's instructions.

For transfection of human PBMCs, 100 nM of the siRNAs were reverse transfected into 1×10^5 PBMCs with 0.3 μ L Lipofectamine 2000 per well of a 96-well plate (N=2) in a total volume of 150 μ L serum-free RPMI medium (Thermo Fisher, cat. no. 11875) for 24 hours. Single-stranded RNA ("R-0006") and DNA ("CpG ODN") oligonucleotides, as well as double-stranded unmodified and 2'-O-methyl modified siRNA ("132/161"), were applied as controls.

IFN α Determination

IFN α protein concentration was quantified in the supernatant of human PBMCs as follows: 25 μ L of the cell culture supernatant was used for measurement of IFN α concentration applying a self-established electrochemiluminescence assay based on MesoScale Discovery's technology, and using a pan IFN α monoclonal capture antibody (MT1/3/5, Mabtech). Alternatively, a human IFN α 2a isoform-specific assay (cat. no. K151VHK) was applied based on MesoScale's U-PLEX platform and according to the supplier's protocol.

Cytotoxicity

siRNA cytotoxicity in human primary hepatocytes was measured 72 hours after incubation of 45,000-50,000 cells per well of a 96-well plate with 1 μ M, 5 μ M and 25 μ M siRNA under free uptake conditions by determining the ratio of cellular viability/toxicity in each sample. Cell viability was measured by determination of the intracellular ATP content using the CellTiter-Glo assay (Promega, cat. no. G7570), and cell toxicity was measured in the supernatant using the LDH assay (Sigma, cat. no. 11644793001) according to the manufacturer's protocols. 25 μ M Ketoconazole and 1% Triton X-100 were used as positive controls.

Nuclease Stability

The GalNAc-conjugated siRNAs were tested for nuclease stability using the method described in Example 1.

In Vivo Assay

To assess the effect of GalNAc-siRNAs targeting human ANGPTL8 in vivo, a transgene expression system based on adeno-associated viral vectors was applied in mice. To this end, an AAV8 vector with liver specific expression of a bicistronic mRNA, simultaneously encoding human ANGPTL3 and ANGPTL8 from an ApoA2 promoter (Vectalys, Toulouse, France), was administered intravenously to female C57BL/6 mice (Charles River, Germany) before siRNA dosing. GalNAc-conjugated siRNAs (including non-targeting control) were administered subcutaneously at 15 mg/kg (n=8) after serum levels of human ANGPTL3 expressed from the AAV vector reached sufficiently high serum levels. Activity of siRNAs targeting human ANGPTL8 was quantified by measuring human ANGPTL3 as a surrogate.

ANGPTL3 ELISA Assay

Serum ANGPTL3 protein levels in mice treated with siRNAs were quantified by applying R&D Systems' human ANGPTL3 Quantikine ELISA kit (cat. no. DANL30). ANGPTL3 serum levels were calculated relative to the group treated with non-targeting control siRNA.

Results

The immune response to 12 GalNAc-siRNAs targeting ANGPTL8 was measured in vitro in human primary cells by examining the production of interferon α secreted from human primary PBMCs isolated from three different healthy donors (FIG. 3) in response to transfection of the siRNAs. No signs of immune stimulation in human PBMCs were observed for any of the tested siRNAs.

The ANGPTL8 GalNAc-siRNAs were also tested for their in vitro nuclease stability in 50% murine serum by determining their relative stability and half-lives (Table 7). Half-lives ranged between 24 h and 72 h.

TABLE 7

In vitro Serum Stability	
Compound	$t_{1/2}$
siRNA#006-c	72 h
siRNA#023-c	72 h
siRNA#040-c	24 h
siRNA#041-c	32 h
siRNA#042-c	32 h
siRNA#043-c	32 h
siRNA#044-c	56 h
siRNA#067-c	24 h
siRNA#100-c	32 h
siRNA#123-c	32 h
siRNA#124-c	32 h
siRNA#131-c	32 h

A cytotoxicity assay was carried out in human primary hepatocytes to exclude GalNAc-siRNAs with any toxic

potential from further selection (FIG. 4). No obvious toxic effects were observed for most molecules. However, mild, dose-dependent assay effects were observed for siRNA#006-c and siRNA#041-c. These results demonstrate that application of our selected siRNAs in the context of GalNAc-conjugates generally does not confer cytotoxicity.

Finally, four selected GalNAc-siRNAs molecules were tested in vivo using the above-described humanized mouse model expressing human ANGPTL8 mRNA (together with ANGPTL3 from a bicistronic construct) (FIG. 6). After subcutaneous administration of the selected compounds, target protein levels were reduced between 80% and 95% (KD_{max}) compared to animals treated with a non-targeting control. Depending on the compound, the levels returned to 50% of the maximum knock-down (KD_{50}) between ~d15 and ~d30 post treatment. All groups had returned to baseline by day 45.

In another study, Hep3B cell line overexpressing human ANGPTL8 were transfected with Lipofectamine RNAiMAX with siRNA#042-c for 72 h at 10 concentrations starting from 25 nM, using 2-10-fold dilution steps. Human ANGPTL8 expression was assayed by Western blot in whole cell protein extracts by using a mouse anti-human ANGPTL8 IgG (R&D System, MAB8548) as a primary antibody, a peroxidase-conjugated AffiniPure Goat anti-mouse IgG (JIR Lab. Inc, 11-035-062) as a secondary antibody, and the ECL detection system (GE Healthcare). The data show that siRNA#042-c inhibited human ANGPTL8 expression in Hep3B in a dose-dependent manner, with a half inhibitory concentration (IC_{50}) at 0.01 nM and a complete inhibition concentration at 0.1 nM. These results further demonstrate the superior ANGPTL8-inhibitory activity of the tested construct.

Example 5: Optimization of GalNAc-Conjugated ANGPTL8 siRNA Sequences

Methods

Production of Modified GalNAc siRNA Sequences

GalNAc siRNA sequences further optimized with modified nucleotides of formula (I) were synthesized as described in PCT Patent Publication WO 2019/170731. All oligonucleotides were synthesized on an ABI 394 synthesizer. Commercially available (Sigma Aldrich) DNA-, RNA-, 2'-OMe-RNA, and 2'-deoxy-F-RNA-phosphoramidites with standard protecting groups, e.g., 5'-O-dimethoxytrityl-thymidine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-tert-butyl dimethylsilyl-uracile-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-tert-butyl dimethylsilyl-N4-cytidine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-tert-butyl dimethylsilyl-N6-benzoyl-adenosine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-tert-butyl dimethylsilyl-N2-isobutryl-guanosine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-methyl-uracile-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-methyl-N4-cytidine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-methyl-N6-benzoyl-adenosine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-O-methyl-N2-isobutryl-guanosine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-desoxy-fluoro-uracile-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-deoxy-fluoro-N4-cytidine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, and 5'-O-dimethoxytrityl-2'-deoxy-fluoro-N2-isobutryl-guanosine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, as well as the corresponding solid support materials (CPG-500 Å, loading 40 mmol/g, ChemGenes) were used for automated oligonucleotide synthesis.

propyl-2-cyanoethyl)-phosphoramidite, 5'-O-dimethoxytrityl-2'-deoxy-fluoro-N6-benzoyl-adenosine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite, and 5'-O-dimethoxytrityl-2'-deoxy-fluoro-N2-isobutryl-guanosine-3'-O—(N,N-diisopropyl-2-cyanoethyl)-phosphoramidite as well as the corresponding solid support materials (CPG-500 Å, loading 40 mmol/g, ChemGenes) were used for automated oligonucleotide synthesis.

Phosphoramidite building blocks were used as 0.1 M solutions in acetonitrile and activated with 5-(bis-3,5-trifluoromethylphenyl)-1H-tetrazole (activator 42, 0.25 M in acetonitrile, Sigma Aldrich). Reaction times of 200 s were used for standard phosphoramidite couplings. In case of phosphoramidites described herein, coupling times of 300 s were applied. As capping reagents, acetic anhydride in THF (capA for ABI, Sigma Aldrich) and N-methylimidazole in THF (capB for ABI, Sigma Aldrich) were used. As oxidizing reagent, iodine in THF/pyridine/water (0.02 M; oxidizer for ABI, Sigma Aldrich) was used. Alternatively, PS-oxidation was achieved with a 0.05 M solution of 3-((N,N-dimethylaminomethylidene)amino)-3H-1,2,4-dithiazole-5-thione (DDTT) in pyridine/acetonitrile (1:1). Deprotection of the DMT-protecting group was done using dichloroacetic acid in DCM (DCA deblock, Sigma Aldrich). Final cleavage from solid support and deprotection (acyl- and cyanoethyl-protecting groups) was achieved with NH_3 (32% aqueous solution/ethanol, v/v 3:1). Treatment with NMP/HF (3:1.5:2) was applied for TBDMS-deprotection.

Oligonucleotides with herein described building blocks at the 3'-end were synthesized on solid support materials shown in Table B or on universal linker-solid support (CPG-500 Å, loading 39 μ mol/g, AM Chemicals LLC) and the corresponding phosphoramidites shown in Table A.

Crude products were analyzed by HPLC and single strand purification was performed using ion exchange or preparative HPLC-methods.

Ion exchange: ÄKTA purifier, (Thermo Fisher Scientific DNAPac PA200 semi prep ion exchange column, 8 μ m particles, width 22 mm×length 250 mm).

Buffer A: 1.5 L H_2O , 2.107 g $NaClO_4$, 438 mg EDTA, 1.818 g TRIS, 540.54 g urea, pH 7.4.

Buffer B: 1.5 L H_2O , 105.34 g $NaClO_4$, 438 mg EDTA, 1.818 g TRIS, 40.54 g urea, pH 7.4.

Isolation of the oligonucleotides was achieved by precipitation induced by the addition of 4 volumes of ethanol and storing at $-20^\circ C$.

Preparative HPLC: Agilent 1100 series prep HPLC (Waters XBridge®BEH C18 OBD™ Prep Column 130 Å, 5 μ m, 10 mm×100 mm); Eluent: Triethylammonium acetate (0.1 M in acetonitrile/water). After lyophilization, the products were dissolved in 1.0 mL 2.5 M NaCl solution and 4.0 mL H_2O . The corresponding Na^+ -salts were isolated after precipitation by adding 20 mL ethanol and storing at $-20^\circ C$ for 18 h.

Final analysis of the single strands was done by LC/MS-TOF methods. For double strand formation, equimolar amounts of sense strands and antisense strands were mixed in 1×PBS buffer and heated to $85^\circ C$ for 10 min. Then it was slowly cooled down to room temperature. Final analysis of the siRNA-double strands was done by LC/MS-TOF methods.

Annealing of siRNA duplexes was performed as described in Example 1. The sequences of each siRNA, including nucleotide modifications, are shown in Tables 3 and 4.

siRNA Stability in Mouse Serum

Stability of optimized ANGPTL8 siRNAs was determined as described in Example 1 with the following exceptions: siRNAs were incubated at 37° C. for 0 h, 24 h, 48 h, 72 h, 96 h, and 168 h. Proteinase K was purchased from Qiagen (cat. no. 19133) and HPLC analysis was performed on an Agilent Technologies 1260 Infinity II instrument using a 1260 DAD detector.

Cell Culture and Cell-Based Assays

Human Hep3B cells, primary human hepatocytes, and primary human PBMCs were isolated and cultivated as described in Examples 2-4. Analysis of mRNA was performed as described in Example 2. Cytotoxicity was measured 72 hours after 5 nM and 50 nM siRNA transfections of human Hep3B cells as described in Example 2. IFN α protein concentration was quantified in the supernatant of human PBMCs as described in Example 4.

In Vivo Assay

In vivo activity of modified GalNAc-ANGPTL8 siRNAs was measured in mice transduced with a bicistronic AAV8 vector encoding for human ANGPTL3 and ANGPTL8 mRNA as described in Example 4. In contrast with Example 4, a single siRNA dose of 6 mg/kg was injected subcutaneously into 5 male C57BL/6 mice per treatment group. Serum ANGPTL3 levels were measured by ELISA.

ANGPTL3 ELISA Assay

Serum ANGPTL3 protein levels in mice treated with modified GalNAc-ANGPTL8 siRNAs were used as surrogate for the determination of hepatic in vivo siRNA activities and quantified as described in Example 4.

Results

Fifty-four different siRNA modification patterns were designed and applied to two GalNAc-conjugated siRNA constructs (siRNA#023-c and siRNA#042-c). The library of 2 $\times$ 54 siRNA molecules (siRNA#023-c-01 to siRNA#023-c-54 and siRNA#042-c-01 to siRNA#042-c-54, Tables 3 and 4) was synthesized using three consecutive modified GalNAc-conjugated nucleotides at the 5'-end of the respective siRNA sense strands.

All of the 108 modified ANGPTL8 siRNAs were tested for their nuclease stability in 50% mouse serum (half-life t_{1/2}). As depicted in Table 8, numerous modified constructs displayed significantly improved stability as compared to their parent constructs having a fixed pattern of 2'-O-methyl and 2'-fluoro modified nucleotides. For the constructs derived from siRNA#023-c, the serum half-lives improved from approximately 72 h for the parental construct to 168 h or more for the modified constructs. For the constructs derived from siRNA#042-c, serum half-lives improved from approximately 32 h to 96 h or more.

TABLE 8

In vitro Serum Stability			
siRNA Construct	t _{1/2}	siRNA Construct	t _{1/2}
siRNA#023-c	=72 h	siRNA#042-c	=32 h
siRNA#023-c-01	=72 h	siRNA#042-c-01	>96 h
siRNA#023-c-02	=72 h	siRNA#042-c-02	>96 h
siRNA#023-c-03	>48 h	siRNA#042-c-03	>72 h
siRNA#023-c-04	>48 h	siRNA#042-c-04	=48 h
siRNA#023-c-05	>48 h	siRNA#042-c-05	>72 h
siRNA#023-c-06	>48 h	siRNA#042-c-06	>72 h
siRNA#023-c-07	>96 h	siRNA#042-c-07	>72 h
siRNA#023-c-08	>48 h	siRNA#042-c-08	>72 h
siRNA#023-c-09	=96 h	siRNA#042-c-09	=48 h
siRNA#023-c-10	=96 h	siRNA#042-c-10	>48 h
siRNA#023-c-11	=96 h	siRNA#042-c-11	>48 h

TABLE 8-continued

In vitro Serum Stability			
siRNA Construct	t _{1/2}	siRNA Construct	t _{1/2}
siRNA#023-c-12	>96 h	siRNA#042-c-12	>48 h
siRNA#023-c-13	>72 h	siRNA#042-c-13	=48 h
siRNA#023-c-14	=168 h	siRNA#042-c-14	=72 h
siRNA#023-c-15	>72 h	siRNA#042-c-15	>48 h
siRNA#023-c-16	=168 h	siRNA#042-c-16	>72 h
siRNA#023-c-17	=96 h	siRNA#042-c-17	>72 h
siRNA#023-c-18	=72 h	siRNA#042-c-18	>48 h
siRNA#023-c-19	=72 h	siRNA#042-c-19	>24 h
siRNA#023-c-20	>48 h	siRNA#042-c-20	>72 h
siRNA#023-c-21	>48 h	siRNA#042-c-21	>96 h
siRNA#023-c-22	=72 h	siRNA#042-c-22	>24 h
siRNA#023-c-23	=72 h	siRNA#042-c-23	>72 h
siRNA#023-c-24	=96 h	siRNA#042-c-24	>72 h
siRNA#023-c-25	>24 h	siRNA#042-c-25	>48 h
siRNA#023-c-26	>48 h	siRNA#042-c-26	>48 h
siRNA#023-c-27	>72 h	siRNA#042-c-27	>48 h
siRNA#023-c-28	=72 h	siRNA#042-c-28	>48 h
siRNA#023-c-29	=72 h	siRNA#042-c-29	>24 h
siRNA#023-c-30	>48 h	siRNA#042-c-30	>48 h
siRNA#023-c-31	>72 h	siRNA#042-c-31	>72 h
siRNA#023-c-32	>72 h	siRNA#042-c-32	>24 h
siRNA#023-c-33	=72 h	siRNA#042-c-33	>48 h
siRNA#023-c-34	>72 h	siRNA#042-c-34	>24 h
siRNA#023-c-35	>48 h	siRNA#042-c-35	>48 h
siRNA#023-c-36	=72 h	siRNA#042-c-36	>24 h
siRNA#023-c-37	>72 h	siRNA#042-c-37	>72 h
siRNA#023-c-38	=72 h	siRNA#042-c-38	>24 h
siRNA#023-c-39	=72 h	siRNA#042-c-39	=48 h
siRNA#023-c-40	>48 h	siRNA#042-c-40	>24 h
siRNA#023-c-41	>96 h	siRNA#042-c-41	>72 h
siRNA#023-c-42	>96 h	siRNA#042-c-42	>48 h
siRNA#023-c-43	>48 h	siRNA#042-c-43	=96 h
siRNA#023-c-44	=72 h	siRNA#042-c-44	=48 h
siRNA#023-c-45	=96 h	siRNA#042-c-45	>48 h
siRNA#023-c-46	>96 h	siRNA#042-c-46	>48 h
siRNA#023-c-47	=72 h	siRNA#042-c-47	>48 h
siRNA#023-c-48	>72 h	siRNA#042-c-48	>48 h
siRNA#023-c-49	=72 h	siRNA#042-c-49	=48 h
siRNA#023-c-50	=96 h	siRNA#042-c-50	>24 h
siRNA#023-c-51	>168 h	siRNA#042-c-51	>72 h
siRNA#023-c-52	>72 h	siRNA#042-c-52	>48 h
siRNA#023-c-53	=72 h	siRNA#042-c-53	>72 h
siRNA#023-c-54	>72 h	siRNA#042-c-54	=48 h

Next, all of the 108 modified GalNAc-siRNAs were evaluated for their knock-down potency in primary human hepatocytes under free uptake conditions using 1 nM, 10 nM, and 100 nM concentrations of the modified siRNAs. The parent constructs siRNA#023-c and siRNA#042-c were used as positive controls. Data are shown in FIGS. 7A and 7B. Surprisingly, most modified variants of siRNA#042-c exhibited strongly improved in vitro knock-down activities as compared to the parent construct (FIG. 7B), while this was not the case for the siRNA#023-c (FIG. 7A).

Based on the in vitro knock-down activity and nuclease stability data, twelve modified variants were selected for each of the two parent constructs. Prior to in vivo activity testing, the 2 $\times$ 12 modified constructs were investigated for their ability to stimulate innate immunity in human PBMCs (FIG. 8) and for their general cytotoxicity in human Hep3B cells (FIG. 9). In both assays, no apparent adverse effects were observed.

Finally, the 2 $\times$ 12 selected modified constructs were tested in vivo as described in Example 4. For this experiment, mice were injected intravenously with a bicistronic AAV8 construct encoding ANGPTL3 and ANGPTL8. Treated mice expressed human ANGPTLs 3 and 8, where knockdown of mRNA led to reduced expression of both proteins. The data show that following the siRNA injection, the protein levels of surrogate biomarker ANGPTL3 were reduced up to 95%

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(KD_{max}) compared to animals treated with PBS (FIGS. 10A and 10B). In fact, the serum level of ANGPTL3 did not return to more than 50% (KD_{50}) of the pre-treatment level even on day 70 in animals treated with siRNA#023-c-02, -06, -07, -10, -15, -16, -38, -42, -46, or 51, or with siRNA#042-c-43, or -53. In contrast, the ANGPTL3 levels in animals treated with the parental constructs were indistinguishable from the control animals at that point in time.

In summary, we have demonstrated successful identification of siRNAs that strongly reduce expression of human ANGPTL8 mRNA and protein translated from it in the context of GalNAc-conjugates in vivo and in vitro. Surprisingly, some identified siRNAs further successfully target cynomolgus ANGPTL8 mRNA despite single base sequence mismatch. We have also demonstrated unexpectedly strong improvement of in vivo efficacy of siRNAs by introduction of optimized modification patterns using modified nucleotides.

ANGPTL8 Sequences

human ANGPTL8 mRNA sequence

(SEQ ID NO: 529)

```

1 atacccttaga cccctcagta tgccagtgcc
   tgcctctgtgc ctgctctggg cccctggcaat
61 ggtgaccccg cctgcctcag cggcccccat
   gggcggccca gaactggcac agcatgagga
121 gctgaccctg ctcttccatg ggaccctgca
   gctgggccag gccctcaacg gtgtgtacag
181 gaccacggag ggacggctga caaaggccag
   gaacagcctg ggtctctatg gccgcacaat
241 agaactcctg gggcaggagg tcagccgggg
   ccgggatgca gccaggaac ttcgggcaag
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   tctgcagctg caggcagagg ccacagctga
361 ggtgctgggg gaggtggccc aggcacagaa
   ggtgctacgg gacagcgtgc agcggctaga
421 agtccagctg aggagcgccg ggctggggcc
   tgccaccga gaatttgagg tcttaaaggc
481 tcacgctgac aagcagagcc acatcctatg
   ggcctcaca ggccacgtgc agcggcagag
541 gcgggagatg gtggcacagc agcatcggtc
   gcgacagatc caggagagac tccacacagc
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   aactgaggac caatcatgct gcaaggaaca
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130

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781 gaggatgtag ccccatggg gaggggtgga
   ggaaggacat gtacccttcc atgcctacac
841 acccctcatt aaagcagagt cgtggcatct
   caaaaaaaaa aaaaaaaaa
human ANGPTL8 polypeptide sequence
                               (SEQ ID NO: 530)
MPVPALCLLW ALAMVTRPAS AAPMGGPELA QHEELTLFLH
GTLQLGQALN GVRTTEGRL TKARNSLGLY GRTIELLGQE
VSRGRDAAQE LRASLLETQM EEDILQLQAE ATAELVGEVA
QAQKVLKRSV QRLEVQLRSA WLGPAYREFE VLKAHADKQS
HILWALTGHV QRQRREMAQ QHRLRQIER LHTAALPA
cynomolgus ANGPTL8 mRNA sequence
                               (SEQ ID NO: 531)
1 atgggggatg ccctccctc ctccctggg
   gtgcagtcac tgaccggcag ggccctggctg
61 gggctcctgc ctgtcatgta ctgcactcgc
   acggcaaggc tgcgcacgga gcctggcgg
121 ctgctgaagt tgaggctgtg cgggtacacg
   tacagcagac cctcagtcac gctagtgcct
181 gctctgtgcc tgctgtgggc cctggcaatg
   gtgatccagc ctgcctcagc ggcctccctg
241 ggcagcccag aactggcaga gcacagaggag
   ctgaccctgc tcttccatgg gaccctgcag
301 ctggggccag ccctcaatgg tgtgtacaag
   accacggagg gacggctgac aaaggccagg
361 aacagcctgg gtctctatgg ccgcacagtg
   gaactcctgg ggcaggaggt cagccggggc
421 cgggatgcag cccaggaact tcgggcaagc
   ctgttgagga ctcatagga ggaggatatt
481 ctgcagctga aggcagaggc catagccgag
   gtgctggagg aggtggccca ggcacagaag
541 gtgctacagg acagcgtgag gcggctagaa
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   cacgctgaca agcagagcca catcctgtgg
661 gccctcacag gccacgtgca gcggcagagg
   cgggagatgg tggcacagca gcacggcgtg
721 cgacagatcc aggagagaat ccacaaagcg
   gcgctcccag cctgaatctg cctggatgga
781 actgaggacc aacctgctg caaggaacac
   tccacgccc catgaggccc ctgaacaggg
841 aggagctgcc tgttcaactg gatcagccag
   ggccggccgc cccacttctg agcacagagc

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-continued

901 agagacagac gcaggcaggg acaaaggcag
 aggacgtagc cccattgggg aggggtggag
 961 gaaggatgtg taccctttca tacctacaca
 ccccccttat taaagcagag tcgtggcatc
 1021 tca

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-continued

cynomolgus ANGPTL8 polypeptide sequence
 (SEQ ID NO: 532)
 MLVPALCLLW ALAMVIQPAS AAPVGSPELA EHEELTLLEPH
 5 GTLQLGQALN GVKTTTEGRL TKARNSLGLY GRTVELLGQE
 VSRGRDAAQE LRASLLETQM EEDILQLKAE AIAEVLEEVA
 QAQKVLQDSV RRLEVQLRSA WLGPAYQEFE VLKAHADKQS
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SEQUENCE LISTING

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<213> ORGANISM: Artificial Sequence

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<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 1

cuuagacccu cagucaugc

19

<210> SEQ ID NO 2

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic oligonucleotide"

<400> SEQUENCE: 2

cagaacuggc acagcauga

19

<210> SEQ ID NO 3

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic oligonucleotide"

<400> SEQUENCE: 3

agaacuggca cagcaugag

19

<210> SEQ ID NO 4

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 4

cugacaaagg ccaggaaca

19

<210> SEQ ID NO 5

<211> LENGTH: 19

<212> TYPE: RNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 5

ugacaaaggc caggaacag                                19

<210> SEQ ID NO 6
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 6

gaacagccug ggucucuaa                                19

<210> SEQ ID NO 7
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 7

aacagccugg gucucuaug                                19

<210> SEQ ID NO 8
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 8

gucucuaugg cgcacaaau                                19

<210> SEQ ID NO 9
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 9

aacuucgggc aagccuguu                                19

<210> SEQ ID NO 10
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 10

caagccuguu ggagacuca                                19

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<210> SEQ ID NO 11
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 11

aagccuguug gagacucag 19

<210> SEQ ID NO 12
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 12

agccuguugg agacucaga 19

<210> SEQ ID NO 13
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 13

gccuguugga gacucagau 19

<210> SEQ ID NO 14
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 14

ccuguuggag acucagaug 19

<210> SEQ ID NO 15
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 15

cuguuggaga cucagaugg 19

<210> SEQ ID NO 16
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:

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Synthetic oligonucleotide"

<400> SEQUENCE: 16

uguuggagac ucagaugga 19

<210> SEQ ID NO 17
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 17

guuggagacu cagauggag 19

<210> SEQ ID NO 18
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 18

uuggagacuc agauggagg 19

<210> SEQ ID NO 19
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 19

gacucagaug gaggaggau 19

<210> SEQ ID NO 20
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 20

cucagaugga ggaggauau 19

<210> SEQ ID NO 21
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 21

ucagauggag gaggauau 19

<210> SEQ ID NO 22
<211> LENGTH: 19

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 22

cagauggagg aggauauuc                                     19

<210> SEQ ID NO 23
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 23

agauggagga ggauauucu                                     19

<210> SEQ ID NO 24
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 24

gauggaggag gauauucug                                     19

<210> SEQ ID NO 25
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 25

auggaggagg auauucugc                                     19

<210> SEQ ID NO 26
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 26

uggaggagga uauucugca                                     19

<210> SEQ ID NO 27
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 27

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ggaggaggau auucugcag 19

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 28

gaggaggaua uucugcagc 19

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<210> SEQ ID NO 29
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 29

aggaggauau ucugcagcu 19

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<210> SEQ ID NO 30
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 30

uaucugcag cugcaggca 19

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<210> SEQ ID NO 31
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 31

cuagaagucc agcugagga 19

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<210> SEQ ID NO 32
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 32

uagaagucca gcugaggag 19

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<210> SEQ ID NO 33
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 33

cccugccuac cgagaauuu 19

<210> SEQ ID NO 34
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 34

ccugccuacc gagaauuug 19

<210> SEQ ID NO 35
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 35

cugccuaccg agaauuuga 19

<210> SEQ ID NO 36
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 36

ugccuaccga gaauuugag 19

<210> SEQ ID NO 37
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 37

gccuaccgag aaauugagg 19

<210> SEQ ID NO 38
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 38

ccuaccgaga auuugaggu 19

<210> SEQ ID NO 39

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<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 39

cuaccgagaa uuugagguc 19

<210> SEQ ID NO 40
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 40

uaccgagaau uugaggucu 19

<210> SEQ ID NO 41
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 41

accgagaaau ugaggucuu 19

<210> SEQ ID NO 42
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 42

ccgagaaauu gaggucuua 19

<210> SEQ ID NO 43
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 43

cgagaaauug aggucuuaa 19

<210> SEQ ID NO 44
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 44

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gagaauuuga ggucuuaaa 19

<210> SEQ ID NO 45
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 45

agaauuugag gucuuaaag 19

<210> SEQ ID NO 46
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 46

gaauuugagg ucuuaaagg 19

<210> SEQ ID NO 47
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 47

aaauugaggu cuuaaaggc 19

<210> SEQ ID NO 48
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 48

auuugagguc uuaaaggcu 19

<210> SEQ ID NO 49
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 49

uuugaggucu uaaaggcuc 19

<210> SEQ ID NO 50
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 50

uugaggucuu aaaggcuca 19

<210> SEQ ID NO 51
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 51

ugaggucuua aaggcucac 19

<210> SEQ ID NO 52
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 52

gaggucuuaa aggcucacg 19

<210> SEQ ID NO 53
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 53

aggucuuaaa ggcucacgc 19

<210> SEQ ID NO 54
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 54

ggucuuaaag gcucacgcu 19

<210> SEQ ID NO 55
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 55

gucuuaaagg cucacgcug 19

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<210> SEQ ID NO 56
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 56

ucuaaaaggc ucacgcuga 19

<210> SEQ ID NO 57
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 57

cuuaaaggcu cacgcugac 19

<210> SEQ ID NO 58
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 58

uuaaaggcuc acgcugaca 19

<210> SEQ ID NO 59
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 59

uaaaggcuca cgcugacaa 19

<210> SEQ ID NO 60
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 60

aaaggcucac gcugacaag 19

<210> SEQ ID NO 61
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

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<400> SEQUENCE: 61

cugacaagca gagccacau

19

<210> SEQ ID NO 62

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 62

ugacaagcag agccacau

19

<210> SEQ ID NO 63

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 63

acaagcagag ccacauccu

19

<210> SEQ ID NO 64

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 64

caagcagagc cacauccua

19

<210> SEQ ID NO 65

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 65

aagcagagcc acauccuau

19

<210> SEQ ID NO 66

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 66

agcagagcca cauccuaug

19

<210> SEQ ID NO 67

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 67

gacagaucca ggagagacu 19

<210> SEQ ID NO 68
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 68

acagauccag gagagacuc 19

<210> SEQ ID NO 69
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 69

agauccagga gagacucca 19

<210> SEQ ID NO 70
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 70

agccugaauc ugccuggau 19

<210> SEQ ID NO 71
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 71

cugaaucugc cuggaugga 19

<210> SEQ ID NO 72
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 72

ugaaucugcc uggauggaa 19

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<210> SEQ ID NO 73
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 73

gaaucugccu ggauggaac 19

<210> SEQ ID NO 74
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 74

aaucugccug gauggaacu 19

<210> SEQ ID NO 75
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 75

aucugccugg auggaacug 19

<210> SEQ ID NO 76
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 76

ucugccugga uggaacuga 19

<210> SEQ ID NO 77
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 77

uggauggaac ugaggacca 19

<210> SEQ ID NO 78
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

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<400> SEQUENCE: 78

gauggaacug aggaccaau

19

<210> SEQ ID NO 79

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 79

auggaacuga ggaccaauc

19

<210> SEQ ID NO 80

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 80

uggaacugag gaccaauca

19

<210> SEQ ID NO 81

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 81

ggaacugagg accaaucan

19

<210> SEQ ID NO 82

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 82

gaacugagga ccaaucaug

19

<210> SEQ ID NO 83

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 83

aacugaggac caaucaugc

19

<210> SEQ ID NO 84

<211> LENGTH: 19

<212> TYPE: RNA

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 84
acugaggacc aaucaugcu                               19

<210> SEQ ID NO 85
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 85
cugaggacca aucaugcug                               19

<210> SEQ ID NO 86
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 86
ugaggaccaa ucaugcugc                               19

<210> SEQ ID NO 87
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 87
gaggaccaau caugcugca                               19

<210> SEQ ID NO 88
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 88
aggaccaauc augcugcaa                               19

<210> SEQ ID NO 89
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 89
ggaccaauca ugcugcaag                               19

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<210> SEQ ID NO 90
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 90

gaccaaucau gcugcaagg 19

<210> SEQ ID NO 91
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 91

accaaucaug cugcaagga 19

<210> SEQ ID NO 92
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 92

ccaaucaugc ugcaaggaa 19

<210> SEQ ID NO 93
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 93

caaucaugcu gcaaggaac 19

<210> SEQ ID NO 94
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 94

aaucaugcug caaggaaca 19

<210> SEQ ID NO 95
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:

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Synthetic oligonucleotide"

<400> SEQUENCE: 95

aucaugcugc aaggaacac 19

<210> SEQ ID NO 96
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 96

ucaugcugca aggaacacu 19

<210> SEQ ID NO 97
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 97

caugcugcaa ggaacacuu 19

<210> SEQ ID NO 98
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 98

augcugcaag gaacacuuc 19

<210> SEQ ID NO 99
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 99

ugcugcaagg aacacuucc 19

<210> SEQ ID NO 100
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 100

gcugcaagga acacuucca 19

<210> SEQ ID NO 101
<211> LENGTH: 19

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 101

ugccuguuca cugggauca                                     19

<210> SEQ ID NO 102
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 102

uguucacugg gaucagcca                                     19

<210> SEQ ID NO 103
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 103

cacuucugag cacagagca                                     19

<210> SEQ ID NO 104
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 104

acuucugagc acagagcag                                     19

<210> SEQ ID NO 105
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 105

ugagcacaga gcagagaca                                     19

<210> SEQ ID NO 106
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
      Synthetic oligonucleotide"

<400> SEQUENCE: 106

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agcacagagc agagacaga 19

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<210> SEQ ID NO 107
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 107

ggacaaaggc agaggaugu 19

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<210> SEQ ID NO 108
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 108

gacaaaggca gaggaugua 19

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<210> SEQ ID NO 109
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 109

acaaaggcag aggauguag 19

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<210> SEQ ID NO 110
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 110

caaaggcaga ggauguagc 19

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<210> SEQ ID NO 111
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 111

aaaggcagag gauguagcc 19

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<210> SEQ ID NO 112
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 112

cagaggauu agcccauu 19

<210> SEQ ID NO 113
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 113

agaggauu gcccuuu 19

<210> SEQ ID NO 114
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 114

uguacuuu caugcuu 19

<210> SEQ ID NO 115
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 115

guacuuu augcuu 19

<210> SEQ ID NO 116
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 116

uacuuu ugcuuu 19

<210> SEQ ID NO 117
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 117

acuuu ucuu gcuu 19

<210> SEQ ID NO 118

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<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 118

cccuuucaug ccuacacac 19

<210> SEQ ID NO 119
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 119

ccuuucaugc cuacacacc 19

<210> SEQ ID NO 120
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 120

cuuucaugcc uacacaccc 19

<210> SEQ ID NO 121
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 121

uuucaugccu acacacccc 19

<210> SEQ ID NO 122
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 122

ccccucauuu aagcagagu 19

<210> SEQ ID NO 123
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 123

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cccucuuuaa agcagaguc 19

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<210> SEQ ID NO 124
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

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<400> SEQUENCE: 124

ccucuuuaaa gcagagucg 19

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<210> SEQ ID NO 125
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

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<400> SEQUENCE: 125

cucauuuaag cagagucgu 19

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<210> SEQ ID NO 126
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

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<400> SEQUENCE: 126

ucauuuaagc agagucgug 19

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<210> SEQ ID NO 127
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

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<400> SEQUENCE: 127

cauuuaagca gagucgugg 19

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<210> SEQ ID NO 128
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

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<400> SEQUENCE: 128

auuaaagcag agucguggc 19

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<210> SEQ ID NO 129
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 129

uuaaagcaga gucguggca                                     19

<210> SEQ ID NO 130
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 130

uaaagcagag ucguggcau                                     19

<210> SEQ ID NO 131
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 131

aaagcagagu cguggcauc                                     19

<210> SEQ ID NO 132
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 132

aagcagaguc guggcaucu                                     19

<210> SEQ ID NO 133
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 133

gcaugacuga gggucuaag                                     19

<210> SEQ ID NO 134
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 134

ucaugcugug ccaguucug                                     19

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<210> SEQ ID NO 135
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 135

cucaugcugu gccaguucu                               19

<210> SEQ ID NO 136
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 136

uguuccuggc cuuugucag                               19

<210> SEQ ID NO 137
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 137

cuguuccugg ccuuuguca                               19

<210> SEQ ID NO 138
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 138

auagagacc aggcuuuc                               19

<210> SEQ ID NO 139
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 139

cauagagacc caggcuguu                               19

<210> SEQ ID NO 140
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

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<400> SEQUENCE: 140

auugugcggc cauagagac

19

<210> SEQ ID NO 141

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 141

aacagguug cccgaaguu

19

<210> SEQ ID NO 142

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 142

ugagucucca acagguug

19

<210> SEQ ID NO 143

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 143

cugagucucc aacagguu

19

<210> SEQ ID NO 144

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 144

ucugagucuc caacaggu

19

<210> SEQ ID NO 145

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 145

aucugagucu ccaacaggc

19

<210> SEQ ID NO 146

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 146

caucugaguc uccaacagg                                19

<210> SEQ ID NO 147
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 147

ccaucugagu cuccaacag                                19

<210> SEQ ID NO 148
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 148

uccaucugag ucuccaaca                                19

<210> SEQ ID NO 149
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 149

cuccaucuga gucccaac                                  19

<210> SEQ ID NO 150
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 150

ccuccaucug agucccaa                                  19

<210> SEQ ID NO 151
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
        Synthetic oligonucleotide"

<400> SEQUENCE: 151

auccuccucc aucugaguc                                19

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<210> SEQ ID NO 152
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 152

auauccuccu ccaucugag 19

<210> SEQ ID NO 153
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 153

aaauccucc uccaucuga 19

<210> SEQ ID NO 154
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 154

gaauauccuc cuccaucug 19

<210> SEQ ID NO 155
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
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Synthetic oligonucleotide"

<400> SEQUENCE: 155

agaauauccu ccuccaucu 19

<210> SEQ ID NO 156
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 156

cagaauaucc uccuccauc 19

<210> SEQ ID NO 157
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
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<400> SEQUENCE: 157

gcagaaauauc cuccuccau

19

<210> SEQ ID NO 158

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 158

ugcagaaauau ccuccucca

19

<210> SEQ ID NO 159

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 159

cugcagaaaua uccuccucc

19

<210> SEQ ID NO 160

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 160

gcugcagaaau auccuccuc

19

<210> SEQ ID NO 161

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 161

agcugcagaa uauccuccu

19

<210> SEQ ID NO 162

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 162

ugccugcagc ugcagaaau

19

<210> SEQ ID NO 163

<211> LENGTH: 19

<212> TYPE: RNA

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<220> FEATURE:
<221> NAME/KEY: source
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<400> SEQUENCE: 163

uccucagcug gacuucuaag                                19

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<212> TYPE: RNA
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<221> NAME/KEY: source
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<400> SEQUENCE: 164

cuccucagcu ggacuucua                                19

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 165

aaaauucucgg uaggcaggg                                19

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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 166

caaaauucucg guaggcagg                                19

<210> SEQ ID NO 167
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
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<400> SEQUENCE: 167

ucaaaauucuc gguaggcag                                19

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<212> TYPE: RNA
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 168

cucaaaauucu cgguaggca                                19

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Synthetic oligonucleotide"

<400> SEQUENCE: 169

ccucaaaauuc ucgguaggc 19

<210> SEQ ID NO 170
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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 170

accucaaauu cucgguagg 19

<210> SEQ ID NO 171
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 171

gaccucaaau ucucgguag 19

<210> SEQ ID NO 172
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Synthetic oligonucleotide"

<400> SEQUENCE: 172

agaccucaa uucucgguag 19

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 173

aagaccucaa auucucggu 19

<210> SEQ ID NO 174
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Synthetic oligonucleotide"

<400> SEQUENCE: 174

uaagaccuca aaauucucgg 19

<210> SEQ ID NO 175
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 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic oligonucleotide"

<400> SEQUENCE: 175

uuaagaccuc aaauucucg 19

<210> SEQ ID NO 176
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 <220> FEATURE:
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 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic oligonucleotide"

<400> SEQUENCE: 176

uuuaagaccu caaaauucuc 19

<210> SEQ ID NO 177
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 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic oligonucleotide"

<400> SEQUENCE: 177

cuuuaagacc ucaaaauucu 19

<210> SEQ ID NO 178
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 <212> TYPE: RNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
 Synthetic oligonucleotide"

<400> SEQUENCE: 178

ccuuuaagac cucaaaauuc 19

<210> SEQ ID NO 179
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 <221> NAME/KEY: source
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<400> SEQUENCE: 179

gccuuuaaga ccucaaauu 19

<210> SEQ ID NO 180
 <211> LENGTH: 19

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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 180

agccuuuaag accucaaau 19

<210> SEQ ID NO 181
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 181

gagccuuuaa gaccucaa 19

<210> SEQ ID NO 182
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 182

ugagccuuua agaccucaa 19

<210> SEQ ID NO 183
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 183

gugagccuuu aagaccuca 19

<210> SEQ ID NO 184
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 184

cgugagccuu uaagaccuc 19

<210> SEQ ID NO 185
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 185

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gcgugagccu uuaagaccu 19

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 186

agcgugagcc uuuaagacc 19

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 187

cagcgugagc cuuaagacc 19

<210> SEQ ID NO 188
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<212> TYPE: RNA
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Synthetic oligonucleotide"

<400> SEQUENCE: 188

ucagcgugag ccuuuaaga 19

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Synthetic oligonucleotide"

<400> SEQUENCE: 189

gucagcguga gccuuuaag 19

<210> SEQ ID NO 190
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 190

ugucagcgug agccuuuaa 19

<210> SEQ ID NO 191
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<220> FEATURE:
<221> NAME/KEY: source

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<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
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Synthetic oligonucleotide"

<400> SEQUENCE: 197

auaggaugug gcucugcuu 19

<210> SEQ ID NO 198
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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 198

cauaggaugu ggcucugcu 19

<210> SEQ ID NO 199
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 199

agucucuccu ggauugcu 19

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<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 200

gagucucucc uggaugcu 19

<210> SEQ ID NO 201
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 201

uggagucucu ccuggaucu 19

<210> SEQ ID NO 202
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 202

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auccaggcag auccaggu 19

<210> SEQ ID NO 203
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 203

uccauccagg cagauucag 19

<210> SEQ ID NO 204
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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Synthetic oligonucleotide"

<400> SEQUENCE: 204

uuccauccag gcagauuca 19

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 205

guuccaucca ggcagauuc 19

<210> SEQ ID NO 206
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 206

aguuccaucc aggcagauu 19

<210> SEQ ID NO 207
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 207

caguuccauc caggcagau 19

<210> SEQ ID NO 208
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 208

ucaguuccau ccaggcaga                                     19

<210> SEQ ID NO 209
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 209

ugguccucag uuccaucca                                     19

<210> SEQ ID NO 210
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 210

auugguccuc aguuccauc                                     19

<210> SEQ ID NO 211
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 211

gauugguccu caguuccau                                     19

<210> SEQ ID NO 212
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 212

ugauuggucc ucaguucca                                     19

<210> SEQ ID NO 213
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
    Synthetic oligonucleotide"

<400> SEQUENCE: 213

augauugguc cucaguucc                                     19

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<210> SEQ ID NO 214
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 214

caugauuggu ccucaguuc 19

<210> SEQ ID NO 215
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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 215

gcaugauugg uccucaguu 19

<210> SEQ ID NO 216
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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 216

agcaugauug guccucagu 19

<210> SEQ ID NO 217
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 217

cagcaugauu gguccucag 19

<210> SEQ ID NO 218
<211> LENGTH: 19
<212> TYPE: RNA
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 218

gcagcaugau ugguccuca 19

<210> SEQ ID NO 219
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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

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<400> SEQUENCE: 219

ugcagcauga uugguccuc

19

<210> SEQ ID NO 220

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 220

uugcagcaug auugguccu

19

<210> SEQ ID NO 221

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 221

cuugcagcau gauuggucc

19

<210> SEQ ID NO 222

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 222

ccuugcagca ugauugguc

19

<210> SEQ ID NO 223

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 223

uccuugcagc augauuggu

19

<210> SEQ ID NO 224

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 224

uuccuugcag caugauugg

19

<210> SEQ ID NO 225

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 225

guuccuugca gcaugauug 19

<210> SEQ ID NO 226
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 226

uguuccuugc agcaugauu 19

<210> SEQ ID NO 227
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 227

guguuccuug cagcaugau 19

<210> SEQ ID NO 228
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 228

aguguuccuu gcagcauga 19

<210> SEQ ID NO 229
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 229

aaguguuccu ugcagcaug 19

<210> SEQ ID NO 230
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 230

gaaguguucc ugcagcau 19

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 231

ggaaguguuc cuugcagca 19

<210> SEQ ID NO 232
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 232

uggaaguguu ccuugcagc 19

<210> SEQ ID NO 233
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 233

ugaucacagu gaacaggca 19

<210> SEQ ID NO 234
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 234

uggcugaucc cagugaaca 19

<210> SEQ ID NO 235
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 235

ugcucuguc ucagaagug 19

<210> SEQ ID NO 236
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
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Synthetic oligonucleotide"

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<400> SEQUENCE: 236

cugcucugug cucagaagu

19

<210> SEQ ID NO 237

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 237

ugucucugcu cugugcuca

19

<210> SEQ ID NO 238

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 238

ucugucucug cucugugcu

19

<210> SEQ ID NO 239

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 239

acauccucug ccuuugucc

19

<210> SEQ ID NO 240

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 240

uacauccucu gccuuuguc

19

<210> SEQ ID NO 241

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 241

cuacauccuc ugccuuugu

19

<210> SEQ ID NO 242

<211> LENGTH: 19

<212> TYPE: RNA

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 242

gcuacauccu cugccuuug                                     19

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<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 243

ggcuacaucc ucugccuuu                                     19

<210> SEQ ID NO 244
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 244

aauggggcua cauccucug                                     19

<210> SEQ ID NO 245
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 245

caauggggcu acauccucu                                     19

<210> SEQ ID NO 246
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 246

guaggcauga aagggguaca                                    19

<210> SEQ ID NO 247
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<221> NAME/KEY: source
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<400> SEQUENCE: 247

uguaggcaug aaaggguac                                    19

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<221> NAME/KEY: source
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Synthetic oligonucleotide"

<400> SEQUENCE: 248

guguaggcau gaaagggua 19

<210> SEQ ID NO 249
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<400> SEQUENCE: 249

uguguaggca ugaaagggg 19

<210> SEQ ID NO 250
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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 250

guguguaggc augaaaggg 19

<210> SEQ ID NO 251
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 251

gguguguagg caugaaagg 19

<210> SEQ ID NO 252
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 252

ggguguguag gcaugaaag 19

<210> SEQ ID NO 253
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 Synthetic oligonucleotide"

<400> SEQUENCE: 253

ggggugugua ggcaugaaa

19

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<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 254

acucugcuu aaugagggg

19

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<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 255

gacucugcuu uaaugaggg

19

<210> SEQ ID NO 256

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 256

cgacucugcu uuaaagagg

19

<210> SEQ ID NO 257

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 257

acgacucugc uuuaaugag

19

<210> SEQ ID NO 258

<211> LENGTH: 19

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 258

cacgacucug cuuaaaga

19

<210> SEQ ID NO 259

<211> LENGTH: 19

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 259

ccacgacucu gcuuuuaug 19

<210> SEQ ID NO 260
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 260

gccacgacuc ugcuuuaau 19

<210> SEQ ID NO 261
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 261

ugccacgacu cugcuuuaa 19

<210> SEQ ID NO 262
<211> LENGTH: 19
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 262

augccacgac ucugcuuua 19

<210> SEQ ID NO 263
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 263

gaugccacga cucugcuuu 19

<210> SEQ ID NO 264
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 264

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agaugccacg acucugcuu 19

<210> SEQ ID NO 265
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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 265

ccacuuagac ccucagucan gct 23

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 266

ccacagaacu ggcacagcnu gat 23

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 267

ccaagaacug gcacagcnu agt 23

<210> SEQ ID NO 268
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 268

ccacugacaa aggccaggaa cat 23

<210> SEQ ID NO 269
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<212> TYPE: DNA
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 269

ccaugacaaa ggccaggaac agt 23

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 270

ccagaacagc cugggucucu aut 23

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<212> TYPE: DNA
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 271

ccaaacagcc ugggucucua ugt 23

<210> SEQ ID NO 272
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<212> TYPE: DNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 272

ccagucucua uggcgcaca aut 23

<210> SEQ ID NO 273
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<212> TYPE: DNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 273

ccaaacuucg ggcaagccug uut 23

<210> SEQ ID NO 274
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 274

ccacaagccu guuggagacu cat 23

<210> SEQ ID NO 275
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 275

ccaaagccug uuggagacuc agt 23

<210> SEQ ID NO 276
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<220> FEATURE:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 276

ccaagccugu uggagacuca gat 23

<210> SEQ ID NO 277
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 277

ccagccuguu ggagacucag aut 23

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 278

ccaccuguug gagacucaga ugt 23

<210> SEQ ID NO 279
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 279

ccacuguugg agacucagau ggt 23

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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 280

ccauguugga gacucagaug gat 23

<210> SEQ ID NO 281
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 281

ccaguuggag acucagaugg agt 23

<210> SEQ ID NO 282
<211> LENGTH: 23
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
    Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 282

ccauuggaga cucagaugga ggt                                     23

<210> SEQ ID NO 283
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 283

ccagacucag auggaggagg aut                                     23

<210> SEQ ID NO 284
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<221> NAME/KEY: source
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 284

ccacucagau ggaggaggau aut                                     23

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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 285

ccaucagaug gaggaggaua uut                                     23

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
    Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 286

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ccacagaugg aggaggauau uct 23

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 287

ccaagaugga ggaggauauu cut 23

<210> SEQ ID NO 288
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 288

ccagauggag gaggauauuc ugt 23

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 289

ccaauaggagg aggauauucu gct 23

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<221> NAME/KEY: source
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 290

ccauggagga ggauauucug cat 23

<210> SEQ ID NO 291
<211> LENGTH: 23
<212> TYPE: DNA

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 291

ccaggaggag gauauucugc agt                                     23

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<400> SEQUENCE: 292

ccagaggagg auauucugca gct                                     23

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<400> SEQUENCE: 293

ccaaggagga uauucugcag cut                                     23

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<400> SEQUENCE: 294

ccauauucug cagcugcagg cat                                     23

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<220> FEATURE:
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<220> FEATURE:
<221> NAME/KEY: source

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 295

ccacuagaag uccagcugag gat                                     23

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

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ccauagaagu ccagcugagg agt                                     23

<210> SEQ ID NO 297
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Synthetic oligonucleotide"
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<400> SEQUENCE: 307

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Molecule: Synthetic oligonucleotide"

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<400> SEQUENCE: 312

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<210> SEQ ID NO 313

<211> LENGTH: 23

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<400> SEQUENCE: 313

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<400> SEQUENCE: 318

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<400> SEQUENCE: 351

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ccaaggacca aucaugcugc aat 23

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 353

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<400> SEQUENCE: 354

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<400> SEQUENCE: 356

ccaccaauca ugcugcaagg aat 23

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<400> SEQUENCE: 358

ccaaaucaug cugcaaggaa cat 23

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:

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<400> SEQUENCE: 361

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<400> SEQUENCE: 362

ccaaugcugc aaggaacacu uct                                     23

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ccaugccugu ucacugggau cat 23

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 366

ccauguuac uggaucagc cat 23

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 367

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 368

ccaacuucug agcacagagc agt 23

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Molecule: Synthetic oligonucleotide"

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 370

ccaagcacag agcagagaca gat                                     23

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<400> SEQUENCE: 371

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<400> SEQUENCE: 372

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<400> SEQUENCE: 373

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 375

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<400> SEQUENCE: 377

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<400> SEQUENCE: 378

ccauguaccc uuucaugccu act                                     23

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ccaguacccu uucaugccua cat 23

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<400> SEQUENCE: 380

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<400> SEQUENCE: 381

ccaacccuuu caugccuaca cat 23

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<400> SEQUENCE: 382

ccacccuuuc augccuacac act 23

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<400> SEQUENCE: 383

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<400> SEQUENCE: 387

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA

Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 388

ccaccucauu aaagcagagu cgt 23

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA

Molecule: Synthetic oligonucleotide"

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Molecule: Synthetic oligonucleotide"

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ccaauuaaag cagagucgug gct 23

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Molecule: Synthetic oligonucleotide"

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 395

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cuguuccugg ccuugucat t 21

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 402

auagagacc aggcuuuct t 21

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 510

guaggcauga aaggguaacat t 21

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uguaggcaug aaaggguaact t 21

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 512

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 513

uguguaggca ugaaagggut t 21

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<400> SEQUENCE: 514

guguguaggc augaaaggg t 21

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<400> SEQUENCE: 516

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<400> SEQUENCE: 517

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<400> SEQUENCE: 519

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<400> SEQUENCE: 521

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<400> SEQUENCE: 522

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<400> SEQUENCE: 523

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 524

gccacgacuc ugcuuuaaut t 21

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<220> FEATURE:
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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 525

ugccacgacu cugcuuuaat t 21

<210> SEQ ID NO 526
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 526

augccacgac ucugcuuuaat t 21

<210> SEQ ID NO 527
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 527

gaugccacga cucugcuuut t 21

<210> SEQ ID NO 528
<211> LENGTH: 21
<212> TYPE: DNA

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 528

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<210> SEQ ID NO 529
<211> LENGTH: 888
<212> TYPE: DNA
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 529

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ggtgacccgg cctgectcag cgcccccat gggcggccca gaactggcac agcatgagga 120
gctgaccctg ctcttccatg ggaccctgca gctgggccag gccctcaacg gtgtgtacag 180
gaccacggag ggacggctga caaaggccag gaacagcctg ggtctctatg gccgcacaat 240
agaactcctg gggcaggagg tcagccgggg ccgggatgca gcccaggaac ttcgggcaag 300
cctgttgagg actcagatgg aggaggatat tctgcagctg caggcagagg ccacagctga 360
ggtgctgggg gaggtggccc aggcacagaa ggtgctacgg gacagcgtgc agcggctaga 420
agtccagctg aggagcgctt ggctgggccc tgcctaccga gaatttgagg tcttaaaggc 480
tcacgctgac aagcagagcc acatcctatg ggccctcaca ggccacgtgc agcggcagag 540
gcgggagatg gtggcacagc agcatcggct gcgacagatc caggagagac tccacacagc 600
ggcgctccca gcctgaatct gcctggatgg aactgaggac caatcatgct gcaaggaaca 660
cttccacgcc ccgtgaggcc cctgtgcagg gaggagctgc ctgttcactg ggatcagcca 720
gggcgcgggg ccccatctct gagcacagag cagagacaga cgcaggcggg gacaaaggca 780
gaggatgtag ccccatggg gaggggtgga ggaaggacat gtaccctttc atgcctacac 840
acccctcatt aaagcagagt cgtggcatct caaaaaaaaa aaaaaaaaa 888

<210> SEQ ID NO 530
<211> LENGTH: 198
<212> TYPE: PRT
<213> ORGANISM: Homo sapiens

<400> SEQUENCE: 530

Met Pro Val Pro Ala Leu Cys Leu Leu Trp Ala Leu Ala Met Val Thr
1          5          10         15

Arg Pro Ala Ser Ala Ala Pro Met Gly Gly Pro Glu Leu Ala Gln His
20         25         30

Glu Glu Leu Thr Leu Leu Phe His Gly Thr Leu Gln Leu Gly Gln Ala
35         40         45

Leu Asn Gly Val Tyr Arg Thr Thr Glu Gly Arg Leu Thr Lys Ala Arg
50         55         60

Asn Ser Leu Gly Leu Tyr Gly Arg Thr Ile Glu Leu Leu Gly Gln Glu
65         70         75         80

Val Ser Arg Gly Arg Asp Ala Ala Gln Glu Leu Arg Ala Ser Leu Leu
85         90         95

Glu Thr Gln Met Glu Glu Asp Ile Leu Gln Leu Gln Ala Glu Ala Thr

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100	105	110	
Ala Glu Val Leu Gly Glu Val	Ala Gln Ala Gln Lys Val Leu Arg Asp		
115	120	125	
Ser Val Gln Arg Leu Glu Val	Gln Leu Arg Ser Ala Trp Leu Gly Pro		
130	135	140	
Ala Tyr Arg Glu Phe Glu Val	Leu Lys Ala His Ala Asp Lys Gln Ser		
145	150	155	160
His Ile Leu Trp Ala Leu Thr	Gly His Val Gln Arg Gln Arg Arg Glu		
165	170	175	
Met Val Ala Gln Gln His Arg	Leu Arg Gln Ile Gln Glu Arg Leu His		
180	185	190	
Thr Ala Ala Leu Pro Ala			
195			
<210> SEQ ID NO 531			
<211> LENGTH: 1023			
<212> TYPE: DNA			
<213> ORGANISM: <i>Macaca fascicularis</i>			
<400> SEQUENCE: 531			
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gggtcctcgc ctgtcatgta ctgcactcgc acggcaaggt tgcgcacgga gccctggcgg	120		
ctgctgaagt tgaggctgtg cgggtacacg tacagcagac cctcagtcac gctagtgcct	180		
gctctgtgcc tgtgtggggc cctggcaatg gtgatccagc ctgcctcagc ggccccctg	240		
ggcagcccag aactggcaga gcatgaggag ctgaccctgc tcttccatgg gaccctgcag	300		
ctgggccagg ccctcaatgg tgtgtacaag accacggagg gacggctgac aaaggccagg	360		
aacagcctgg gtctctatgg ccgcacagtg gaactcctgg ggcaggaggt cagccggggc	420		
cgggatgcag cccaggaact tcgggcaagc ctgttgagga ctcagatgga ggaggatatt	480		
ctgcagctga aggagaggc catagccgag gtgctggagg aggtggccca ggcacagaag	540		
gtgtacagc acagcgtgcg gcggctagaa gtccagctga ggagcgcctg gctgggacct	600		
gcctaccaag aatttgaggt cttaaaggct caccgtgaca agcagagcca catcctgtgg	660		
gccctcacag gccacgtgca gcggcagagg cgggagatgg tggcacagca gcatcggtg	720		
cgacagatcc aggagagaat ccacaaagcg gcgctcccag cctgaatctg cctggatgga	780		
actgaggacc aaccatgctg caaggaacac ttccacgccc catgaggccc ctgaacaggg	840		
aggagctgcc tgttccactg gatcagccag ggcgcccggc cccacttctg agcacagagc	900		
agagacagac gcaggcaggg acaaaggcag aggacgtagc cccattgggg aggggtggag	960		
gaaggatgtg taccctttca tacctacaca ccccccttat taaagcagag tcgtggcatc	1020		
tca	1023		

<210> SEQ ID NO 532
 <211> LENGTH: 198
 <212> TYPE: PRT
 <213> ORGANISM: *Macaca fascicularis*
 <400> SEQUENCE: 532

Met Leu Val Pro Ala Leu Cys Leu Leu Trp Ala Leu Ala Met Val Ile	
1 5 10 15	
Gln Pro Ala Ser Ala Ala Pro Val Gly Ser Pro Glu Leu Ala Glu His	
20 25 30	
Glu Glu Leu Thr Leu Leu Phe His Gly Thr Leu Gln Leu Gly Gln Ala	

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35	40	45
Leu Asn Gly Val Tyr Lys Thr Thr Glu Gly Arg	Leu Thr Lys Ala Arg	
50	55	60
Asn Ser Leu Gly Leu Tyr Gly Arg Thr Val Glu	Leu Leu Gly Gln Glu	
65	70	75
Val Ser Arg Gly Arg Asp Ala Ala Gln Glu Leu Arg	Ala Ser Leu Leu	
	85	90
Glu Thr Gln Met Glu Glu Asp Ile Leu Gln Leu Lys	Ala Glu Ala Ile	
	100	105
Ala Glu Val Leu Glu Glu Val Ala Gln Ala Gln Lys	Val Leu Gln Asp	
	115	120
Ser Val Arg Arg Leu Glu Val Gln Leu Arg Ser	Ala Trp Leu Gly Pro	
	130	135
Ala Tyr Gln Glu Phe Glu Val Leu Lys Ala His	Ala Asp Lys Gln Ser	
	145	150
His Ile Leu Trp Ala Leu Thr Gly His Val Gln Arg	Gln Arg Arg Glu	
	165	170
Met Val Ala Gln Gln His Arg Leu Arg Gln Ile	Gln Glu Arg Ile His	
	180	185
Lys Ala Ala Leu Pro Ala		
	195	

<210> SEQ ID NO 533
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<400> SEQUENCE: 533

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16

<210> SEQ ID NO 534
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 <213> ORGANISM: Artificial Sequence
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<400> SEQUENCE: 534

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21

<210> SEQ ID NO 535
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<400> SEQUENCE: 535

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22

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 537

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 538

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<400> SEQUENCE: 539

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:

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Synthetic oligonucleotide"
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<400> SEQUENCE: 540

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<400> SEQUENCE: 541

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<220> FEATURE:
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<400> SEQUENCE: 543

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<400> SEQUENCE: 545

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<220> FEATURE:
<221> NAME/KEY: source

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<211> LENGTH: 24

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<400> SEQUENCE: 575

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Molecule: Synthetic oligonucleotide"

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Molecule: Synthetic oligonucleotide"

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<400> SEQUENCE: 588

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<212> TYPE: RNA

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<220> FEATURE:

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<212> TYPE: RNA

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 590

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<212> TYPE: RNA

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<220> FEATURE:

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<212> TYPE: DNA

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

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<212> TYPE: RNA

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<220> FEATURE:

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 594

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 596

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<400> SEQUENCE: 597

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<212> TYPE: RNA
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 598

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<212> TYPE: RNA
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 599

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 600

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 601

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<210> SEQ ID NO 602
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 602

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<210> SEQ ID NO 603
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 603

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<210> SEQ ID NO 604
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 605

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<212> TYPE: DNA

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<221> NAME/KEY: source

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Molecule: Synthetic oligonucleotide"

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<210> SEQ ID NO 607

<211> LENGTH: 21

<212> TYPE: RNA

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 607

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<210> SEQ ID NO 609

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<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

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<400> SEQUENCE: 615

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 617

agaauauccu ccuccaucua a                                21

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 618

agaauauccu ccuccaucua a                                21

<210> SEQ ID NO 619
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 619

agaauauccu ccuccaucut t                                21

<210> SEQ ID NO 620
<211> LENGTH: 21
<212> TYPE: RNA
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 620

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<210> SEQ ID NO 621
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 622

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<400> SEQUENCE: 623

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 624

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 625

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 agaauauccu ccuccaucua a 21

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 <223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 628

agaauauccu ccuccaucut t 21

<210> SEQ ID NO 629
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<400> SEQUENCE: 629

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<400> SEQUENCE: 630

agaauauccu ccuccaucua a 21

<210> SEQ ID NO 631
 <211> LENGTH: 21

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<212> TYPE: RNA
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<220> FEATURE:
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<400> SEQUENCE: 631

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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 632

agaauauccu ccuccaucut t 21

<210> SEQ ID NO 633
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 633

agaauauccu ccuccaucua a 21

<210> SEQ ID NO 634
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 634

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 635

agaauauccu ccuccaucua a 21

<210> SEQ ID NO 636
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Synthetic oligonucleotide"

<400> SEQUENCE: 636

agaauauccu ccuccaucua a 21

<210> SEQ ID NO 637
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 637

agaauauccu ccuccaucut t 21

<210> SEQ ID NO 638
<211> LENGTH: 21
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 638

agaauauccu ccuccaucua a 21

<210> SEQ ID NO 639
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 639

agaauauccu ccuccaucua a 21

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 640

agaauauccu ccuccaucua a 21

<210> SEQ ID NO 641
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 641

agaauauccu ccuccaucua a 21

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 642

agaauauccu ccuccaucut t 21

<210> SEQ ID NO 643
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<212> TYPE: DNA
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 643

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 644
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<212> TYPE: DNA
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 644

tttccgagaa uuugaggucu ua 22

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<212> TYPE: DNA
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<221> NAME/KEY: source
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 645

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 646
<211> LENGTH: 22
<212> TYPE: DNA
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<220> FEATURE:

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 646

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 647
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<212> TYPE: DNA
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 647

tttccgagaa uuugaggucu ua 22

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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 648

tttccgagaa uuugaggucu ua 22

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<212> TYPE: DNA
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 649

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 650
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

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<400> SEQUENCE: 650

tttccgagaa uuugaggucu ua

22

<210> SEQ ID NO 651

<211> LENGTH: 22

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 651

tttccgagaa uuugaggucu ua

22

<210> SEQ ID NO 652

<211> LENGTH: 22

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 652

tttccgagaa uuugaggucu ua

22

<210> SEQ ID NO 653

<211> LENGTH: 22

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 653

tttccgagaa uuugaggucu ua

22

<210> SEQ ID NO 654

<211> LENGTH: 22

<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 654

tttccgagaa uuugaggucu ua

22

<210> SEQ ID NO 655

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 655

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 656
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 656

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 657
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 657

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 658
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 658

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 659
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 659

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 660
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 660

tttccgagaa uuugaggucu ua 22

<210> SEQ ID NO 661
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 661

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 662
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 662

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 663
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 663

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tttccgagaa uuugaggucu tt 22

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 664

tttccgagaa uuugaggucu tt 22

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 665

tttccgagaa uuugaggucu tt 22

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<212> TYPE: DNA
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 666

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 667
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 667

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 668
<211> LENGTH: 22
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 668

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 669
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 669

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 670
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 670

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 671
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 671

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 672
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 672

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 673
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<212> TYPE: DNA
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 673

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 674
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 674

tttccgagaa uuugaggucu tt 22

<210> SEQ ID NO 675
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"
<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

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Molecule: Synthetic oligonucleotide"

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tttccgagaa uuugaggucu tt 22

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Molecule: Synthetic oligonucleotide"

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 678

tttccgagaa uuugaggucu tt 22

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Molecule: Synthetic oligonucleotide"

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Molecule: Synthetic oligonucleotide"

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<400> SEQUENCE: 681

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<400> SEQUENCE: 682

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tttccgagaa uuugaggucu uatt 24

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tttccgagaa uuugaggucu uatt 24

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tttccgagaa uuugaggucu uatt 24

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<400> SEQUENCE: 687

tttccgagaa uuugaggucu uatt 24

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 688

tttccgagaa uuugaggucu uatt 24

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 689

tttccgagaa uuugaggucu uatt 24

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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tttccgagaa uuugaggucu uatt                                     24

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<400> SEQUENCE: 693

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
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<400> SEQUENCE: 694

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 695

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Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 696

tttccgagaa uuugaggucu uatt                                     24

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<400> SEQUENCE: 697

uaagaccuca aaucucgga a                                       21

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 698

uaagaccuca aaucucgga a                                       21

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:

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Synthetic oligonucleotide"

<400> SEQUENCE: 699

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 700

uaagaccuca aaucucggt t 21

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<221> NAME/KEY: source
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<400> SEQUENCE: 701

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Synthetic oligonucleotide"

<400> SEQUENCE: 702

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<211> LENGTH: 21
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 703

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

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<400> SEQUENCE: 704

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<211> LENGTH: 21

<212> TYPE: RNA

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<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 705

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<212> TYPE: RNA

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<212> TYPE: RNA

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 707

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21

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<212> TYPE: RNA

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<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 708

uaagaccuca aaucucgga a

21

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<212> TYPE: DNA

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<221> NAME/KEY: source

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<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

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uaagaccuca aaucucgga a 21

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 711

uaagaccuca aaucucgga a 21

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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 712

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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 713

uaagaccuca aaucucgga a 21

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<221> NAME/KEY: source
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 714

uaagaccuca aaucucggt t 21

<210> SEQ ID NO 715
<211> LENGTH: 21
<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 715

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 716
<211> LENGTH: 21
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 716

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 717
<211> LENGTH: 21
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 717

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 718
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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 718

uaagaccuca aaucucggt t 21

<210> SEQ ID NO 719
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<212> TYPE: RNA
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<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 719

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<212> TYPE: RNA
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

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<400> SEQUENCE: 720

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 721

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<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 721

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 722

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<212> TYPE: DNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 722

uaagaccuca aaucucggt t 21

<210> SEQ ID NO 723

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<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 723

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<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 724

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 725

<211> LENGTH: 21

<212> TYPE: RNA

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<221> NAME/KEY: source

<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 725

uaagaccuca aaucucgga a 21

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<210> SEQ ID NO 726
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<213> ORGANISM: Artificial Sequence
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<221> NAME/KEY: source
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Synthetic oligonucleotide"

<400> SEQUENCE: 726

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 727
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<212> TYPE: DNA
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<221> NAME/KEY: source
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Synthetic oligonucleotide"

<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 727

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<210> SEQ ID NO 728
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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 728

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 729
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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 729

uaagaccuca aaucucgga a 21

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
Synthetic oligonucleotide"

<400> SEQUENCE: 730

uaagaccuca aaucucgga a 21

<210> SEQ ID NO 731
<211> LENGTH: 21
<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:

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<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 731

uaagaccuca aaucucgga a                                     21

<210> SEQ ID NO 732
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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
    Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 732

uaagaccuca aaucucggt t                                     21

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<220> FEATURE:
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<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 733

uaagaccuca aaucucgga a                                     21

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<212> TYPE: RNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 734

uaagaccuca aaucucgga a                                     21

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<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<400> SEQUENCE: 735

uaagaccuca aaucucgga a                                     21

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<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<223> OTHER INFORMATION: /note="Description of Artificial Sequence:
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<220> FEATURE:
<221> NAME/KEY: source

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<223> OTHER INFORMATION: /note="Description of Combined DNA/RNA
Molecule: Synthetic oligonucleotide"

<400> SEQUENCE: 736

uaagaccuca aaucucggt t                                     21

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<213> ORGANISM: Artificial Sequence
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The invention claimed is:

1. A double-stranded ribonucleic acid (dsRNA) that inhibits expression of a human angiopoietin-like protein 8 (ANGPTL8) gene by targeting a target sequence on an RNA transcript of the ANGPTL8 gene, wherein the dsRNA comprises a sense strand comprising a sense sequence, and an antisense strand comprising an antisense sequence, wherein the sense sequence is at least 90% identical to the target sequence, wherein the target sequence is nucleotides 457-475 of SEQ ID NO: 529, and wherein the dsRNA is conjugated to one or more ligands with or without a linker.

2. The dsRNA of claim 1, wherein the sense strand and the antisense strand are complementary to each other over a region of 15-25 contiguous nucleotides, and/or the sense strand and the antisense strand are no more than 30 nucleotides in length.

3. The dsRNA of claim 1, wherein the dsRNA comprises an antisense sequence that is at least 90% identical to the nucleotide sequence of SEQ ID NO: 174.

4. The dsRNA of claim 1, wherein the sense sequence and the antisense sequence are complementary, wherein:

- a) the sense sequence comprises the nucleotide sequence of SEQ ID NO: 42; and/or
- b) the antisense sequence comprises the nucleotide sequence of SEQ ID NO: 174.

5. The dsRNA of claim 1, wherein one or both of the sense strand and the antisense strand further comprise:

- a) a 5' overhang comprising one or more nucleotides; and/or
- b) a 3' overhang comprising one or more nucleotides.

6. The dsRNA of claim 5, wherein an overhang in the dsRNA comprises two or three nucleotides, and/or one or more thymines.

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7. The dsRNA of claim 1, wherein the ligand is a cholesterol derivative, a lipophilic moiety, or N-acetylgalactosamine (GalNAc) and the dsRNA is conjugated to one or more GalNAc.

8. The dsRNA of claim 1, wherein the dsRNA is a small interfering RNA (siRNA), or a short hairpin RNA (shRNA).

9. The dsRNA of claim 1, comprising one or more internucleoside linking groups independently selected from the group consisting of phosphodiester, phosphotriester, phosphorothioate, phosphorodithioate, alkyl-phosphonate and phosphoramidate backbone linking groups, or a pharmaceutically acceptable salt thereof.

10. A pharmaceutical composition comprising the dsRNA of claim 1 and a pharmaceutically acceptable excipient.

11. The dsRNA of claim 1, wherein the dsRNA comprises one or more modified nucleotides,

wherein at least one of the one or more modified nucleotides is 2'-deoxy-2'-fluoro-ribonucleotide, 2'-deoxyribonucleotide, or 2'-O-methyl-ribonucleotide, and/or

wherein the dsRNA comprises an inverted 2'-deoxyribonucleotide at the 3'-end of its sense or antisense strand, and/or

wherein the sense sequence and the antisense sequence comprise alternating 2'-O-methyl ribonucleotides and 2'-deoxy-2'-fluoro ribonucleotides.

12. The dsRNA of claim 1, wherein:

a) the sense sequence comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 643-696; and/or

b) the antisense sequence comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 697-750.

13. A double-stranded ribonucleic acid (dsRNA) that inhibits expression of a human angiopoietin-like protein 8 (ANGPTL8) gene by targeting a target sequence on an RNA transcript of the ANGPTL8 gene, wherein the dsRNA com-

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prises a sense strand comprising a sense sequence, and an antisense strand comprising an antisense sequence, wherein the sense sequence is at least 90% identical to the target sequence, wherein the target sequence is nucleotides 457-475 of SEQ ID NO: 529, and wherein the dsRNA comprises one or more modified nucleotides,

wherein at least one of the one or more modified nucleotides is 2'-deoxy-2'-fluoro-ribonucleotide, 2'-deoxyribonucleotide, or 2'-O-methyl-ribonucleotide, and/or

wherein the dsRNA comprises an inverted 2'-deoxyribonucleotide at the 3'-end of its sense or antisense strand, and/or

wherein the sense sequence and the antisense sequence comprise alternating 2'-O-methyl ribonucleotides and 2'-deoxy-2'-fluoro ribonucleotides.

14. A pharmaceutical composition comprising the dsRNA of claim 13 and a pharmaceutically acceptable excipient.

15. A double-stranded ribonucleic acid (dsRNA) that inhibits expression of a human angiopoietin-like protein 8 (ANGPTL8) gene by targeting a target sequence on an RNA transcript of the ANGPTL8 gene, wherein the dsRNA comprises a sense strand comprising a sense sequence, and an antisense strand comprising an antisense sequence, wherein the sense sequence is at least 90% identical to the target sequence, wherein the target sequence is nucleotides 457-475 of SEQ ID NO: 529, and wherein:

a) the sense sequence comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 643-696; and/or

b) the antisense sequence comprises a nucleotide sequence selected from the group consisting of SEQ ID NOs: 697-750.

16. A pharmaceutical composition comprising the dsRNA of claim 15 and a pharmaceutically acceptable excipient.

* * * * *